

कार्यालय : बी-9, प्रथम एवं द्वितीय तल, कुतुब इंस्टीट्यूशनल एरिया, कटवारिया सराय, नई दिल्ली - 110016
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संदर्भ संख्या:- GRID-INDIA/NLDC/MR/

दिनांक: 24.08.2023

सेवा में,

वितरण सूची के अनुसार

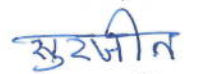
विषय:- निष्पादन रिपोर्ट – जुलाई 2023

महोदय,

आई०ई०जी०सी०-2010 के प्रावधान के अनुसार, जुलाई 2023 माह की अखिल भारतीय प्रणाली की निष्पादन रिपोर्ट संलग्न है।

धन्यवाद

भवदीय



(सुरजीत बनर्जी)

मुख्य महाप्रबन्धक (प्रणाली प्रचालन)

(रा.भा.प्रे.के.)

संलग्नक: मासिक प्रणाली रिपोर्ट

वितरण सूची Distribution List

- 1 सचिव, के.वि.नि.आ. तीसरा एवं चौथा तल, चंद्रलोक भवन, 36, जनपथ, नई दिल्ली-110001
Secretary, CERC, 3rd & 4th Floor, Chanderlok Building, 36, Janpath, New Delhi-110001
- 2 मुख्य अभियंता (जी एम), के.वि.प्रा., सेवा भवन, आर. के. पुरम, नई दिल्ली-110066
Chief Engineer (GM), CEA, Sewa Bhavan, R.K.Puram, New Delhi-110066
- 3 सदस्य सचिव, उ. क्षेत्र. वि. स., 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Member Secretary, NRPC, 18/A, SJSS Marg, Katwaria Sarai, New Delhi-110016
- 4 सदस्य सचिव, प. क्षेत्र. वि. स., एफ-3, एम आई डी सी क्षेत्र, अंधेरी (पूर्व), मुंबई - 400093
Member Secretary, WRPC, F-3, MIDC Area, Andheri (East), Mumbai-400093
- 5 सदस्य सचिव, द. क्षेत्र. वि. स., 29, रेस कोर्स क्रॉस रोड, बंगलूरु - 560009
Member Secretary, SRPC, 29, Race Course Cross Road, Bangalore-560009
- 6 सदस्य सचिव, पू. क्षेत्र. वि. स., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Member Secretary, ERPC, 14, Golf Club Road, Kolkata-700033
- 7 सदस्य सचिव, उ. पू. क्षेत्र. वि. स., मेघालय राज्य आवासीय वित्त सहकारी समिति लिमिटेड भवन, नॉग्रिम हिल्स, शिलोंग - 793003
Member Secretary, NERPC, Meghalaya State Housing Finance Cooperative Society Ltd. Building, Nongrim Hills, Shillong -793003
- 8 मुख्य अभियंता, राष्ट्रीय विद्युत समिति, एनआरपीसी भवन, तृतीय तल, 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Chief Engineer, National Power Committee, NRPC Building, 3rd floor, 18/A, SJSS Marg, Katwaria Sarai, New Delhi-110016
- 9 कार्यपालक निदेशक, पू. क्षेत्र. भा. प्रे. के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14, Golf Club Road, Kolkata-700033
- 10 कार्यपालक निदेशक, प. क्षेत्र. भा. प्रे. के., एफ-3, एम आई डी सी क्षेत्र, अंधेरी (पूर्व), मुंबई - 400093
Executive Director, WRLDC, F-3, MIDC Area, Andheri (East), Mumbai-400093
- 11 कार्यपालक निदेशक, उ. क्षेत्र. भा. प्रे. के., 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRPC, 18/A, SJSS Marg, Katwaria Sarai, New Delhi-110016
- 12 कार्यपालक निदेशक, द. क्षेत्र. भा. प्रे. के., 29, रेस कोर्स क्रॉस रोड, बंगलूरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009
- 13 मुख्य महाप्रबंधक, पू. क्षेत्र. भा. प्रे. के., डोंगतिह, लोअर नॉग्रह, लापलंग, शिलोंग- 793006
Chief General Manager, NERLDC, Dongtiah, Lower Nongrah, Laplang, Shillong-793006



ग्रिड-इंडिया
GRID-INDIA

GRID CONTROLLER OF INDIA LIMITED

Formerly Power System Operation Corporation Limited

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

NATIONAL LOAD DESPATCH CENTRE –राष्ट्रीय भार प्रेषण केन्द्र

मासिक ऑपरेशन रिपोर्ट

Monthly
Operation Report



जुलाई 2023

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड
राष्ट्रीय भार प्रेषण केंद्र, नई दिल्ली

GRID CONTROLLER OF INDIA LIMITED

Formerly Power System Operation Corporation Limited

NATIONAL LOAD DESPATCH CENTRE, NEW DELHI



ग्रिड-इंडिया
GRID-INDIA

माह जुलाई 2023 के लिए प्रचालन निष्पादन रिपोर्ट

OPERATIONAL PERFORMANCE REPORT FOR THE

MONTH OF JULY-2023

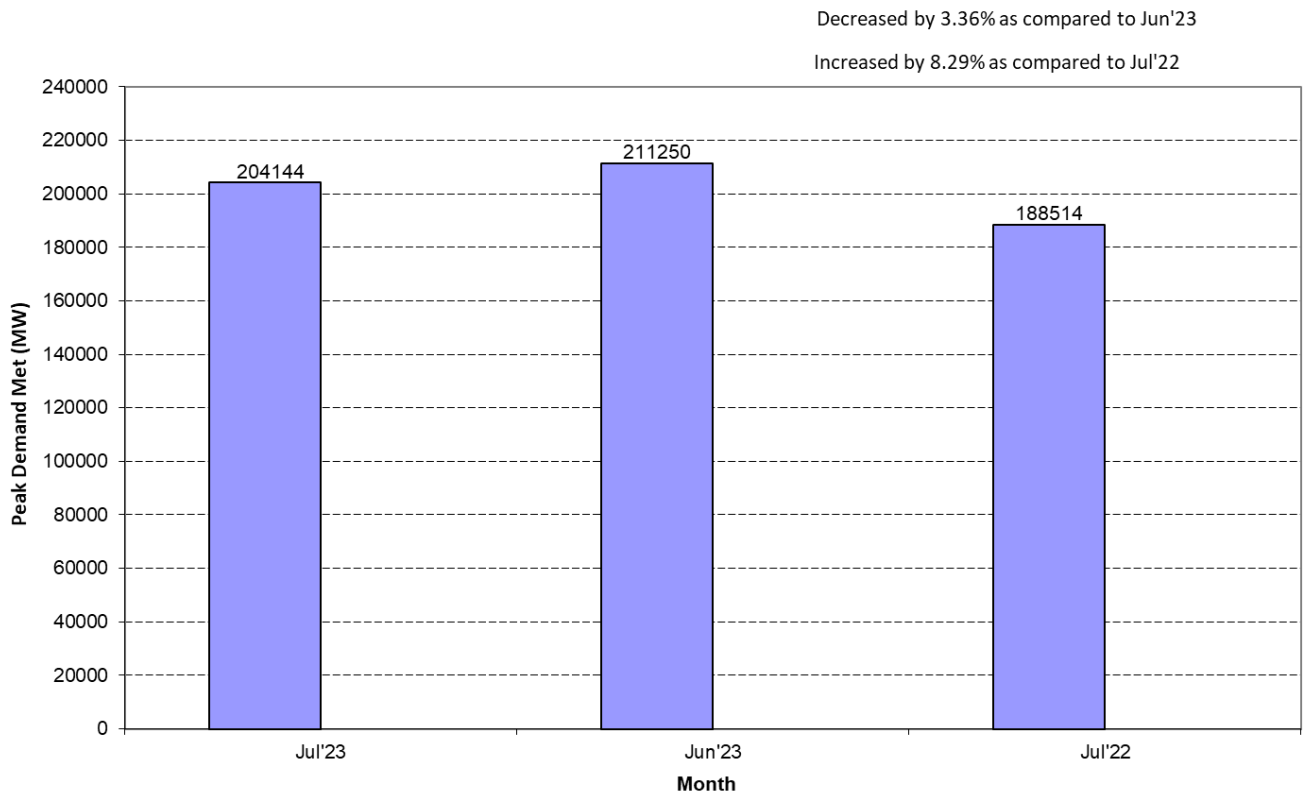
पृष्ठसूची CONTENTS

1	माह जुलाई 2023 की सारांश रिपोर्ट Summary of report for the month of JULY 2023	1-5
2	क्षेत्रवार स्थापित विद्युत क्षमता Region wise Installed Capacity	6
3	राष्ट्रीय स्तर पर सांध्यकालीन शिखर मांगपूर्ति Peak demand met at National level	7-8
4	राष्ट्रीय स्तर पर विद्युत ऊर्जा खपत Energy consumption at National level	9-10
5	फ्रिक्वेंसी रूपरेखा Frequency profile	11-12
6	राष्ट्रीय स्तर पर जल विद्युत ऊर्जा उत्पादन Hydro generation at National level	13-14
7	राष्ट्रीय स्तर पर पवन ऊर्जा उत्पादन Wind generation at National level	15
8	राष्ट्रीय स्तर पर सौर ऊर्जा उत्पादन Solar generation at National level	16
9	राष्ट्रीय स्तर पर दैनिक अधिकतम विद्युत मांग पूर्ति Daily maximum demand met at National level	17
10	21 जुलाई के माँग एवं उत्पादन प्लॉट (अधिकतम मांग पूर्ति) व माँग का डाटा (15 मिनट अंतराल) Demand and generation plots for 21-JULY (Max Demand Met) and Demand data (15-minute interval)	18-22
11	ऊर्जा आपूर्ति की वास्तविक स्थिति Actual power supply position	23
12	ऊर्जा आवश्यकता एवं आपूर्ति की पिछले वर्ष माह जुलाई 2022 से तुलना Energy requirement and met comparison with JULY 2022	24
13	शिखर मांग आवश्यकता एवं आपूर्ति की पिछले वर्ष माह जुलाई 2022 से तुलना Peak demand requirement and met comparison with JULY 2022	25

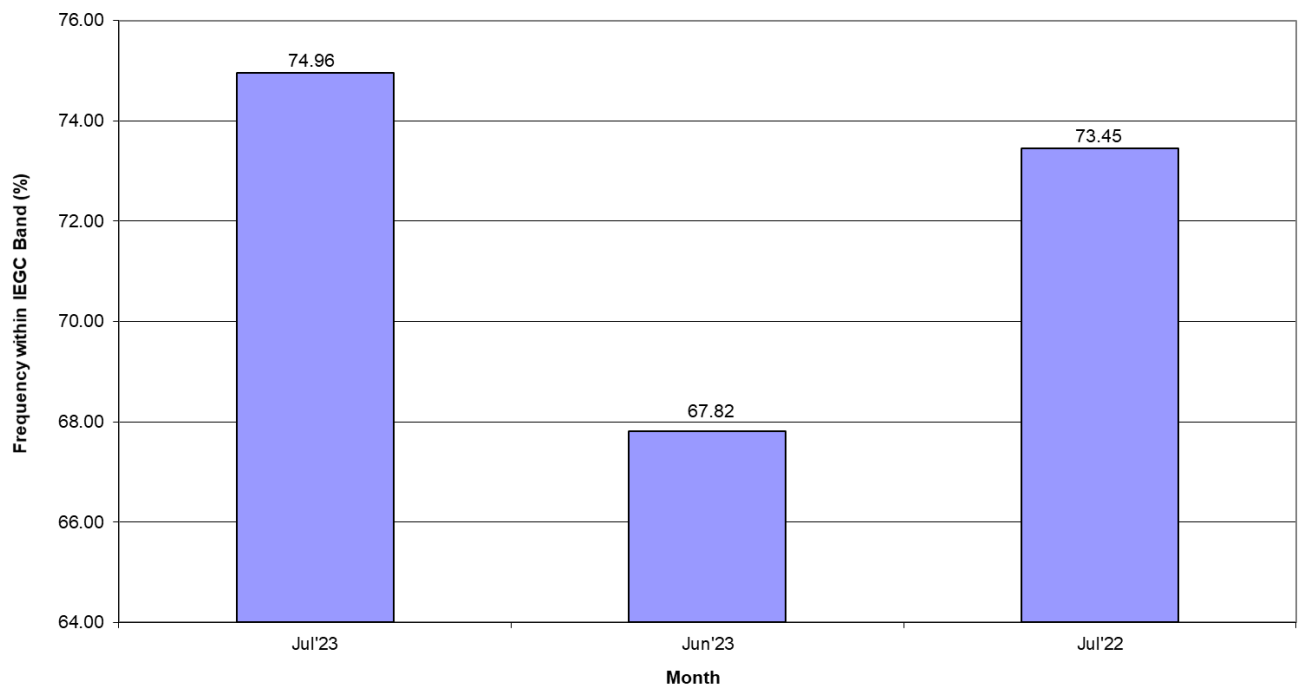
14	क्षेत्रीय स्तर पर शिड्यूल एवं ड्रावल Schedule & drawal at regional level	26
15	अंतर क्षेत्रीय विनिमय Inter-regional exchanges	27-36
16	अंतर राष्ट्रीय विद्युत विनिमय Inter-national exchanges	37-39
17	एसटीओए (द्विपक्षीय एवं सामूहिक) एवं यू आइ का ब्यौरा Bilateral and collective short-term open access & UI details	40
18	पावर मार्केट की सूचना Power market information	41-44
19	अक्षय ऊर्जा प्रमाण पत्र प्रक्रिया Renewable energy certificate mechanism	45-47
20	मुख्य घटनाएँ एवं डिस्टर्बेंस Important events/disturbances	48-63
21	नए कमीशंड पारेषण तत्व New elements commissioned	64
22	विभव रूपरेखा Voltage profile	65
23	जुलाई 2023 के दौरान रा. भा. प्रे. के. द्वारा निगरानी में तत्वों के आउटेज की सूची List of outage of elements monitored by NLDC for the month of JULY 2023	66-80
24	अब तक उच्चतम (मांग व ऊर्जा) आँकड़े : माह जुलाई 2023 तक All Time Highest (Demand & Energy) figures: Till JULY 2023	81
25	जुलाई 2023 के लिए सिस्टम रिलाइबिलिटी सूचकांक System Reliability Indices for JULY 2023	82

1. SUMMARY OF REPORT FOR THE MONTH OF JULY-2023

EVENING PEAK HOUR (at 2000hrs) DEMAND MET AT NATIONAL LEVEL (MW)



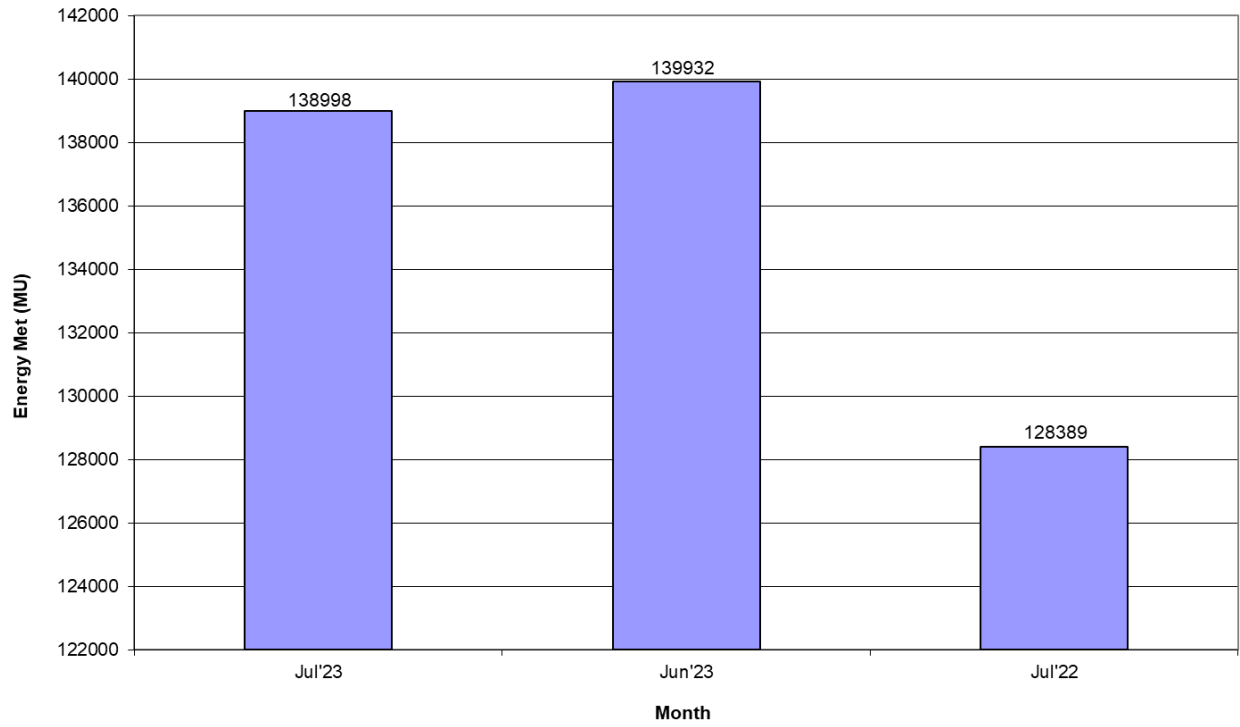
Frequency within IEGC Band



ENERGY MET AT NATIONAL LEVEL (MU)

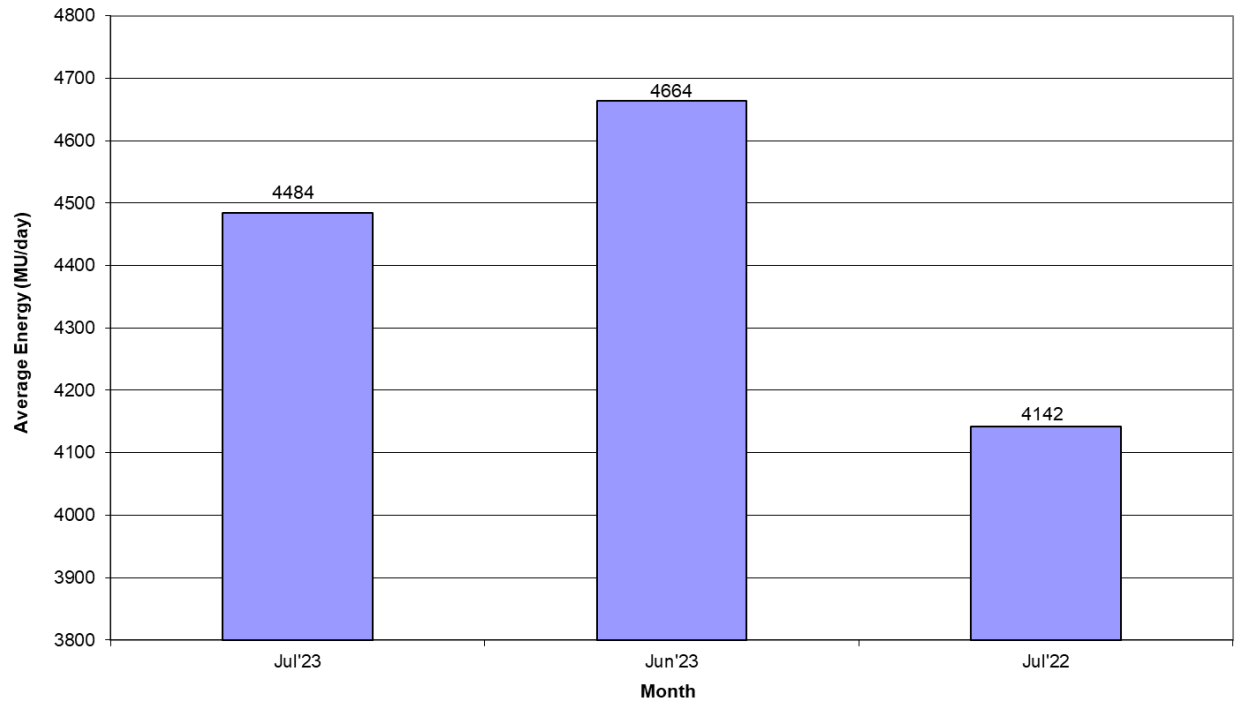
Decreased by 0.67% as compared to Jun'23

Increased by 8.26% as compared to Jul'22



AVERAGE ENERGY MET AT NATIONAL LEVEL (MU/Day)

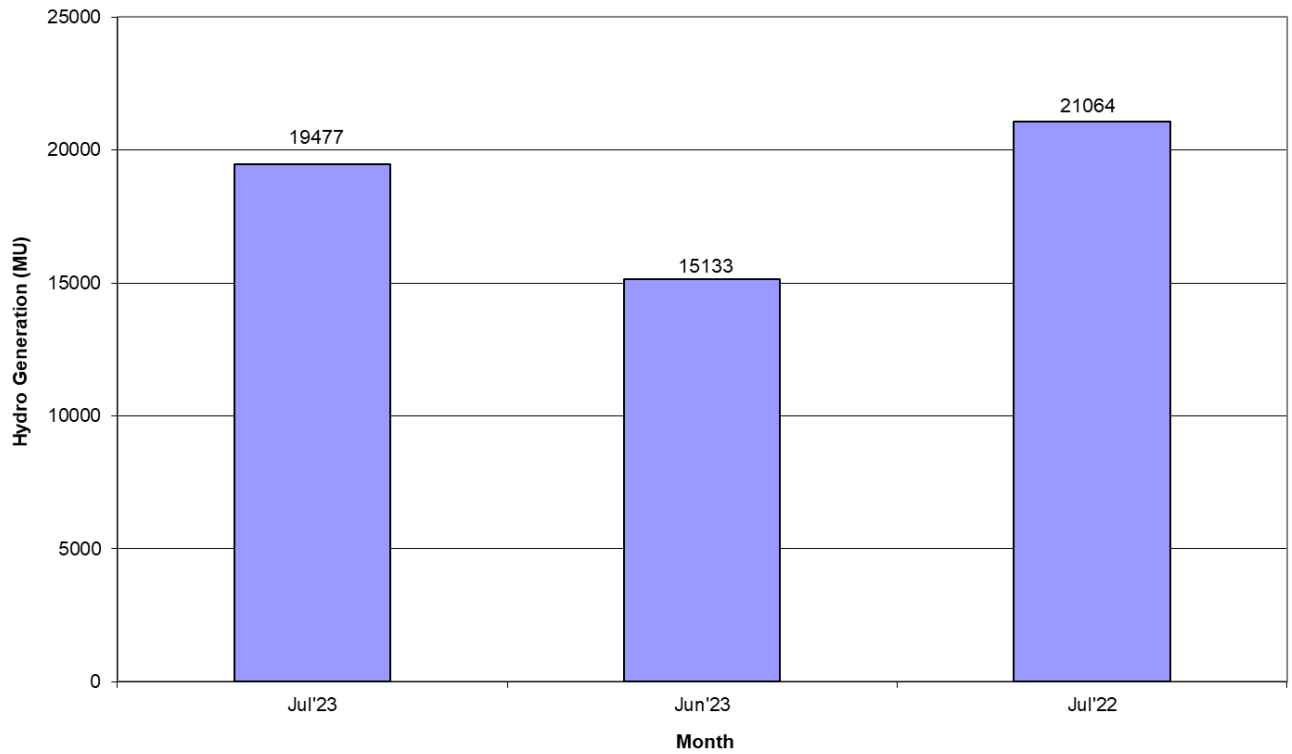
Increased by 8.26% as compared to Jul'22



HYDRO GENERATION AT NATIONAL LEVEL (MU)

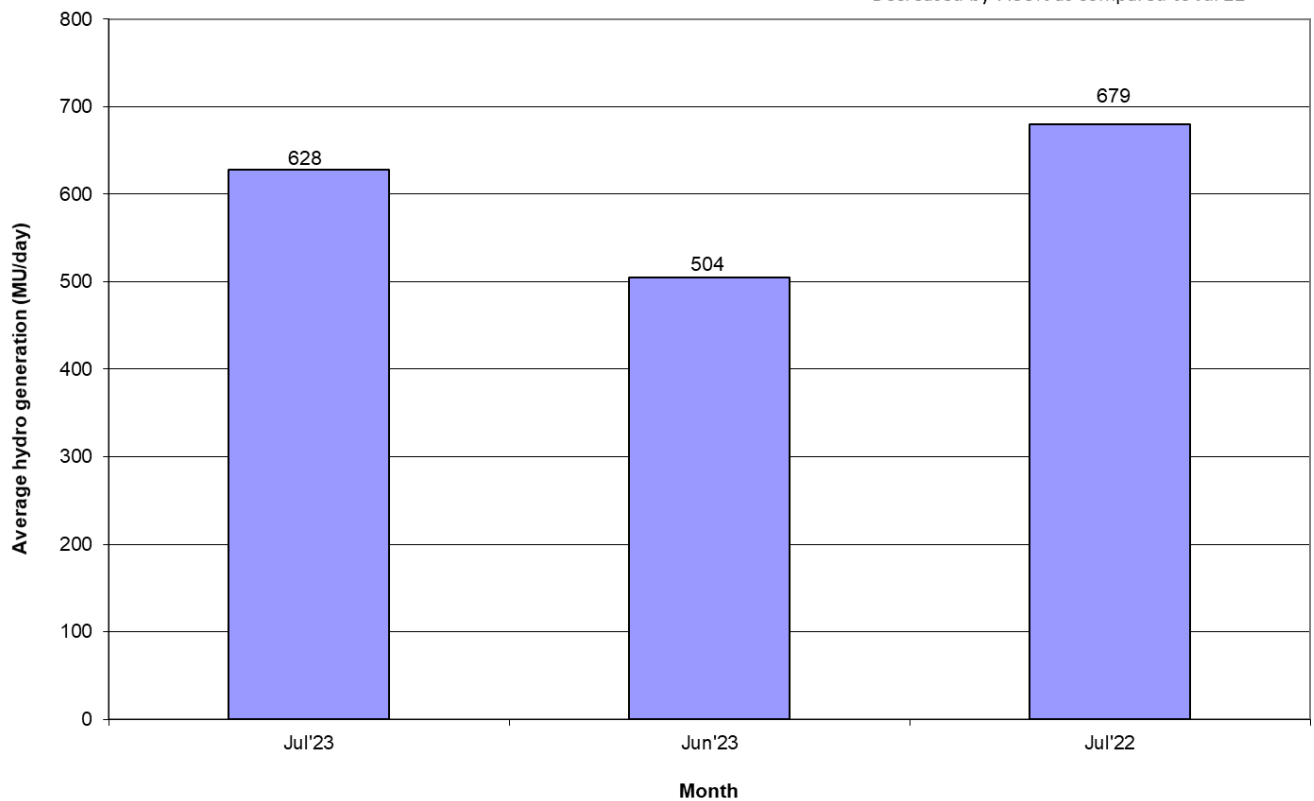
Increased by 28.70% as compared to Jun'23

Decreased by 7.53% as compared to Jul'22



AVERAGE HYDRO GENERATION AT NATIONAL LEVEL (MU/Day)

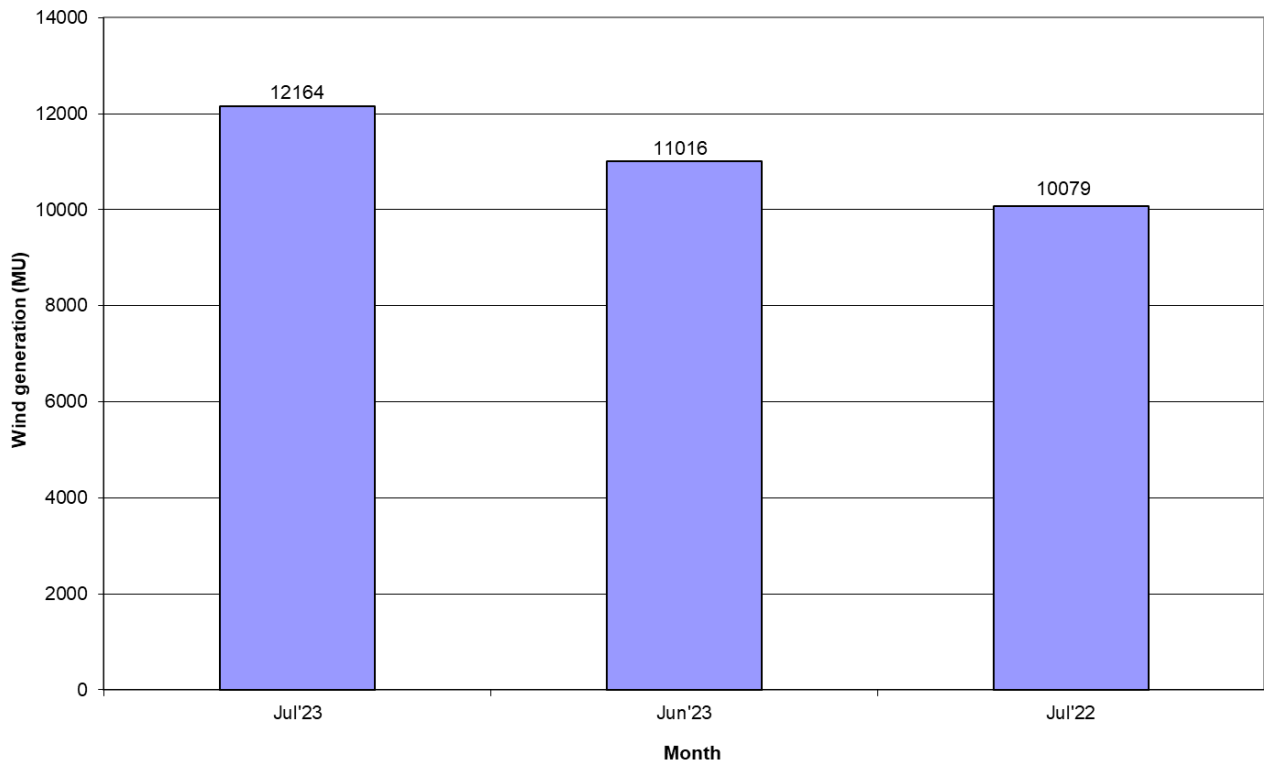
Decreased by 7.53% as compared to Jul'22



WIND GENERATION AT NATIONAL LEVEL (MU)

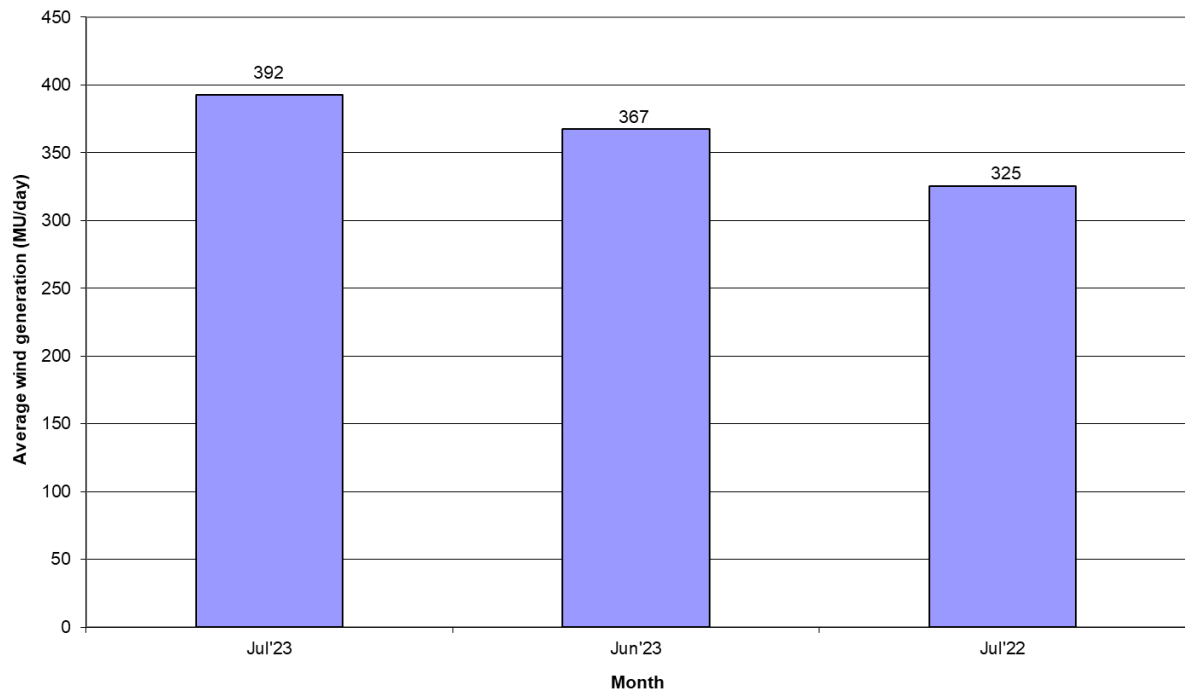
Increased by 10.42% as compared to Jun'23

Increased by 20.68% as compared to Jul'22



AVERAGE WIND GENERATION AT NATIONAL LEVEL (MU/Day)

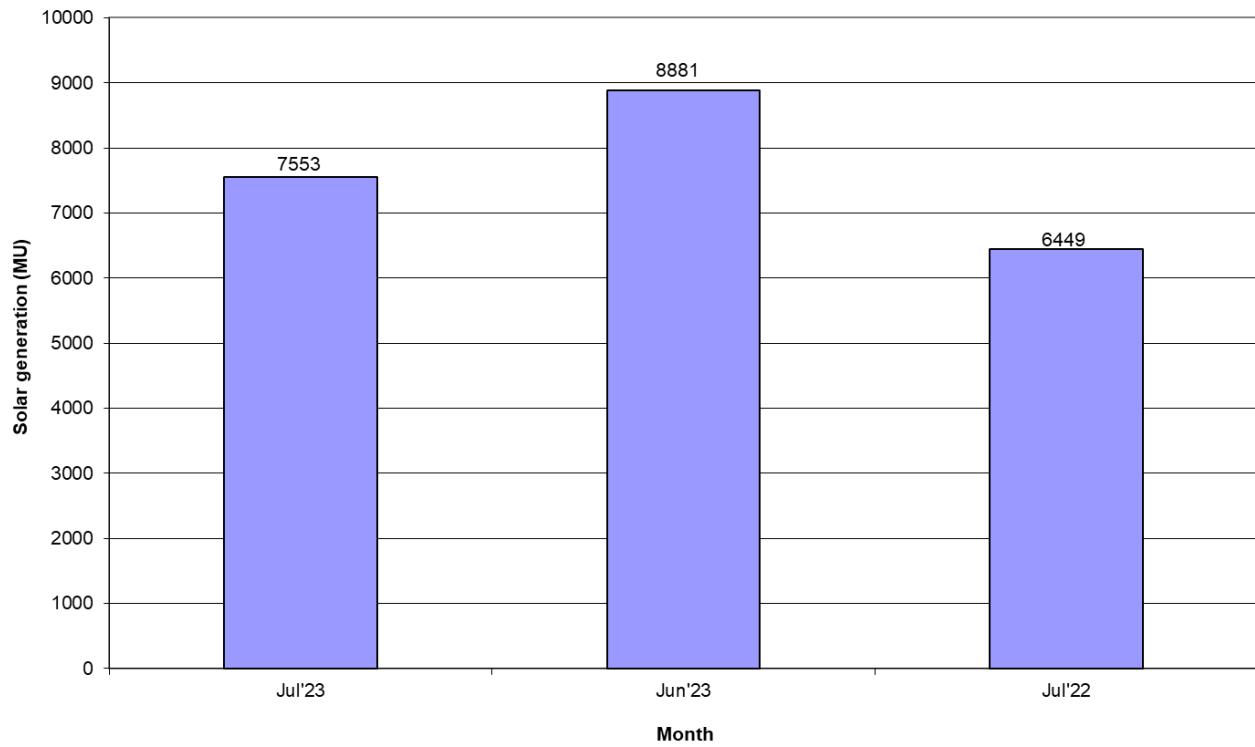
Increased by 20.68% as compared to Jul'22



SOLAR GENERATION AT NATIONAL LEVEL (MU)

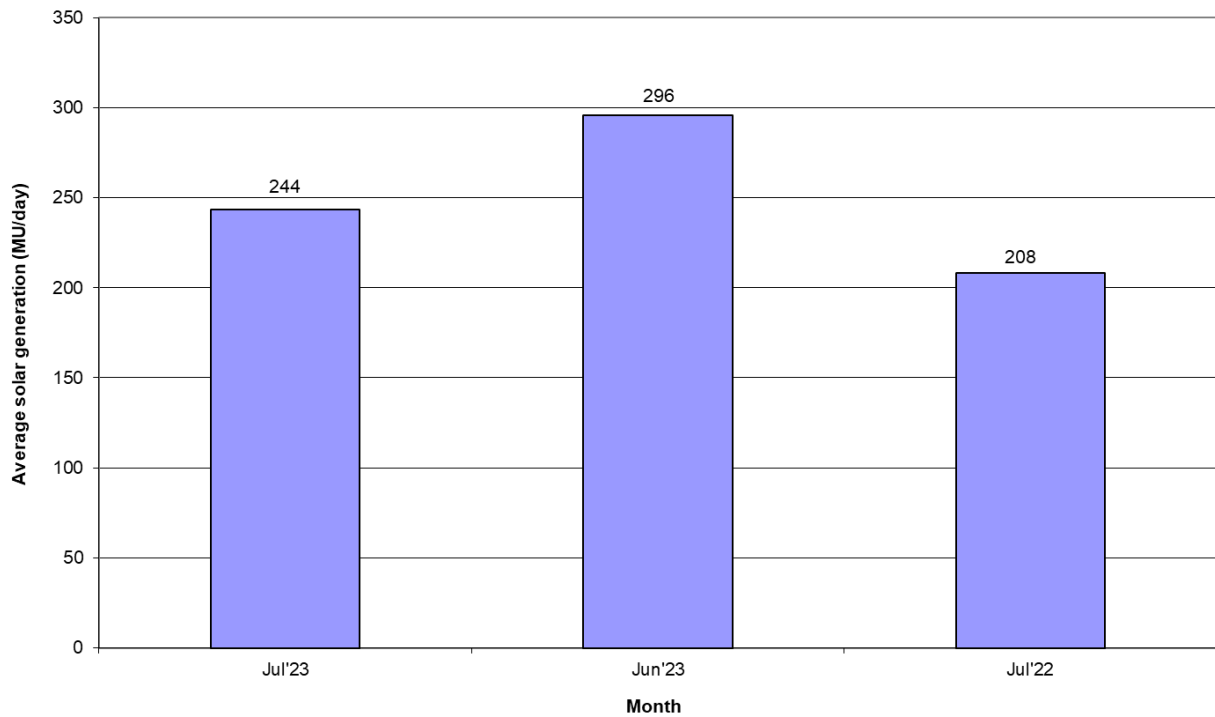
Decreased by 14.96% as compared to Jun'23

Increased by 17.12% as compared to Jul'22



AVERAGE SOLAR GENERATION AT NATIONAL LEVEL (MU/Day)

Increased by 17.12% as compared to Jul'22



2. ALL INDIA REGIONWISE INSTALLED CAPACITY

As on 31.07.2023
(All figures are in MW)

S No	Region	THERMAL					NUCLEAR	HYDRO	RES @ MNRE	GRAND TOTAL
		COAL	LIGNITE	GAS	DIESEL	TOTAL				
1	NR	55437	1580	5995	0	63012	1620	20752	34403	119786
2	WR	74602	1400	10806	0	86808	2540	7563	42065	138976
3	SR	46858	3640	6492	434	57424	3320	11827	51934	124505
4	ER	28229	0	80	0	28309	0	4764	1872	34945
5	NER	770	0	1665	36	2471	0	1944	574	4989
6	ISLANDS	0	0	0	120	120	0	0	38	158
	ALL INDIA	205895	6620	25038	589	238143	7480	46850	130886	423359

Remarks: Data from CEA website

3. राष्ट्रीय स्तर पर संध्याकालीन शिखर विद्युत मांग पूर्ति
EVENING PEAK HOUR DEMAND MET AT NATIONAL LEVEL

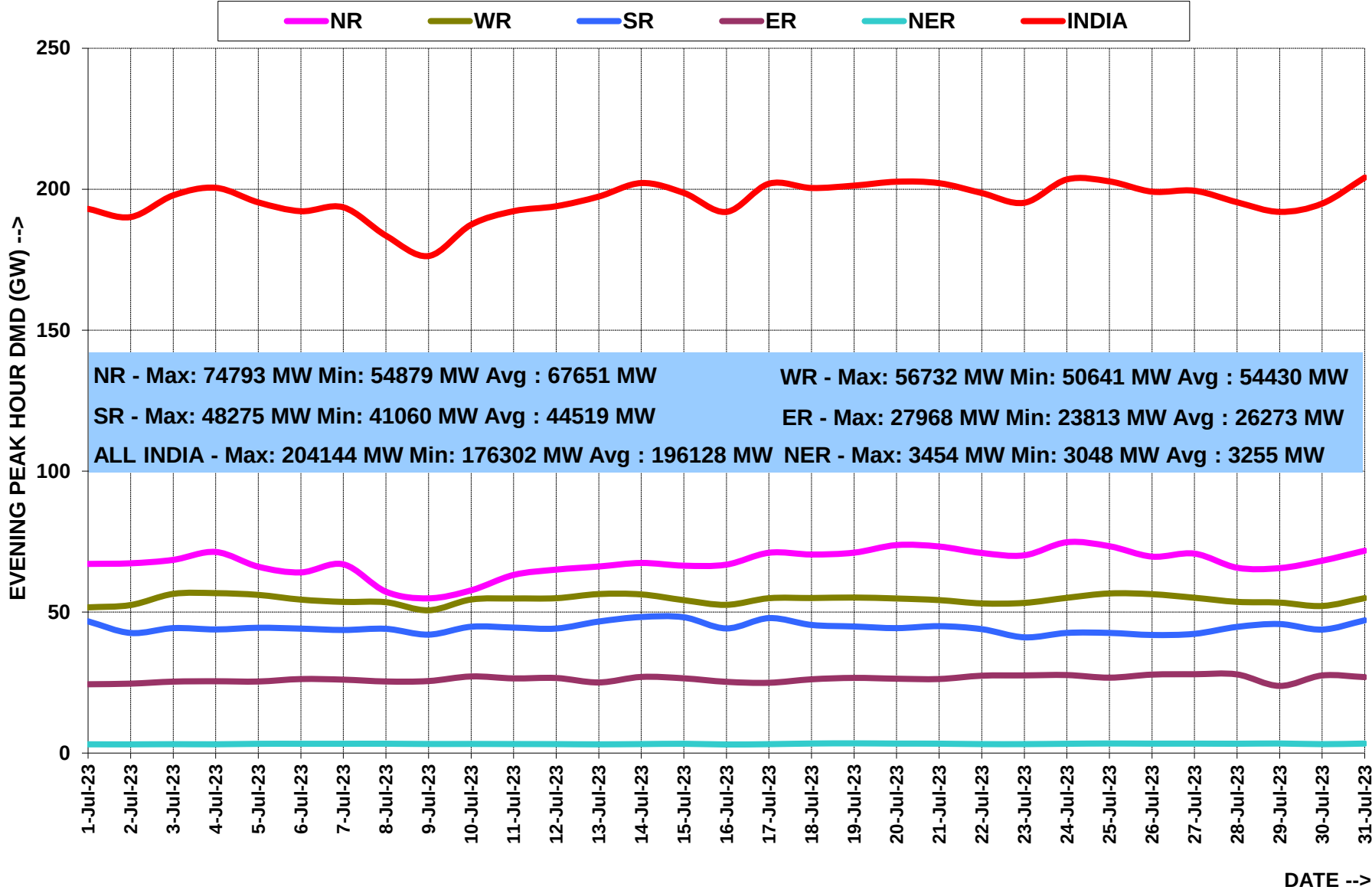
माह: जुलाई 2023 MONTH:- JUL 2023

सभी आंकड़े मेगावाट में All figures in MW

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
1-Jul-23	67100	51690	46722	24407	3111	193030
2-Jul-23	67317	52503	42599	24630	3086	190135
3-Jul-23	68529	56468	44321	25347	3155	197820
4-Jul-23	71334	56732	43853	25477	3115	200511
5-Jul-23	66103	56104	44448	25387	3295	195337
6-Jul-23	64081	54429	44142	26259	3299	192210
7-Jul-23	66943	53624	43659	26048	3308	193582
8-Jul-23	57236	53509	44075	25392	3318	183530
9-Jul-23	54879	50641	42012	25547	3223	176302
10-Jul-23	57721	54469	44813	27177	3239	187419
11-Jul-23	63185	54836	44508	26486	3206	192221
12-Jul-23	65066	54929	44177	26628	3173	193973
13-Jul-23	66179	56405	46660	25054	3094	197392
14-Jul-23	67437	56281	48275	27012	3193	202198
15-Jul-23	66470	54240	48175	26515	3298	198698
16-Jul-23	66841	52586	44215	25270	3048	191960
17-Jul-23	71061	54935	47876	24937	3160	201969
18-Jul-23	70438	55001	45446	26168	3389	200442
19-Jul-23	71085	55171	44905	26669	3454	201284
20-Jul-23	73774	54831	44320	26378	3367	202670
21-Jul-23	73272	54258	45012	26271	3337	202150
22-Jul-23	70987	53089	43960	27444	3173	198653
23-Jul-23	70144	53268	41060	27555	3178	195205
24-Jul-23	74793	55077	42620	27674	3309	203473
25-Jul-23	73381	56624	42624	26748	3399	202776
26-Jul-23	69690	56364	41894	27838	3351	199137
27-Jul-23	70749	55095	42298	27968	3358	199468
28-Jul-23	65750	53638	44766	27902	3323	195379
29-Jul-23	65610	53382	45771	23813	3398	191974
30-Jul-23	68237	52176	43816	27529	3161	194919
31-Jul-23	71785	54963	47082	26938	3376	204144
उच्चतम MAXIMUM	74793	56732	48275	27968	3454	204144
निम्नतम MINIMUM	54879	50641	41060	23813	3048	176302
औसत AVERAGE	67651	54430	44519	26273	3255	196128
अब तक का उच्चतम All Time Max.	76072	63593	52366	28579	3510	211250
दिनांक Date	23.06.23	18.04.23	18.04.23	18.04.23	12.08.22	22.06.23

नोट : सभी आंकड़े 2000 बजे की मांगपूर्ति पर आधारित है ।
 Note: The above figures correspond to value at 2000 Hrs

EVENING PEAK HOUR DEMAND(at 2000hrs)MET DURING THE MONTH OF JULY 2023



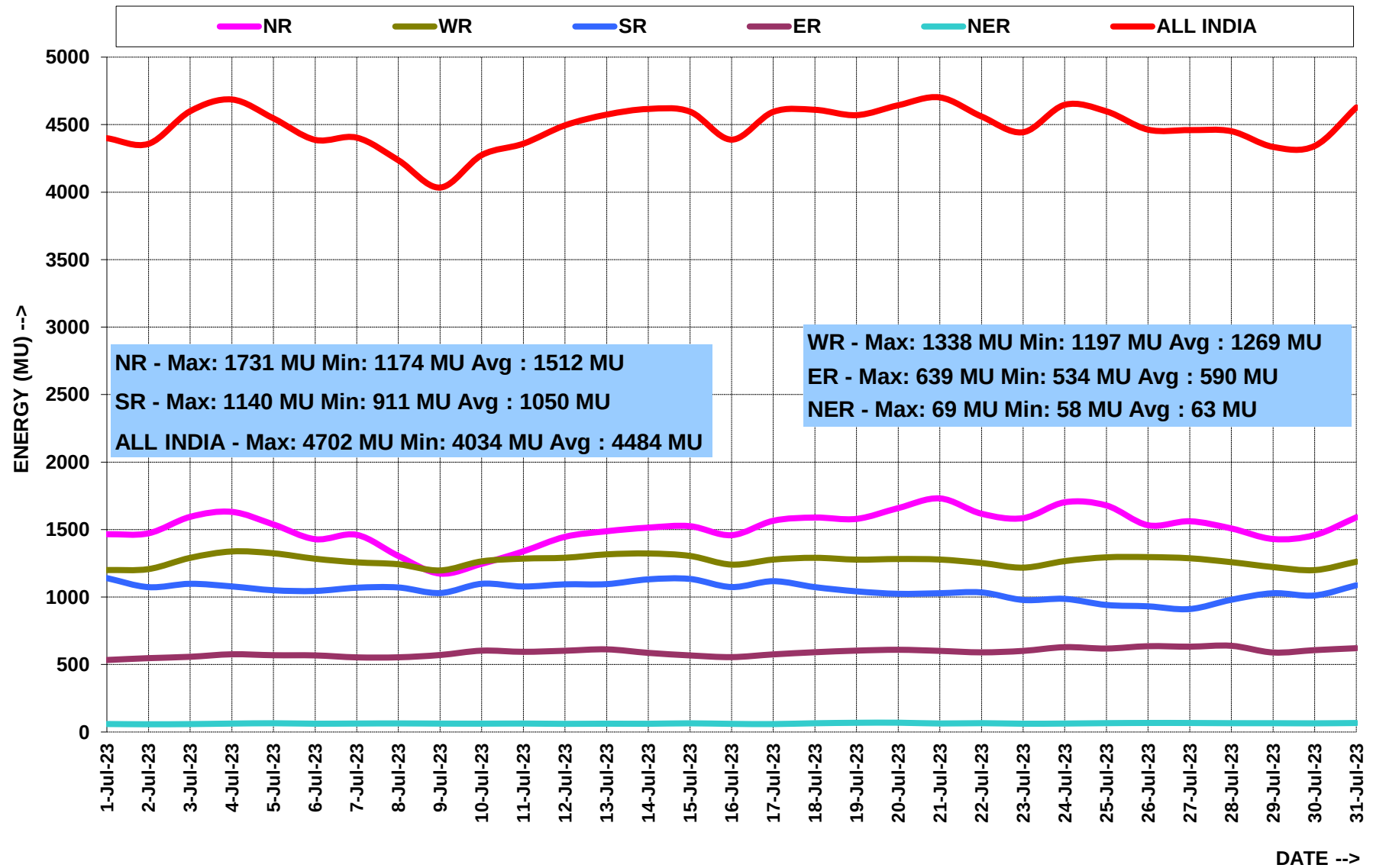
4. राष्ट्रीय स्तर पर विद्युत ऊर्जा आपूर्ति ENERGY MET AT NATIONAL LEVEL

माह: जुलाई 2023 MONTH:- JUL 2023

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jul-23	1466	1200	1140	534	60	4399
02-Jul-23	1472	1207	1073	547	58	4358
03-Jul-23	1594	1291	1098	557	59	4599
04-Jul-23	1631	1338	1079	576	63	4686
05-Jul-23	1538	1324	1050	569	65	4546
06-Jul-23	1429	1284	1045	568	62	4387
07-Jul-23	1461	1257	1069	553	63	4404
08-Jul-23	1304	1243	1071	554	64	4236
09-Jul-23	1174	1197	1030	571	62	4034
10-Jul-23	1246	1265	1098	603	62	4274
11-Jul-23	1339	1285	1078	594	63	4359
12-Jul-23	1446	1291	1094	602	61	4495
13-Jul-23	1488	1316	1096	613	62	4574
14-Jul-23	1514	1323	1131	586	61	4616
15-Jul-23	1525	1305	1134	568	65	4596
16-Jul-23	1458	1241	1074	555	60	4389
17-Jul-23	1565	1278	1117	575	59	4595
18-Jul-23	1589	1291	1073	591	65	4610
19-Jul-23	1579	1277	1042	603	68	4570
20-Jul-23	1659	1282	1024	609	69	4643
21-Jul-23	1731	1278	1028	601	63	4702
22-Jul-23	1617	1253	1034	590	66	4560
23-Jul-23	1585	1218	978	601	62	4444
24-Jul-23	1702	1266	987	629	63	4647
25-Jul-23	1678	1294	941	619	66	4598
26-Jul-23	1532	1296	931	636	67	4463
27-Jul-23	1561	1287	911	633	67	4459
28-Jul-23	1508	1258	980	639	66	4451
29-Jul-23	1430	1222	1028	589	65	4335
30-Jul-23	1459	1200	1012	607	65	4342
31-Jul-23	1591	1262	1087	621	67	4627
कुल TOTAL	46873	39327	32538	18294	1967	138998
उच्चतम MAXIMUM	1731	1338	1140	639	69	4702
निम्नतम MINIMUM	1174	1197	911	534	58	4034
औसत AVERAGE	1512	1269	1050	590	63	4484
संचयी 2023-24 Cumulative 2023-24	163794	167645	138045	68775	6794	545052
अब तक का उच्चतम All Time Max.	1737	1539	1329	639	69	4914
दिनांक Date	28.06.22	18.04.23	19.04.23	08.06.23	15.07.22	13.06.23

ENERGY MET DURING THE MONTH OF JULY' 2023



5. वर्ष 2023-24 के लिए आवृत्ति रूपरेखा FREQUENCY PROFILE FOR YEAR 2023-24

राष्ट्रीय ग्रीड NATIONAL GRID								
फ्रिक्वेंसी रूपरेखा (Hz)			<49.9	49.9-50.05	>50.05	उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी
% समय	Apr-23	All India Grid	10.54	67.90	21.56	50.33	49.49	49.99
	May-23	All India Grid	9.83	68.48	21.69	50.43	49.48	49.99
	Jun-23	All India Grid	6.47	67.82	25.71	50.41	49.51	50.01
	Jul-23	All India Grid	4.60	74.96	20.44	50.23	49.76	50.01
	2023-24 (upto Jul)	All India Grid	7.85	69.82	22.33	50.43	49.48	50.00

5.1 जुलाई 2023 के लिए आवृत्ति रूपरेखा FREQUENCY PROFILE FOR JULY 2023

फ्रिक्वेंसी रूपरेखा (Hz)		<49.9	49.9-50.05	>50.05	उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी	एफ.वी.आई.
% समय	01-Jul-23	8.84	72.60	18.55	50.23	49.75	49.99	0.04
	02-Jul-23	12.53	71.55	15.91	50.27	49.62	49.98	0.07
	03-Jul-23	11.26	71.89	16.85	50.20	49.58	49.98	0.07
	04-Jul-23	2.49	71.81	25.71	50.26	49.83	50.02	0.03
	05-Jul-23	4.47	67.51	28.02	50.23	49.77	50.01	0.05
	06-Jul-23	5.21	68.34	26.45	50.23	49.77	50.01	0.05
	07-Jul-23	4.93	72.86	22.21	50.25	49.79	50.00	0.04
	08-Jul-23	5.38	69.90	24.72	50.30	49.78	50.01	0.04
	09-Jul-23	4.27	79.83	15.90	50.20	49.74	50.00	0.03
	10-Jul-23	7.36	62.80	29.84	50.31	49.70	50.01	0.06
	11-Jul-23	4.51	74.68	20.81	50.20	49.73	50.01	0.04
	12-Jul-23	7.51	74.46	18.03	50.21	49.71	50.00	0.05
	13-Jul-23	4.71	70.84	24.44	50.15	49.73	50.01	0.04
	14-Jul-23	1.40	71.74	26.86	50.26	49.83	50.03	0.05
	15-Jul-23	3.44	76.98	19.58	50.18	49.83	50.01	0.03
	16-Jul-23	2.95	83.83	13.22	50.15	49.82	50.00	0.02
	17-Jul-23	3.44	79.87	16.69	50.19	49.73	50.00	0.03
	18-Jul-23	5.47	81.15	13.38	50.20	49.76	50.00	0.03
	19-Jul-23	3.47	79.39	17.14	50.19	49.76	50.01	0.03
	20-Jul-23	2.89	79.99	17.12	50.25	49.83	50.00	0.03
	21-Jul-23	6.12	83.47	10.41	50.16	49.79	49.99	0.03
	22-Jul-23	3.55	77.49	18.96	50.16	49.79	50.01	0.03
	23-Jul-23	3.53	76.24	20.23	50.26	49.77	50.01	0.03
	24-Jul-23	3.01	79.25	17.74	50.22	49.75	50.00	0.03
	25-Jul-23	5.51	79.24	15.25	50.15	49.66	50.00	0.03
	26-Jul-23	1.59	71.55	26.86	50.31	49.82	50.02	0.05
	27-Jul-23	1.83	77.60	20.57	50.19	49.84	50.01	0.03
	28-Jul-23	0.46	83.48	16.05	50.19	49.87	50.01	0.02
	29-Jul-23	2.93	75.63	21.45	50.39	49.76	50.02	0.08
	30-Jul-23	2.64	66.79	30.57	50.42	49.70	50.04	0.10
	31-Jul-23	4.98	71.03	23.99	50.16	49.74	50.01	0.04

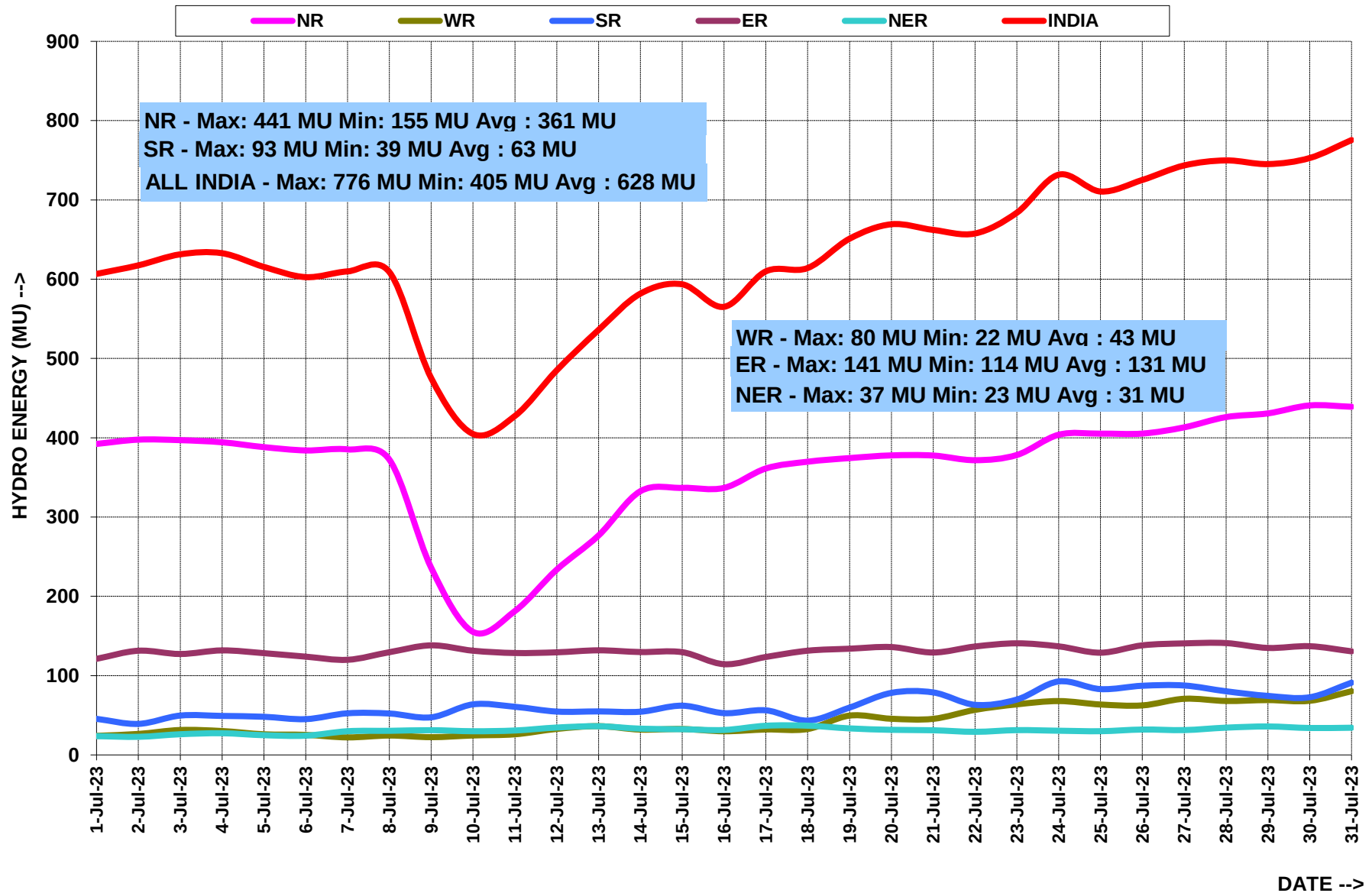
6.राष्ट्रीय स्तर पर जल विद्युत उत्पादन
HYDRO GENERATION AT NATIONAL LEVEL

माह: जुलाई 2023 MONTH:- JUL 2023

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तरी क्षे. NER	कुल TOTAL
01-Jul-23	392	24	46	121	24	607
02-Jul-23	398	26	39	131	23	617
03-Jul-23	397	31	50	127	26	631
04-Jul-23	394	30	49	132	27	633
05-Jul-23	388	26	48	128	25	616
06-Jul-23	384	25	45	124	24	603
07-Jul-23	385	22	53	120	30	610
08-Jul-23	372	24	52	130	30	609
09-Jul-23	236	22	47	138	31	475
10-Jul-23	155	25	64	131	30	405
11-Jul-23	181	26	61	129	31	427
12-Jul-23	234	33	55	129	35	485
13-Jul-23	277	36	55	132	36	536
14-Jul-23	333	32	54	130	33	582
15-Jul-23	337	33	62	130	32	594
16-Jul-23	337	30	53	114	31	565
17-Jul-23	361	32	56	124	37	610
18-Jul-23	370	33	43	131	37	614
19-Jul-23	374	50	60	134	33	651
20-Jul-23	378	46	78	136	32	669
21-Jul-23	378	46	79	129	31	662
22-Jul-23	372	57	63	137	29	658
23-Jul-23	378	64	70	141	31	684
24-Jul-23	404	68	93	137	31	732
25-Jul-23	405	64	83	129	30	711
26-Jul-23	405	62	87	138	32	725
27-Jul-23	413	71	88	141	31	743
28-Jul-23	426	68	80	141	34	750
29-Jul-23	431	69	74	135	36	745
30-Jul-23	441	68	73	137	34	753
31-Jul-23	439	80	91	131	34	776
कुल TOTAL	11176	1323	1950	4067	961	19477
उच्चतम MAXIMUM	441	80	93	141	37	776
निम्नतम MINIMUM	155	22	39	114	23	405
औसत AVERAGE	361	43	63	131	31	628
संचयी 2023-24 Cumulative 2023-24	32104	4119	7658	9731	2239	55851
अब तक का उच्चतम All Time Max.	409	167	208	157	41	877
दिनांक Date	22.08.22	18.12.14	31.08.18	14.09.22	01.08.22	30.08.22

HYDRO ENERGY DURING THE MONTH OF JULY' 2023



7. राष्ट्रीय स्तर पर पवन ऊर्जा उत्पादन
WIND GENERATION AT NATIONAL LEVEL
माह: जुलाई 2023 MONTH:- JUL 2023

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jul-23	29	150	199	-----	-----	378
02-Jul-23	29	88	209	-----	-----	325
03-Jul-23	26	61	227	-----	-----	313
04-Jul-23	45	72	248	-----	-----	364
05-Jul-23	35	73	250	-----	-----	358
06-Jul-23	13	71	268	-----	-----	352
07-Jul-23	12	94	240	-----	-----	347
08-Jul-23	5	128	210	-----	-----	342
09-Jul-23	8	135	170	-----	-----	313
10-Jul-23	10	151	148	-----	-----	309
11-Jul-23	27	143	161	-----	-----	331
12-Jul-23	40	130	160	-----	-----	330
13-Jul-23	52	110	192	-----	-----	354
14-Jul-23	41	124	215	-----	-----	380
15-Jul-23	58	178	182	-----	-----	418
16-Jul-23	43	153	169	-----	-----	364
17-Jul-23	24	117	223	-----	-----	364
18-Jul-23	27	143	272	-----	-----	442
19-Jul-23	15	154	296	-----	-----	465
20-Jul-23	4	144	295	-----	-----	443
21-Jul-23	9	126	298	-----	-----	433
22-Jul-23	7	129	298	-----	-----	433
23-Jul-23	21	103	285	-----	-----	410
24-Jul-23	9	129	263	-----	-----	401
25-Jul-23	7	123	245	-----	-----	374
26-Jul-23	6	116	251	-----	-----	373
27-Jul-23	9	184	255	-----	-----	449
28-Jul-23	32	227	238	-----	-----	497
29-Jul-23	51	246	243	-----	-----	541
30-Jul-23	29	209	229	-----	-----	466
31-Jul-23	31	215	246	-----	-----	493
कुल TOTAL	753	4227	7184	-----	-----	12164
उच्चतम MAXIMUM	58	246	298	-----	-----	541
निम्नतम MINIMUM	4	61	148	-----	-----	309
औसत AVERAGE	24	136	232	-----	-----	392
संचयी 2023-24 Cumulative 2023-24	3439	15060	15986	-----	-----	34485
अब तक का उच्चतम All Time Max.	78	272	298	-----	-----	555
दिनांक Date	09.06.22	22.05.22	21.07.23	-----	-----	22.05.22

***Source:** As reported by SLDCs. Limited visibility of embedded wind generator data.

8. राष्ट्रीय स्तर पर सौर ऊर्जा उत्पादन
SOLAR GENERATION AT NATIONAL LEVEL

माह: जुलाई 2023 MONTH:- JUL 2023

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jul-23	129	37	118	2.8	0.8	288
02-Jul-23	136	48	100	2.6	0.7	288
03-Jul-23	141	52	100	2.3	0.9	296
04-Jul-23	135	50	92	2.7	1.1	282
05-Jul-23	128	46	87	5.2	1.2	268
06-Jul-23	109	39	83	2.2	0.9	234
07-Jul-23	103	34	89	2.3	0.9	229
08-Jul-23	100	38	89	2.2	1.1	230
09-Jul-23	117	30	108	2.7	1.0	259
10-Jul-23	121	40	94	2.7	0.7	259
11-Jul-23	133	41	82	2.3	0.6	258
12-Jul-23	138	43	98	5.4	0.6	285
13-Jul-23	138	49	102	5.4	0.8	295
14-Jul-23	129	47	115	2.1	0.9	294
15-Jul-23	135	44	109	1.2	1.1	290
16-Jul-23	130	43	93	1.8	0.7	268
17-Jul-23	126	39	90	1.6	0.7	257
18-Jul-23	125	36	76	1.8	1.4	240
19-Jul-23	106	36	74	2.4	1.4	219
20-Jul-23	118	41	81	5.0	1.1	245
21-Jul-23	115	40	76	5.0	0.9	237
22-Jul-23	108	31	83	2.6	1.1	226
23-Jul-23	118	38	77	2.6	1.0	237
24-Jul-23	111	44	66	2.5	0.9	225
25-Jul-23	80	40	60	2.4	1.0	183
26-Jul-23	95	48	55	1.9	0.9	201
27-Jul-23	66	38	76	1.8	1.0	182
28-Jul-23	86	42	102	4.6	0.8	235
29-Jul-23	103	43	112	4.7	0.9	264
30-Jul-23	120	44	111	1.9	1.2	278
कुल TOTAL	3499	1241	2698	87	28	7553
उच्चतम MAXIMUM	141	52	118	5.4	1.4	296
निम्नतम MINIMUM	66	30	55	1.2	0.6	182
औसत AVERAGE	117	41	90	2.9	0.9	252
संचयी 2023-24 Cumulative 2023-24	11812	5447	10697	336	89.3	28381
अब तक का उच्चतम All Time Max.	147	71	143	6.7	2.4	351
दिनांक Date	10.05.23	12.05.23	21.03.23	23.03.23	22.06.22	10.05.23

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

9. राष्ट्रीय स्तर पर दैनिक अधिकतम विद्युत मांग पूर्ति
DAILY MAXIMUM DEMAND MET AT NATIONAL LEVEL

माह: जुलाई 2023 MONTH:- JUL 2023

सभी आंकड़े मेगावाट में All figures in MW

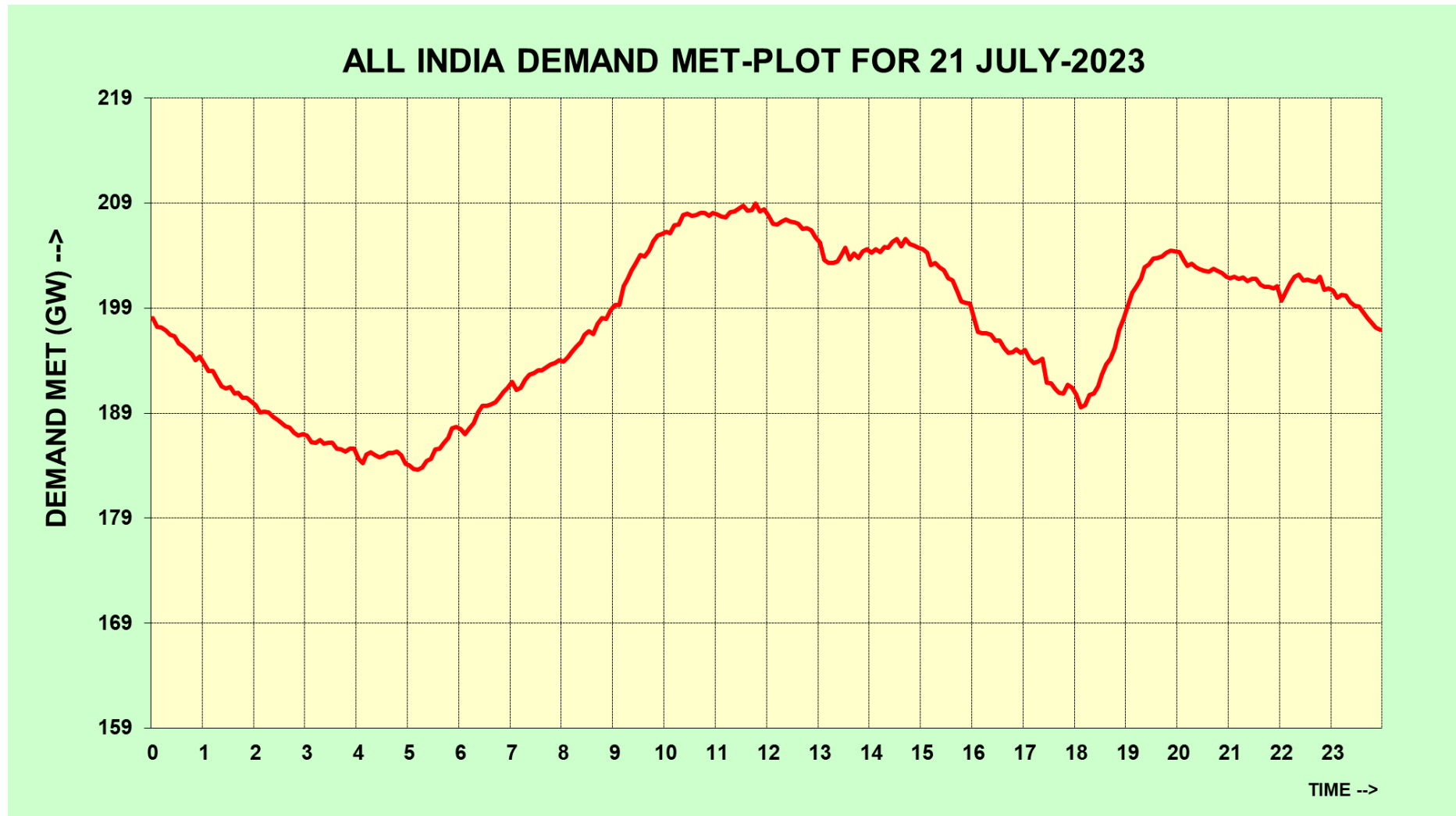
दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL	विभिन्नता फैक्टर Diversity Factor*
01-Jul-23	68504	54047	53428	25625	3151	196142	1.044
02-Jul-23	69384	53575	50039	26290	3098	194770	1.039
03-Jul-23	73223	57484	53546	27086	3156	207982	1.031
04-Jul-23	73465	59049	51050	27611	3166	206300	1.039
05-Jul-23	71326	58280	49273	27394	3380	198089	1.058
06-Jul-23	65536	56085	49122	27142	3354	193038	1.042
07-Jul-23	67568	55561	50534	26573	3365	193700	1.051
08-Jul-23	64235	55931	50720	26653	3309	185373	1.083
09-Jul-23	55565	52538	47902	28475	3229	179616	1.045
10-Jul-23	58633	56856	53125	28891	3268	191860	1.046
11-Jul-23	64899	56753	50656	28379	3299	193441	1.055
12-Jul-23	65596	57905	52208	28885	3235	200357	1.037
13-Jul-23	68460	58531	51461	29625	3202	203212	1.040
14-Jul-23	68316	58691	53302	29190	3219	205712	1.034
15-Jul-23	67994	57283	53196	27359	3420	203098	1.030
16-Jul-23	69059	54053	50408	27289	3087	195668	1.042
17-Jul-23	72520	56790	54160	27503	3162	204718	1.046
18-Jul-23	71773	57067	49927	28436	3408	202935	1.038
19-Jul-23	72185	56566	49104	28558	3490	201837	1.040
20-Jul-23	75076	57127	47876	28962	3376	205282	1.035
21-Jul-23	77569	56958	48255	28573	3347	208825	1.028
22-Jul-23	74022	55808	48472	28212	3197	198788	1.055
23-Jul-23	74361	54211	45740	28503	3212	196625	1.048
24-Jul-23	76426	57312	47918	29428	3385	205996	1.041
25-Jul-23	75233	57657	43759	29132	3409	202014	1.036
26-Jul-23	71285	58056	43439	29458	3421	200402	1.026
27-Jul-23	71557	57570	44042	29492	3364	202085	1.019
28-Jul-23	69241	56032	45629	29641	3392	197665	1.032
29-Jul-23	66447	55056	48622	29049	3504	192758	1.051
30-Jul-23	69584	53619	46558	28751	3257	195133	1.034
31-Jul-23	73563	56820	52200	28667	3405	206362	1.040
उच्चतम MAXIMUM	77569	59049	54160	29641	3504	208825	1.083
निम्नतम MINIMUM	55565	52538	43439	25625	3087	179616	1.019
औसत AVERAGE	69761	56428	49538	28220	3299	199025	1.042
अब तक का उच्चतम All Time Max.	77569	71321	64314	29641	3596	223235	
दिनांक Date	21.07.23	27.12.22	15.03.23	28.07.23	17.08.22	09.06.23	

* Diversity factor = (Sum of regional max demands) / All India max demand

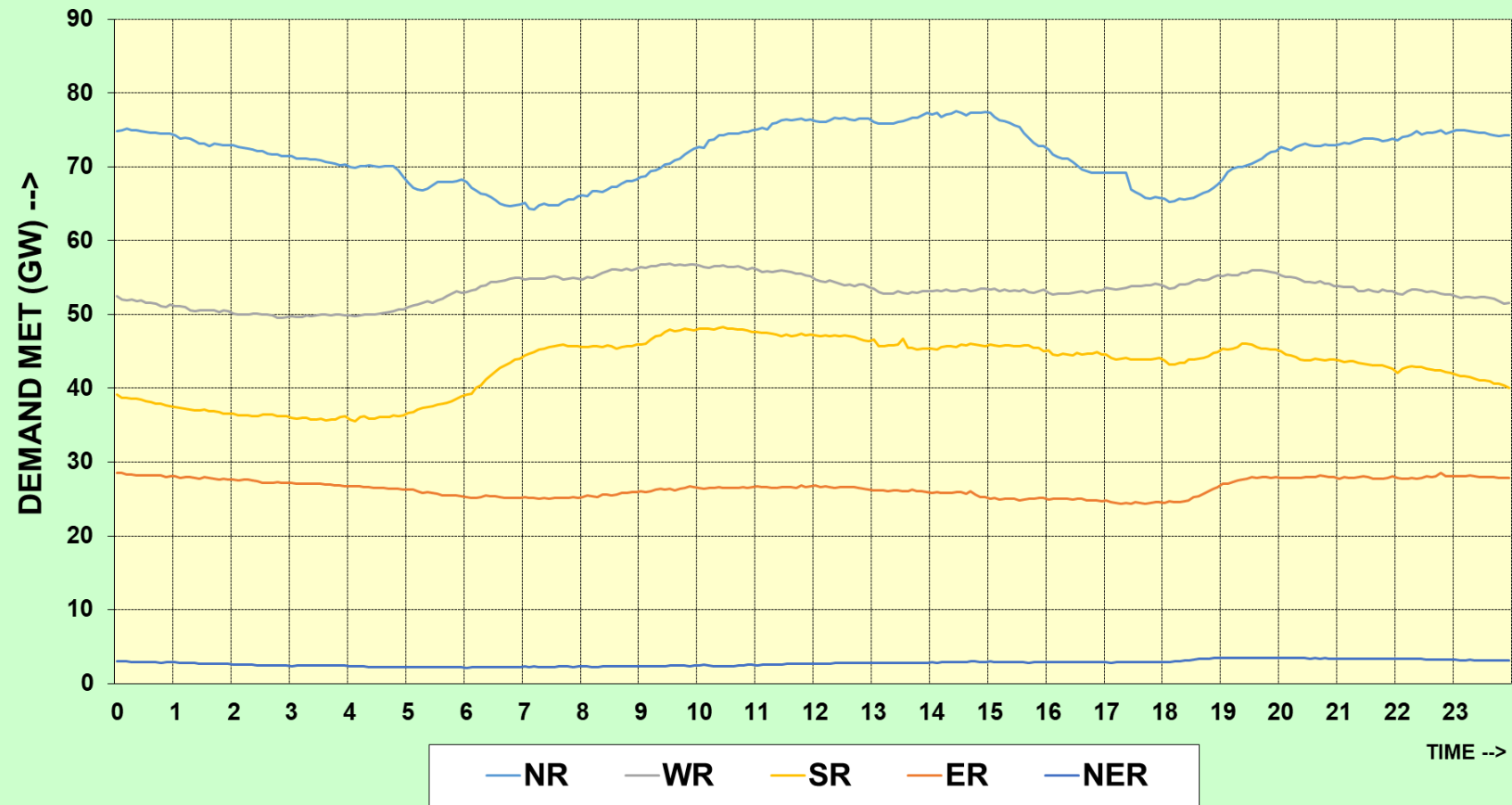
नोट : यह आंकड़े एनएलडीसी स्काडा प्रणाली में दर्ज दैनिक अधिकतम मांगपूर्ति दर्शाते हैं ।

Note: The above figures denote daily maximum demand met recorded in NLDC SCADA.

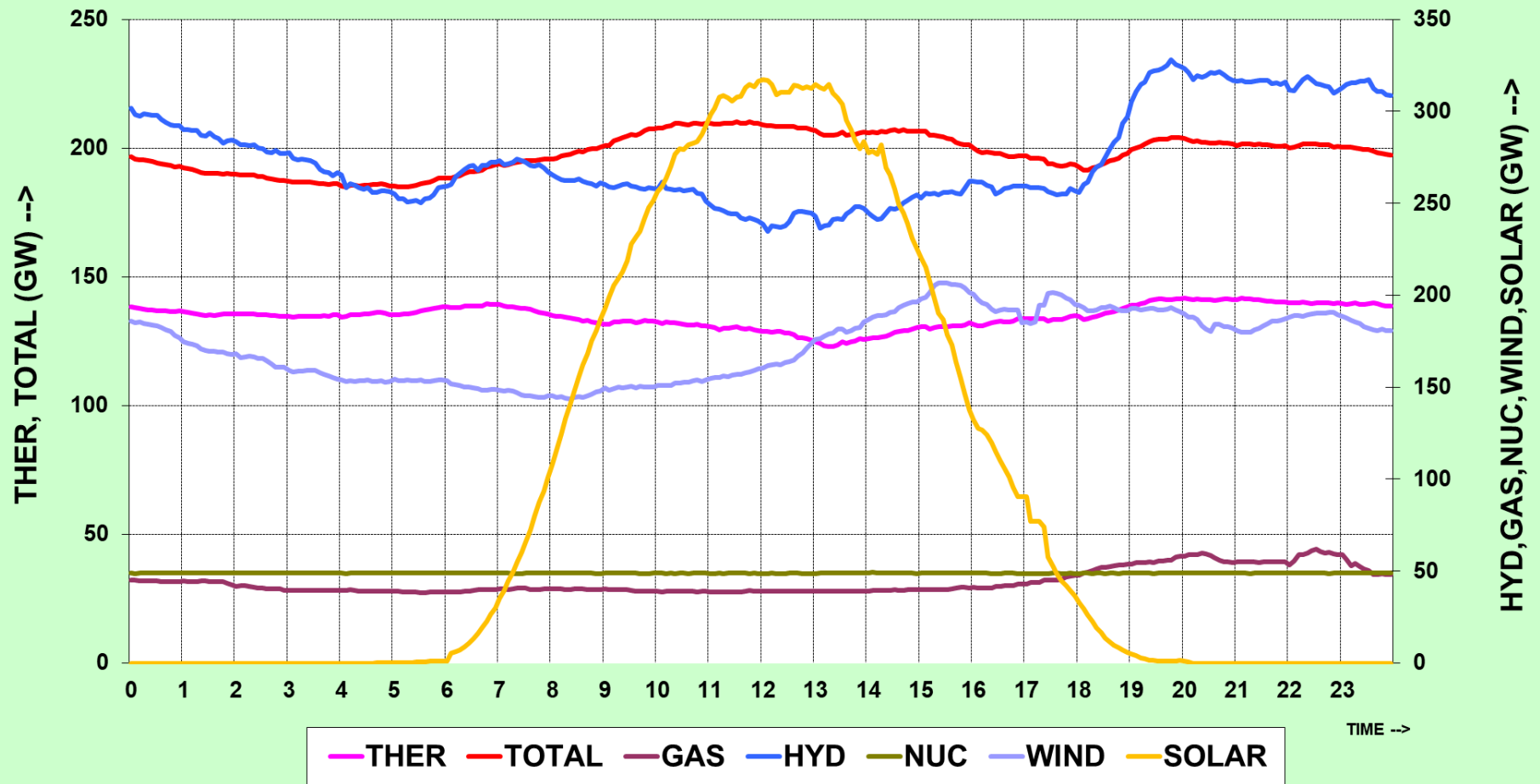
10. DEMAND AND GENERATION PLOTS FOR 21 JULY (MAXIMUM DEMAND MET)



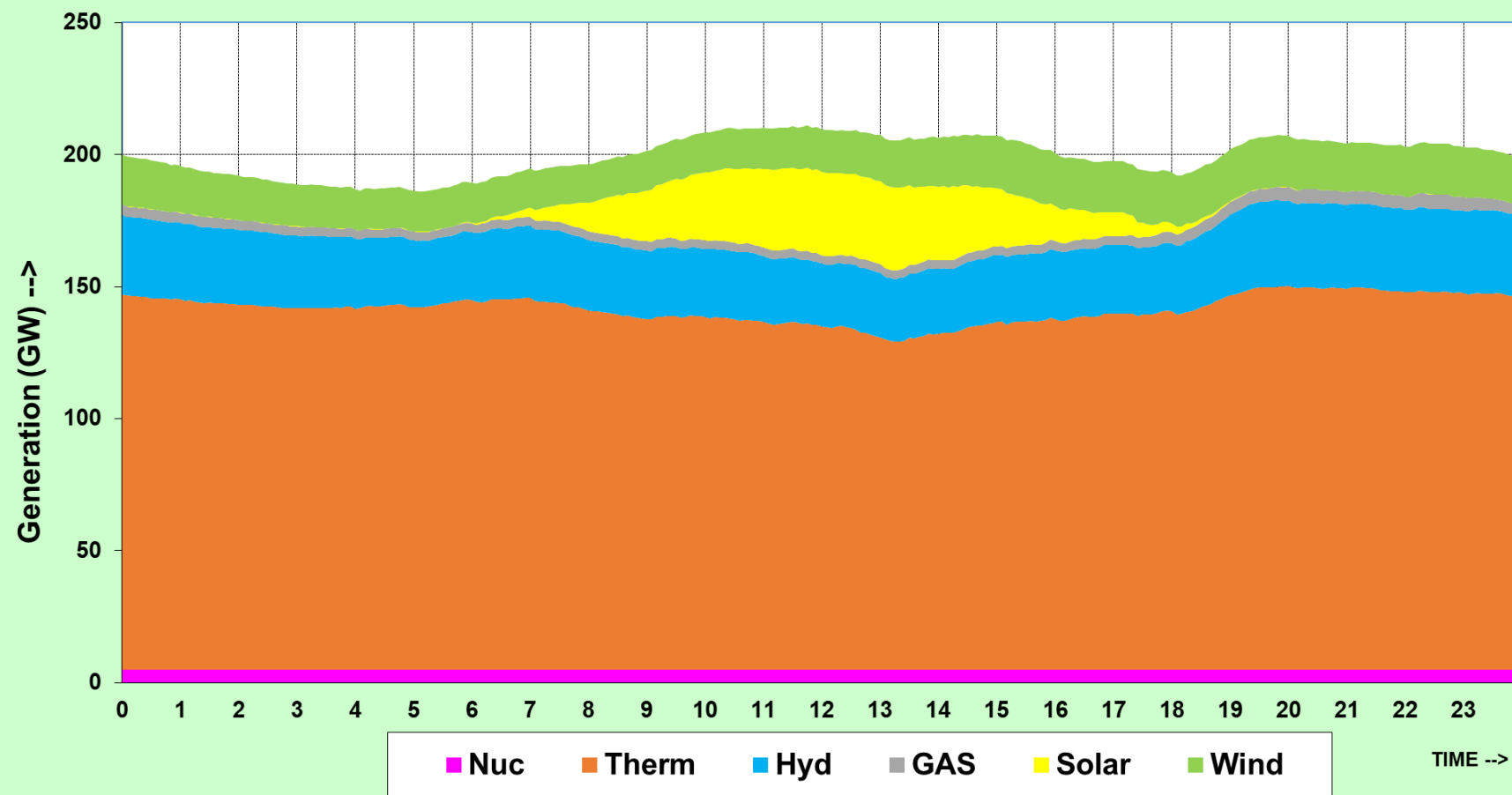
REGIONAL DMD MET PLOT FOR 21-JULY-2023



ALL INDIA GENERATION PLOT FOR 21-JULY-2023



All India Fuel-wise Generation Patterns for 21-JULY-2023



21 जुलाई 2023 (अधिकतम माँग) की आल इंडिया माँग आपूर्ति
ALL INDIA DEMAND MET FOR 21 JUL 2023 (MAXIMUM DEMAND*)

Time (hrs)	All India Demand (MW)		Time (hrs)	All India Demand (MW)
00:05	197194		12:05	207033
00:20	196461		12:20	207419
00:35	195356		12:35	207046
00:50	194032		12:50	206378
01:05	193033		13:05	203541
01:20	191542		13:20	203422
01:35	190897		13:35	203671
01:50	190432		13:50	204408
02:05	189086		14:05	204612
02:20	188670		14:20	204739
02:35	187791		14:35	204883
02:50	186830		14:50	204940
03:05	186246		15:05	204297
03:20	186097		15:20	202887
03:35	185617		15:35	201612
03:50	185645		15:50	199488
04:05	184205		16:05	196768
04:20	184973		16:20	196466
04:35	185197		16:35	195248
04:50	184998		16:50	195112
05:05	183697		17:05	194150
05:20	184426		17:20	194209
05:35	185628		17:35	191278
05:50	187570		17:50	191692
06:05	186973		18:05	189578
06:20	189093		18:20	190859
06:35	189813		18:35	193661
06:50	191016		18:50	196922
07:05	191243		19:05	200434
07:20	192641		19:20	202860
07:35	193110		19:35	203752
07:50	193781		19:50	204450
08:05	194353		20:05	203550
08:20	195781		20:20	202887
08:35	196558		20:35	202483
08:50	197965		20:50	202307
09:05	199316		21:05	201984
09:20	202593		21:20	201585
09:35	203956		21:35	201223
09:50	205902		21:50	200858
10:05	206104		22:05	200528
10:20	207856		22:20	202194
10:35	207846		22:35	201577
10:50	207787		22:50	200769
11:05	207743		23:05	199985
11:20	208235		23:20	199566
11:35	208291		23:35	198701
11:50	208210		23:50	197159
Maximum Demand of 208825 MW met@11:45 hrs (from 1 min. interval SCADA DATA)				

* 15 minute interval SCADA DATA of instantaneous demand

11. ACTUAL POWER SUPPLY POSITION - JULY 2023

REGION	STATE	Requirement/Availability in MU/DAY			Requirement/Availability in MU				Peak Demand/Peak Met in MW			
		Requirement	Energy met	Surplus(+)/ Deficit(-)	Requirement	Energy met	Surplus(+)/ Deficit(-)	% Shortage	Requirement	Peak Demand Met	Surplus(+)/ Deficit(-)	% Shortage
NR	Chandigarh	6	6	0.0	196	196	0	0.0	371	371	0	0.0
	Delhi	126	126	0.0	3912	3911	0	0.0	7398	7398	0	0.0
	Haryana	217	217	-0.4	6737	6725	-12	-0.2	12227	12227	0	0.0
	Himachal Pradesh	32	32	-0.1	1007	1004	-3	-0.3	1775	1775	0	0.0
	J&K(UT) and Ladakh(UT)	51	51	-0.5	1591	1574	-17	-1.0	2628	2543	-85	-3.2
	Punjab	269	269	0.0	8332	8332	0	0.0	14831	14831	0	0.0
	Rajasthan	270	269	-0.8	8375	8351	-24	-0.3	14204	14204	0	0.0
	Uttar Pradesh	495	494	-0.7	15338	15315	-23	-0.2	28704	28284	-420	-1.5
	Uttarakhand	44	44	-0.4	1375	1361	-14	-1.0	2298	2223	-75	-3.3
	Railway	3	3	0.0	106	106	0	0.0	191	191	0	0.0
	NFL	0	0	0.0	2	2	0	0.0	12	12	0	0.0
	TOTAL	1515	1512	-3.0	46971	46878	-93	-0.2	77145	77145	0	0.0
WR	Gujarat	358	358	0.0	11093	11093	0	0.0	17715	17672	-43	-0.2
	Madhya Pradesh	236	236	0.0	7309	7309	0	0.0	12006	12006	0	0.0
	Chhattisgarh	108	108	-0.2	3362	3356	-6	-0.2	5314	5216	-98	-1.8
	Maharashtra	494	494	-0.1	15308	15304	-4	0.0	24544	24535	-9	0.0
	Goa	12	12	0.0	379	379	0	0.0	647	646	0	-0.1
	DNHDDPDCL	30	30	0.0	920	920	0	0.0	1307	1307	0	0.0
	AMNSIL	19	19	0.0	577	577	0	0.0	930	930	0	0.0
	BALCO*	12	12	0.0	384	384	0	0.0	523	523	0	0.0
	TOTAL	1269	1268	-0.3	39332	39322	-10	0.0	58165	58064	-102	-0.2
SR	Andhra Pradesh	204	204	0.0	6334	6334	0	0.0	11619	11584	-35	-0.3
	Telangana	207	209	2.6	6407	6489	82	1.3	12629	12610	-19	-0.1
	Karnataka	197	200	2.6	6116	6198	82	1.3	12359	12359	0	0.0
	Kerala	74	74	0.9	2281	2307	27	1.2	3999	3999	0	0.0
	Tamil Nadu	348	352	3.4	10800	10907	107	1.0	17774	17774	0	0.0
	Puducherry	10	10	0.1	301	304	3	1.0	492	492	0	0.0
	TOTAL	1040	1050	9.7	32239	32540	301	0.9	54211	54211	0	0.0
ER	Bihar	141	139	-1.4	4362	4318	-44	-1.0	7578	7578	0	0.0
	Jharkhand	37	37	-0.3	1158	1149	-9	-0.8	1857	1857	0	0.0
	DVC	75	75	0.0	2339	2339	0	0.0	3687	3687	0	0.0
	Odisha	124	124	0.0	3846	3846	0	0.0	7104	7104	0	0.0
	West Bengal	213	213	0.0	6591	6591	0	0.0	10675	10675	0	0.0
	Sikkim	1	1	0.0	37	37	0	0.0	96	96	0	0.0
	Railways_ER ISTS**	0	0	0.0	5	5	0	0.0	26	26	0	0.0
	TOTAL	592	590	-1.7	18339	18286	-53	-0.3	29757	29633	-124	-0.4
NER	Arunachal Pradesh	3	3	0.0	83	83	0	0.0	151	151	0	0.0
	Assam	42	42	0.0	1317	1317	0	0.0	2390	2390	0	0.0
	Manipur	3	3	0.0	82	82	0	0.0	202	202	0	0.0
	Meghalaya	6	5	-0.8	182	159	-23	-12.9	330	330	0	0.0
	Mizoram	2	2	0.0	53	53	0	0.0	122	122	0	0.0
	Nagaland	3	3	0.0	89	89	0	0.0	167	167	0	0.0
	Tripura	6	6	0.0	179	179	0	0.0	343	343	0	0.0
	TOTAL	64	63	-0.8	1986	1963	-24	-1.2	3516	3508	-8	-0.2
ALL INDIA		4480	4483	3.9	138867	138988	121	0.1				

* BALCO has been added as new entity wef 20.12.2022

** Railways_ER ISTS has been added as new entity wef 29.04.2023

NOTE: BASED ON THE DATA FURNISHED BY ENTITIES ON DAILY BASIS

12. ENERGY COMPARISON OF JULY 2023 vs JULY 2022

REGION	STATE	Energy Requirement (MU)					Energy Met (MU)				
		Jul-22	Jul-23	Difference	% Change	Average MU/day for Jul-23	Jul-22	Jul-23	Difference	% Change	Average MU/day for Jul-23
NR	Chandigarh	203	196	-8	-4	6	203	196	-8	-4	6
	Delhi	3876	3912	35	1	126	3876	3911	35	1	126
	Haryana	6506	6737	231	4	217	6491	6725	234	4	217
	Himachal Pradesh	1000	1007	7	1	32	997	1004	7	1	32
	J&K(UT) and Ladakh(UT)	1535	1591	56	4	51	1534	1574	40	3	51
	Punjab	8192	8332	140	2	269	8192	8332	140	2	269
	Rajasthan	7263	8375	1112	15	270	7244	8351	1107	15	269
	Uttar Pradesh	15014	15338	324	2	495	14909	15315	406	3	494
	Uttarakhand	1481	1375	-106	-7	44	1445	1361	-84	-6	44
	Railway/NFL	2	108	106	4906	3	2	108	106	4906	3
	TOTAL	45166	46971	1805	4	1515	44986	46878	1891	4	1512
WR	Gujarat	10263	11093	830	8	358	10256	11093	837	8	358
	Madhya Pradesh	6594	7309	714	11	236	6594	7309	714	11	236
	Chhattisgarh	3093	3362	269	9	108	3093	3356	263	9	108
	Maharashtra	13877	15308	1431	10	494	13877	15304	1427	10	494
	Goa	367	379	12	3	12	367	379	12	3	12
	DNHDDPDCL	821	920	99	12	30	821	920	99	12	30
	AMNSIL	564	577	14	2	19	564	577	14	2	19
	BALCO*		384	-	-	12		384	-	-	12
	TOTAL	35579	39332	3753	11	1269	35572	39322	3750	11	1268
SR	Andhra Pradesh	5725	6334	609	11	204	5722	6334	612	11	204
	Telangana	5325	6407	1082	20	207	5325	6489	1164	22	209
	Karnataka	5371	6116	746	14	197	5371	6198	828	15	200
	Kerala	2116	2281	165	8	74	2116	2307	192	9	74
	Tamil Nadu	10142	10800	658	6	348	10142	10907	765	8	352
	Puducherry	292	301	9	3	10	292	304	12	4	10
	TOTAL	28971	32239	3268	11	1040	28968	32540	3571	12	1050
ER	Bihar	4166	4362	196	5	141	4033	4318	285	7	139
	Jharkhand	1094	1158	64	6	37	1037	1149	112	11	37
	DVC	2353	2339	-13	-1	75	2353	2339	-13	-1	75
	Odisha	3853	3846	-8	0	124	3853	3846	-8	0	124
	West Bengal	5960	6591	632	11	213	5957	6591	634	11	213
	Sikkim	43	37	-6	-15	1	43	37	-6	-15	1
	Railways_ER ISTS**		5	-	-	0		5	-	-	0
	TOTAL	17469	18339	870	5	592	17277	18286	1009	6	590
NER	Arunachal Pradesh	86	83	-3	-3	3	86	83	-3	-3	3
	Assam	1242	1317	75	6	42	1242	1317	75	6	42
	Manipur	83	82	-1	-1	3	83	82	-1	-1	3
	Meghalaya	183	182	-1	0	6	176	159	-17	-10	5
	Mizoram	52	53	1	1	2	52	53	1	1	2
	Nagaland	84	89	5	6	3	84	89	5	6	3
	Tripura	259	179	-80	-31	6	259	179	-80	-31	6
	TOTAL	1989	1986	-3	0	64	1982	1963	-20	-1	63
ALL INDIA		129175	138867	9692	8	4480	128786	138988	10202	8	4483

* BALCO has been added as new entity wef 20.12.2022

** Railways_ER ISTS has been added as new entity wef 29.04.2023

13. DEMAND COMPARISON OF JULY 2023 vs JULY 2022

REGION	STATE	Peak Requirement (MW)				Peak Met (MW)			
		Jul-22	Jul-23	Difference	% Change	Jul-22	Jul-23	Difference	% Change
NR	Chandigarh	401	371	-30	-7.5	401	371	-30	-7.5
	Delhi	7517	7398	-119	-1.6	7517	7398	-119	-1.6
	Haryana	12327	12227	-100	-0.8	12327	12227	-100	-0.8
	Himachal Pradesh	1732	1775	43	2.5	1732	1775	43	2.5
	J&K(UT) and Ladakh(UT)	2718	2628	-90	-3.3	2718	2543	-175	-6.4
	Punjab	14058	14831	773	5.5	14058	14831	773	5.5
	Rajasthan	12332	14204	1872	15.2	12332	14204	1872	15.2
	Uttar Pradesh	26361	28704	2343	8.9	25951	28284	2333	9.0
	Uttarakhand	2342	2298	-44	-1.9	2342	2223	-119	-5.1
	Railway/NFL	5	203	198	3887.4	5	203	198	3887.4
WR	Gujarat	17428	17715	287	2	17428	17672	244	1.4
	Madhya Pradesh	10244	12006	1762	17	10237	12006	1769	17.3
	Chhattisgarh	4831	5314	483	10	4831	5216	385	8.0
	Maharashtra	22540	24544	2004	9	22540	24535	1996	8.9
	Goa	624	647	23	4	623	646	23	3.7
	DNHDDPDCL	1198	1307	109	9.1	1198	1307	109	9.1
	AMNSIL	924	930	6	0.7	924	930	6	0.7
	BALCO*		523	-	-		523	-	-
SR	Andhra Pradesh	9805	11619	1814	18.5	9763	11584	1821	18.7
	Telangana	12475	12629	154	1.2	12468	12610	142	1.1
	Karnataka	10855	12359	1504	13.9	10855	12359	1504	13.9
	Kerala	3662	3999	337	9.2	3662	3999	337	9.2
	Tamil Nadu	16243	17774	1531	9.4	16206	17774	1568	9.7
	Puducherry	469	492	23	4.9	469	492	23	4.9
ER	Bihar	3325	7578	4253	127.9	6830	7578	748	10.9
	Jharkhand	719	1857	1138	158.3	1724	1857	133	7.7
	DVC	-1221	3687	4908	-401.9	3443	3687	244	7.1
	Odisha	1955	7104	5149	263.3	6262	7104	842	13.4
	West Bengal	1733	10675	8942	515.8	9766	10675	909	9.3
	Sikkim	44	96	52	117.2	101	96	-5	-5.0
	Railways_ER ISTS**		26	26	-		26	26	-
NER	Arunachal Pradesh	119	151	32	27.3	119	151	32	27.3
	Assam	2231	2390	159	7.1	2231	2390	159	7.1
	Manipur	201	202	1	0.4	201	202	1	0.4
	Meghalaya	332	330	-2	-0.6	332	330	-2	-0.6
	Mizoram	119	122	3	2.6	119	122	3	2.6
	Nagaland	159	167	8	4.9	159	167	8	4.9
	Tripura	319	343	24	7.6	319	343	24	7.6

* BALCO has been added as new entity wef 20.12.2022

** Railways_ER ISTS has been added as new entity wef 29.04.2023

14. SCHEDULE AND DRAWAL OF CONSTITUENTS - JULY 2023

REGION	STATE	SCHEDULE (MU)	ACTUAL DRAWAL (MU)	Over drawal(+)/ Under Drawal(-) (MU)	% OD/UD	SCHEDULE (MU/DAY)	ACTUAL DRAWAL (MU/DAY)	Over drawal(+)/ Under Drawal(-) (MU/DAY)
NR	Chandigarh	198.92	199.42	0.50	0.25	6.42	6.43	0.02
	Delhi	3685.12	3624.86	-60.27	-1.64	118.87	116.93	-1.94
	Haryana	5117.86	5093.98	-23.88	-0.47	165.09	164.32	-0.77
	Himachal Pradesh	31.69	54.33	22.64	71.42	1.02	1.75	0.73
	J&K(UT) and Ladakh(UT)	781.21	739.54	-41.67	-5.33	25.20	23.86	-1.34
	Punjab	5054.05	4942.94	-111.11	-2.20	163.03	159.45	-3.58
	Rajasthan	2814.70	2763.85	-50.84	-1.81	90.80	89.16	-1.64
	Uttar Pradesh	7287.25	7233.08	-54.18	-0.74	235.07	233.33	-1.75
	Uttarakhand	729.55	739.35	9.80	1.34	23.53	23.85	0.32
	Railway	107.64	106.14	-1.50	-1.39	3.47	3.42	-0.05
	NFL	2.31	1.91	-0.39	-17.06	0.07	0.06	-0.01
	Total	25810.30	25499.38	-310.91	-1.20	832.59	822.56	-10.03
WR	Gujarat	5128.28	5064.51	-63.78	-1.24	165.43	163.37	-2.06
	Madhya Pradesh	3691.57	3602.49	-89.08	-2.41	119.08	116.21	-2.87
	Chhattisgarh	1855.62	1820.46	-35.16	-1.89	59.86	58.72	-1.13
	Maharashtra	5348.75	5235.51	-113.24	-2.12	172.54	168.89	-3.65
	Goa	378.44	367.82	-10.62	-2.81	12.21	11.87	-0.34
	DNHDDPDCL	919.19	919.86	0.66	0.07	29.65	29.67	0.02
	AMNSIL	316.08	314.02	-2.06	-0.65	10.20	10.13	-0.07
	BALCO*	386.09	384.36	-1.74	-0.45	12.45	12.40	-0.06
	Total	18024.02	17709.02	-315.00	-1.75	581.42	571.26	-10.16
SR	Andhra Pradesh	1266.18	1262.29	-3.89	-0.31	40.84	40.72	-0.13
	Telangana	2971.45	2989.39	17.94	0.60	95.85	96.43	0.58
	Karnataka	1364.29	1358.01	-6.28	-0.46	44.01	43.81	-0.20
	Kerala	1664.45	1708.03	43.58	2.62	53.69	55.10	1.41
	Tamil Nadu	4603.35	4493.54	-109.81	-2.39	148.50	144.95	-3.54
	Puducherry	290.95	282.59	-8.36	-2.87	9.39	9.12	-0.27
	Goa (SR)	63.09	62.06	-1.03	-1.63	2.04	2.00	-0.03
	Total	12223.76	12155.91	-67.85	-0.56	394.31	392.13	-2.19
ER	Bihar	4079.63	4071.33	-8.30	-0.20	131.60	131.33	-0.27
	Jharkhand	973.44	958.34	-15.10	-1.55	31.40	30.91	-0.49
	DVC	-1103.82	-1098.05	5.77	-0.52	-35.61	-35.42	0.19
	Odisha	1353.30	1340.37	-12.93	-0.96	43.65	43.24	-0.42
	West Bengal	2926.41	2888.74	-37.67	-1.29	94.40	93.19	-1.22
	Sikkim	38.06	36.46	-1.60	-4.21	1.23	1.18	-0.05
	Railways_ER ISTS*	8.11	4.23	-3.88	-47.89	0.26	0.14	-0.13
	Total	8275.14	8201.41	-73.72	-0.89	266.94	264.56	-2.38
NER	Arunachal Pradesh	78.37	74.20	-4.17	-5.32	2.53	2.39	-0.13
	Assam	1089.76	1102.61	12.86	1.18	35.15	35.57	0.41
	Manipur	83.35	82.17	-1.18	-1.42	2.69	2.65	-0.04
	Meghalaya	30.84	26.77	-4.07	-13.19	0.99	0.86	-0.13
	Mizoram	49.47	43.95	-5.52	-11.16	1.60	1.42	-0.18
	Nagaland	79.46	75.91	-3.55	-4.47	2.56	2.45	-0.11
	Tripura	169.52	175.77	6.24	3.68	5.47	5.67	0.20
	Total	1580.78	1581.38	0.60	0.04	50.99	51.01	0.02
All India Total		65914.00	65147.11	-766.89	-1.16	2126.26	2101.52	-24.74

* BALCO has been added as new entity wef 20.12.2022

** Railways_ER ISTS has been added as new entity wef 29.04.2023

15. INTER REGIONAL EXCHANGES 2023-24

(All figures in MU)

	Apr'23	May'23	June'23	July'23	Fin. Year 2023-24
Name of Line					
Import of NR from WR (WR-NR)					
WR - NR 765 kV Gwalior - Agra 2xS/C	635.95	996.71	863.68	972.77	3469.11
WR-NR 765 kV Gwalior - Phagi 2xS/C	553.05	453.93	342.56	671.61	2021.15
WR-NR 765 kV Jabalpur-Orai D/C	430.29	804.53	778.33	964.77	2977.92
WR-NR 765 kV Satna-Orai	505.37	598.85	574.83	615.77	2294.82
WR-NR 765 kV Gwalior-Orai	0.00	0.00	0.00	0.00	0.00
WR - NR HVDC Champa - Kurukshetra	804.26	1126.68	2069.11	2235.18	6235.23
WR - NR HVDC Vindhyachal	0.49	3.23	3.13	94.78	101.63
WR - NR 400 kV Zerda Kankroli	0.00	0.00	1.48	15.36	16.84
WR - NR 400 kV Zerda Bhinmal	0.00	0.00	0.00	12.54	12.54
WR - NR 400 kV RAPP C - Shujalpur D/C	2.28	31.04	16.05	164.49	213.86
WR - NR 220 kV Bhanpura - Ranpur	9.67	65.53	68.13	70.02	213.35
WR - NR 220 kV Bhanpura - Modak	0.00	0.00	0.00	0.00	0.00
WR - NR 220 kV Auraiya - Malanpur/Mehgaon	0.00	0.15	0.00	0.00	0.15
WR - NR 400 kV Vindhyachal-Rihand D/C	0.00	0.00	0.00	0.00	0.00
WR - NR HVDC Mundra - Mohindergarh	132.55	340.75	234.27	124.89	832.46
WR-NR 765 kV Banaskata-Chittorgarh D/C	0.00	3.26	16.02	169.31	188.59
WR-NR 765 kV Varanasi-Vindhyachal	1030.05	1663.34	1854.30	1854.38	6402.07
Total WR - NR	4103.96	6088.00	6821.89	7965.87	24979.72
Export of NR to WR (NR-WR)					
NR - WR 765 kV Gwalior - Agra 2xS/C	0.00	0.00	0.00	0.00	0.00
NR - WR 765 kV Gwalior - Phagi 2xS/C	0.00	5.41	0.00	0.00	5.41
NR - WR 765 kV Jabalpur-Orai D/C	0.00	0.00	0.00	0.00	0.00
NR - WR 765 kV Satna-Orai	0.00	0.00	0.00	0.00	0.00
NR - WR 765 kV Gwalior-Orai	440.87	385.11	301.32	367.88	1495.18
NR - WR HVDC Champa- Kurukshetra	0.00	0.00	0.00	0.00	0.00
NR - WR HVDC Vindhyachal	289.83	273.88	267.04	184.43	1015.18
NR - WR 400 kV Zerda Kankroli	185.75	94.43	108.34	24.78	413.30
NR - WR 400 kV Zerda Bhinmal	282.42	181.64	200.56	73.76	738.38
NR - WR 400 kV RAPP C - Shujalpur D/C	85.74	45.64	45.64	0.00	177.02
NR - WR 220 kV Bhanpura - Ranpur	0.00	0.00	0.00	0.00	0.00
NR - WR 220 kV Bhanpura - Modak	0.00	0.00	0.00	0.00	0.00
NR - WR 220 kV Auraiya - Malanpur/Mehgaon	56.56	28.58	50.82	47.16	183.12
NR-WR 400 kV Rihand-Vindhyachal D/C	644.30	672.49	642.32	664.33	2623.44
NR - WR HVDC Mundra - Mohindergarh	243.86	54.36	36.25	94.16	428.63
NR - WR 765 kV Banaskata-Chittorgarh D/C	976.83	428.19	421.52	94.53	1921.07
NR-WR 765 kV Varanasi-Vindhyachal	0.00	0.00	0.00	0.00	0.00
Total NR - WR	3206.16	2169.73	2073.81	1551.03	9000.73

	Apr'23	May'23	June'23	July'23	Fin. Year 2023-24
Name of Line					
Import of NR from ER (ER-NR)					
ER - NR 765 kV Sasaram - Fatehpur	166.02	114.36	82.57	74.16	437.11
ER - NR 765 kV Gaya - Varanasi 2*S/C	224.56	107.46	28.46	47.20	407.68
ER - NR 765 kV Gaya - Balia	246.38	335.46	321.28	300.74	1203.86
ER - NR HVDC Alipurduar - Agra	0.00	0.00	375.05	784.99	1160.04
ER-NR 400 kV Patna - Balia D/C	219.16	246.32	242.35	242.23	950.06
ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	127.03	223.63	277.35	376.44	1004.45
ER - NR 400 kV Biharshariff - Balia D/C	22.11	72.32	102.97	92.77	290.17
ER - NR 400 kV Motihari - Gorakhpur D/C	156.88	188.70	180.09	228.59	754.26
ER - NR 400 kV Biharshariff - Varanasi D/C	67.56	47.21	34.66	24.65	174.08
ER - NR 400 kV Sasaram -Varanasi	74.99	38.68	46.09	55.64	215.40
ER - NR 400 kV Sasaram -Allahabad	51.91	33.39	21.29	28.77	135.36
ER - NR 220 kV Sahupuri-Karamnasa	78.64	82.22	82.32	43.45	286.63
ER - NR 132 kV Sahupuri - Karamnasa	1.92	0.00	0.96	1.12	4.00
ER - NR 132 kV 132 kV Rihand - Nagar Untari	0.00	0.00	0.00	0.00	0.00
ER - NR 132 kV Rihand - Garhwa	0.46	0.00	0.00	0.00	0.46
ER-NR 400 kV Naubatpur - Balia D/C	211.13	243.59	239.56	236.53	930.81
Total ER-NR	1648.75	1733.34	2035.00	2537.28	7954.37
Import of NR from NER (NER-NR)					
NER-NR HVDC Biswanath Chariali - Agra	0.00	0.00	153.62	339.28	492.90
Total NER - NR	0.00	0.00	153.62	339.28	492.90
Export of NR to ER (NR-ER)					
NR - ER 765 kV Sasaram - Fatehpur	0.00	2.65	7.74	3.45	13.84
NR - ER 765 kV Gaya - Varanasi 2*S/C	0.30	38.74	102.98	45.94	187.96
NR - ER 765 kV Gaya - Balia	0.00	0.00	0.00	0.00	0.00
NR - ER HVDC Alipurduar - Agra	0.00	0.00	0.00	0.00	0.00
NR - ER 400 kV Patna - Balia 2xD/C	0.00	0.00	0.00	0.00	0.00
NR - ER 400 kV Muzaffarpur - Gorakhpur D/C	3.47	0.00	0.00	0.00	3.47
NR - ER 400 kV Biharshariff - Balia D/C	17.15	4.18	13.06	0.00	34.39
ER - NR 400 kV Motihari - Gorakhpur D/C	0.00	0.00	0.00	0.00	0.00
NR - ER 400 kV Biharshariff - Varanasi D/C	2.90	7.31	20.53	10.39	41.13
NR - ER 400 kV Sasaram -Varanasi	0.00	0.00	0.00	0.00	0.00
NR - ER 400 kV Sasaram -Allahabad	2.25	0.00	0.00	0.82	3.07
NR - ER 220 kV Sahupuri-Karamnasa	0.00	0.00	0.00	0.00	0.00
NR - ER 132 kV Sahupuri - Karamnasa	0.01	0.96	0.48	0.96	2.41
NR - ER 132 kV 132 kV Rihand - Nagar Untari	1.74	2.20	1.08	3.37	8.39
ER - NR 132 kV Rihand - Garhwa	10.23	8.48	19.67	18.35	56.73
NR-ER 400 kV Balia- Naubatpur D/C	0.00	0.00	0.00	0.00	0.00
Total NR - ER	38.05	64.52	165.54	83.28	351.39
Export of NR to NER (NER-NR)					
NR-NER HVDC Biswanath Chariali - Agra	261.93	291.71	104.47	0.00	658.11
Total NR - NER	261.93	291.71	104.47	0.00	658.11

	Apr'23	May'23	June'23	July'23	Fin. Year 2023-24
Name of Line					
Export of WR to ER (WR-ER)					
WR - ER 765 kV Ranchi - Dharamjaygarh 2xS/C	139.81	504.96	673.15	734.00	2051.92
WR - ER 765 kV Dharamjaygarh - Jharsuguda 2xD/C	652.31	469.82	490.99	593.61	2206.73
WR - ER 765 kV Durg - Jharsuguda D/C	0.00	0.64	1.75	26.78	29.16
WR - ER 400 kV Sipat - Ranchi D/C	15.97	93.84	118.29	133.75	361.85
WR - ER 400 kV Raigarh - Jharsuguda 2xD/C	0.44	6.23	8.79	6.26	21.72
WR - ER 220 kV Korba - Budhipadar D/C	36.08	80.04	73.19	22.96	212.27
WR - ER 220 kV Raigarh - Budhipadar	0.15	0.57	1.17	0.00	1.89
Total WR - ER	844.75	1156.11	1367.33	1517.34	4885.54
Import of WR from ER (ER - WR)					
ER - WR 765 kV Ranchi - Dharamjaygarh 2xS/C	144.10	11.95	15.57	10.81	182.44
ER - WR 765 kV Dharamjaygarh - Jharsuguda 2XD/C	25.46	58.29	33.08	15.65	132.49
ER - WR 765 kV Durg - Jharsuguda D/C	394.08	204.42	148.02	71.84	818.36
ER - WR 400 kV Sipat - Ranchi D/C	55.56	5.71	9.07	7.22	77.55
ER - WR 400 kV Raigarh - Jharsuguda 2xD/C	248.78	148.94	125.69	125.00	648.40
ER - WR 220 kV Korba - Budhipadar D/C	1.45	0.00	0.14	6.23	7.82
ER - WR 220 kV Raigarh - Budhipadar	55.98	38.46	41.41	66.29	202.13
Total ER - WR	925.41	467.76	372.98	303.04	2069.19
Export of ER to NER (ER - NER)					
ER - NER 400 kV Binaguri - Bongaigaon D/C	6.99	16.15	105.81	142.19	271.14
ER - NER 400 kV Alipurduar - Bongaigaon D/C	4.62	34.31	135.94	83.03	257.90
ER - NER 220 kV Birpara - Salakati D/C	0.54	3.12	31.19	37.25	72.10
Total ER - NER	12.16	53.58	272.93	262.48	601.14
Import of ER from NER (NER - ER)					
NER - ER 400 kV Binaguri - Bongaigaon D/C	52.83	40.19	37.75	2.10	132.87
NER - ER 400 kV Alipurduar - Bongaigaon D/C	182.37	135.22	72.81	20.59	410.98
NER - ER 220 kV Birpara - Salakati D/C	34.59	31.32	10.78	0.62	77.30
Total NER - ER	269.78	206.73	121.33	23.31	621.14
Export of ER to SR (ER - SR)					
ER - SR HVDC Gazuwaka	375.93	395.23	208.61	178.44	1158.21
ER - SR HVDC Talchar - Kolar	1233.20	1204.28	984.75	977.61	4399.83
ER - SR 765 kV Angul- Srikakulam D/C	1683.42	1561.33	1412.42	1184.37	5841.54
Total ER - SR	3292.55	3160.85	2605.77	2340.42	11399.58
Import of ER from SR (SR - ER)					
SR - ER HVDC Gazuwaka	0.00	0.00	55.34	114.11	169.45
SR - ER HVDC Talchar - Kolar	0.00	0.00	0.00	0.00	0.01
SR - ER 765 kV Angul- Srikakulam D/C	0.00	0.00	0.00	0.00	0.00
Total SR - ER	0.00	0.00	55.34	114.11	169.46

	Apr'23	May'23	June'23	July'23	Fin. Year 2023-24
Name of Line					
Export of WR to SR (WR-SR)					
WR - SR HVDC Bhadrawati	544.41	411.52	171.68	139.64	1267.26
WR - SR 765 kV Sholapur - Raichur 2xS/C	417.69	222.76	146.07	83.93	870.46
WR - SR 765 kV Wardha - Nizamabad D/C	1323.51	873.44	1018.25	852.05	4067.26
WR - SR 400KV Kolhapur-Kudgi D/C	0.00	0.00	0.00	0.00	0.00
WR-SR HVDC Raigarh-Pugalur	3004.35	1791.86	1182.91	885.62	6864.74
WR-SR 220kV Xeldem - Ambewadi S/C	77.16	70.18	68.17	61.74	277.25
WR-SR 220kV Ponda - Ambewadi S/C	0.29	1.54	0.28	0.16	2.26
Total WR - SR	5367.41	3371.30	2587.37	2023.14	13349.22
Import of WR from SR (SR - WR)					
SR - WR HVDC Bhadrawati	4.11	25.09	178.09	308.60	515.90
SR - WR 765 kV Raichur - Sholapur 2xS/C	29.13	205.27	344.82	478.51	1057.73
SR - WR 765 kV Wardha - Nizamabad D/C	0.00	3.90	0.32	18.33	22.55
SR - WR 400KV Kolhapur-Kudgi D/C	623.59	719.30	784.98	796.02	2923.90
SR-WR HVDC Pugalur-Raigarh	0.00	3.73	14.87	16.65	35.24
WR-SR 220kV Ambewadi - Xeldem S/C	0.00	0.00	0.00	0.00	0.00
WR-SR 220kV Ambewadi-Ponda S/C	0.00	0.00	0.01	0.00	0.01
Total SR - WR	656.84	957.30	1323.09	1618.10	4555.33
TOTAL ALL INDIA	20628	19721	20060	20679	81088

*In case of mutiple ckt / DC pole year corres. to the commissioning of final element

Date	15.1 Import-Export of NR with WR during July 2023																	
	Import of NR from WR (WR-NR)																	
	WR - NR 765 kV Gwalior - Agra 2x5/C	WR-NR 765 kV Gwalior-Phagi 2x5/C	WR-NR 765 kV Jabalpur- Orai D/C	WR-NR 765 kV Satna- Orai	WR-NR 765 kV Gwalior-Orai	WR - NR HVDC Champa- Kurukshetra	WR - NR HVDC Vindhyachal	WR - NR 400 kV Zerd- Kankroli	WR - NR 400 kV Zerd- Bhinmal	WR-NR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura Ranpur	WR - NR 220 kV Bhanpura - Modak	WR - NR 220 kV Auraiya - Malanpur / Mehgaon	WR-NR 400 KV Rihand Vindhyachal D/C	WR - NR HVDC Mundra - M'garh	WR -NR 765 kV Banaskata - Chittorgarh D/C	WR-NR 765 kV Varanasi- Vindhyachal	Total WR - NR
1-Jul-23	31.21	18.64	22.54	18.90	0.00	65.00	0.00	0.00	0.00	3.13	2.43		0.00	0.00	0.00	0.00	58.47	220.32
2-Jul-23	29.25	22.85	29.52	21.27	0.00	40.95	0.00	0.00	0.00	4.48	2.36		0.00	0.00	0.00	0.00	57.91	208.59
3-Jul-23	32.65	27.43	34.94	22.71	0.00	63.16	0.00	0.00	0.00	6.14	2.68		0.00	0.00	0.00	0.00	57.63	247.34
4-Jul-23	32.43	26.03	35.54	22.15	0.00	72.01	0.00	0.00	0.00	5.33	2.72		0.00	0.00	0.00	0.00	56.53	252.74
5-Jul-23	26.97	22.90	32.09	19.69	0.00	69.52	0.00	0.00	0.00	5.52	2.47		0.00	0.00	0.00	0.00	60.85	240.01
6-Jul-23	20.17	20.99	24.59	18.63	0.00	80.95	0.00	0.00	0.00	3.50	2.35		0.00	0.00	0.00	0.00	67.56	238.74
7-Jul-23	23.84	22.61	27.70	19.41	0.00	88.70	0.00	0.00	0.00	3.98	2.26		0.00	0.00	0.00	0.00	64.81	253.31
8-Jul-23	15.38	16.37	19.21	15.02	0.00	74.04	0.00	0.00	0.00	1.83	1.97		0.00	0.00	0.00	0.00	55.37	199.19
9-Jul-23	21.31	17.24	22.00	15.95	0.00	64.14	0.00	0.00	0.00	3.06	2.18		0.00	0.00	0.00	0.00	66.30	212.18
10-Jul-23	30.74	21.73	28.40	19.18	0.00	75.76	0.00	0.00	0.00	5.00	2.38		0.00	0.00	0.00	3.51	65.13	251.83
11-Jul-23	30.20	18.72	30.00	20.87	0.00	82.41	0.00	0.10	0.00	4.55	2.10		0.00	0.00	0.00	5.21	62.43	256.59
12-Jul-23	28.58	20.48	27.79	20.38	0.00	69.96	0.00	0.00	0.00	4.91	1.57		0.00	0.00	16.11	0.00	57.89	247.67
13-Jul-23	24.68	21.95	28.20	24.24	0.00	53.11	0.00	0.00	0.00	4.14	2.53		0.00	0.00	19.05	0.00	50.63	228.53
14-Jul-23	27.95	21.57	29.08	20.83	0.00	77.21	0.00	0.00	0.00	4.32	2.26		0.00	0.00	3.77	0.76	59.83	247.58
15-Jul-23	31.49	18.80	28.37	21.72	0.00	71.20	0.00	0.59	0.00	4.10	2.30		0.00	0.00	0.00	11.50	55.10	245.17
16-Jul-23	29.86	17.28	28.70	20.25	0.00	60.70	0.00	0.55	0.00	4.39	2.04		0.00	0.00	0.00	8.51	57.30	229.58
17-Jul-23	31.01	24.71	32.73	21.73	0.00	77.41	0.00	0.00	0.00	5.50	2.04		0.00	0.00	0.00	0.00	53.03	248.16
18-Jul-23	34.12	23.14	35.03	20.96	0.00	87.60	0.00	0.00	0.00	6.08	2.38		0.00	0.00	0.00	2.11	63.39	274.81
19-Jul-23	28.69	21.44	33.25	19.60	0.00	95.18	0.06	1.23	0.93	5.22	2.34		0.00	0.00	0.00	10.32	61.56	279.82
20-Jul-23	37.98	27.38	41.90	21.70	0.00	79.88	2.59	1.85	1.61	8.20	2.46		0.00	0.00	0.00	10.32	66.97	302.84
21-Jul-23	42.50	33.51	46.04	24.11	0.00	75.28	12.01	1.89	1.48	10.77	2.61		0.00	0.00	0.00	11.52	65.01	326.73
22-Jul-23	38.63	28.04	36.82	22.28	0.00	88.07	9.48	1.13	0.44	8.89	2.50		0.00	0.00	0.00	7.88	64.69	308.85
23-Jul-23	39.20	22.37	31.06	18.79	0.00	81.72	4.39	0.57	0.00	7.15	2.24		0.00	0.00	0.00	2.94	69.10	279.53
24-Jul-23	41.98	28.54	35.96	21.94	0.00	95.20	11.94	1.35	1.47	8.65	2.16		0.00	0.00	4.69	10.63	59.80	324.31
25-Jul-23	44.08	32.98	44.11	22.93	0.00	76.54	8.49	2.36	2.53	10.72	2.22		0.00	0.00	13.09	13.39	56.50	329.94
26-Jul-23	33.01	24.71	34.86	20.40	0.00	63.70	4.49	1.08	0.29	6.86	2.50		0.00	0.00	9.01	6.79	56.93	264.63
27-Jul-23	34.48	24.53	37.13	19.36	0.00	67.45	12.01	1.62	1.64	7.05	2.27		0.00	0.00	24.21	11.03	59.70	302.48
28-Jul-23	33.58	12.34	32.98	16.33	0.00	53.43	10.70	0.00	0.85	2.83	2.11		0.00	0.00	11.93	15.97	64.92	257.97
29-Jul-23	32.48	7.38	24.64	14.48	0.00	51.00	7.22	0.33	0.00	1.30	2.01		0.00	0.00	7.35	13.56	54.63	216.38
30-Jul-23	34.96	13.81	26.33	14.78	0.00	50.50	5.97	0.00	1.30	4.72	1.50		0.00	0.00	7.35	11.59	48.95	221.76
31-Jul-23	29.36	11.14	23.26	15.18	0.00	83.40	5.43	0.71	0.00	2.17	2.08		0.00	0.00	8.33	11.77	55.46	248.29
Total	972.77	671.61	964.77	615.77	0.00	2235.18	94.78	15.36	12.54	164.49	70.02	0.00	0.00	0.00	124.89	169.31	1854.38	7965.87
Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day																		

Date	15.1 Import-Export of NR with WR during July 2023																	
	Export of NR to WR (NR- WR)																	
	NR - WR 765 kV Gwalior - Agra 2xS/C	NR-WR 765 kV Gwalior- Phagi 2xS/C	NR-WR 765 kV Jabalpur- Orai D/C	NR-WR 765 kV Satna- Orai	NR-WR 765 kV Gwalior-Orai	NR - WR HVDC Champa- Kurukshetra	NR - WR HVDC Vindhyachal	NR - WR 400 kV Zerda- Kankroli	NR - WR 400 kV Zerda- Bhinmal	NR-WR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura - Ranpur	WR - NR 220 kV Bhanpura - Modak	NR - WR 220 kV Auraiya - Malanpur / Mehgaon	NR-WR 400 kV Rihand- Vindhyachal D/C	NR - WR HVDC Mundra - M'garh	NR - WR 765 kV Banaskata - Chittorgarh D/C	NR-WR 765 kV Varanasi- Vindhyachal	Total NR - WR
1-Jul-23	0.00	0.00	0.00	0.00	13.24	0.00	12.10	4.36	1.38	0.00	0.00		0.69	21.89	6.02	2.25	0.00	61.93
2-Jul-23	0.00	0.00	0.00	0.00	10.26	0.00	12.21	2.11	5.26	0.00	0.00		0.98	21.78	6.02	10.32	0.00	68.94
3-Jul-23	0.00	0.00	0.00	0.00	11.37	0.00	12.16	2.36	5.11	0.00	0.00		1.47	21.94	6.04	12.44	0.00	72.89
4-Jul-23	0.00	0.00	0.00	0.00	11.49	0.00	12.18	2.23	6.09	0.00	0.00		1.45	21.60	6.03	11.00	0.00	72.07
5-Jul-23	0.00	0.00	0.00	0.00	11.22	0.00	12.18	1.30	2.80	0.00	0.00		1.53	21.55	5.88	7.32	0.00	63.78
6-Jul-23	0.00	0.00	0.00	0.00	10.84	0.00	11.43	2.50	3.73	0.00	0.00		2.00	21.46	3.69	17.21	0.00	72.86
7-Jul-23	0.00	0.00	0.00	0.00	10.64	0.00	6.04	1.24	3.06	0.00	0.00		2.43	22.06	4.23	8.87	0.00	58.57
8-Jul-23	0.00	0.00	0.00	0.00	11.87	0.00	6.07	1.56	3.45	0.00	0.00		2.11	20.53	5.97	7.48	0.00	59.04
9-Jul-23	0.00	0.00	0.00	0.00	12.56	0.00	6.06	1.48	3.40	0.00	0.00		1.73	20.25	5.93	6.90	0.00	58.31
10-Jul-23	0.00	0.00	0.00	0.00	11.27	0.00	8.01	0.10	1.70	0.00	0.00		1.70	21.78	5.93	0.00	0.00	50.49
11-Jul-23	0.00	0.00	0.00	0.00	10.01	0.00	11.62	0.00	3.02	0.00	0.00		1.78	21.98	4.96	0.00	0.00	53.37
12-Jul-23	0.00	0.00	0.00	0.00	10.31	0.00	7.30	0.70	6.38	0.00	0.00		1.87	22.12	0.00	0.05	0.00	48.73
13-Jul-23	0.00	0.00	0.00	0.00	11.04	0.00	7.60	1.96	7.44	0.00	0.00		3.20	22.10	0.00	6.50	0.00	59.84
14-Jul-23	0.00	0.00	0.00	0.00	10.83	0.00	12.20	0.57	4.58	0.00	0.00		2.91	21.96	0.00	0.00	0.00	53.05
15-Jul-23	0.00	0.00	0.00	0.00	9.82	0.00	12.14	0.00	4.20	0.00	0.00		2.53	22.09	6.00	0.00	0.00	56.78
16-Jul-23	0.00	0.00	0.00	0.00	9.63	0.00	12.15	0.00	2.20	0.00	0.00		2.44	20.59	3.00	0.00	0.00	50.01
17-Jul-23	0.00	0.00	0.00	0.00	12.23	0.00	11.71	1.05	2.50	0.00	0.00		2.84	22.16	2.25	4.19	0.00	58.93
18-Jul-23	0.00	0.00	0.00	0.00	11.34	0.00	7.07	0.08	1.29	0.00	0.00		2.63	21.72	5.94	0.00	0.00	50.07
19-Jul-23	0.00	0.00	0.00	0.00	10.26	0.00	4.20	0.00	0.00	0.00	0.00		3.17	22.10	5.77	0.00	0.00	45.50
20-Jul-23	0.00	0.00	0.00	0.00	12.90	0.00	0.00	0.00	0.00	0.00	0.00		2.92	22.11	5.92	0.00	0.00	43.85
21-Jul-23	0.00	0.00	0.00	0.00	13.99	0.00	0.00	0.00	0.00	0.00	0.00		2.89	22.27	4.58	0.00	0.00	43.73
22-Jul-23	0.00	0.00	0.00	0.00	12.77	0.00	0.00	0.00	0.00	0.00	0.00		1.89	22.26	0.00	0.00	0.00	36.92
23-Jul-23	0.00	0.00	0.00	0.00	14.95	0.00	0.00	0.00	1.60	0.00	0.00		0.00	20.77	0.00	0.00	0.00	37.32
24-Jul-23	0.00	0.00	0.00	0.00	14.97	0.00	0.00	0.00	0.00	0.00	0.00		0.00	22.10	0.00	0.00	0.00	37.07
25-Jul-23	0.00	0.00	0.00	0.00	13.58	0.00	0.00	0.00	0.00	0.00	0.00		0.00	22.20	0.00	0.00	0.00	35.78
26-Jul-23	0.00	0.00	0.00	0.00	12.17	0.00	0.00	0.00	0.00	0.00	0.00		0.00	22.17	0.00	0.00	0.00	34.34
27-Jul-23	0.00	0.00	0.00	0.00	11.82	0.00	0.00	0.00	0.00	0.00	0.00		0.00	22.41	0.00	0.00	0.00	34.23
28-Jul-23	0.00	0.00	0.00	0.00	10.24	0.00	0.00	1.09	0.00	0.00	0.00		0.00	20.12	0.00	0.00	0.00	31.45
29-Jul-23	0.00	0.00	0.00	0.00	12.81	0.00	0.00	0.00	3.04	0.00	0.00		0.00	18.13	0.00	0.00	0.00	33.98
30-Jul-23	0.00	0.00	0.00	0.00	14.25	0.00	0.00	0.09	0.00	0.00	0.00		0.00	17.38	0.00	0.00	0.00	31.72
31-Jul-23	0.00	0.00	0.00	0.00	13.20	0.00	0.00	0.00	1.53	0.00	0.00		0.00	20.75	0.00	0.00	0.00	35.48
Total	0.00	0.00	0.00	0.00	367.88	0.00	184.43	24.78	73.76	0.00	0.00	0.00	47.16	664.33	94.16	94.53	0.00	1551.03
	Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day																	

15.2 Import-Export of NR with ER & NER during July 2023

Date	Import of NR from ER (ER-NR)																		Import of NR from NER (NER-NR)	
	ER - NR 765 kV Sasaram - Fatehpur	ER - NR 765 kV Gaya - Varanasi 2*S/C	ER - NR 765 kV Gaya Ballia	ER - NR HVDC Alipurduar - Agra	ER-NR 400 kV Patna - Ballia D/C	ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Ballia D/C	ER - NR 400 kV Motihari - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Varanasi D/C	ER-NR 400 kV Sasaram - Varanasi	ER-NR 400 kV Sasaram - Allahabad	ER - NR 220 kV Sahupuri Karamnasa	ER - NR 132 kV Sahupuri Karamnasa	ER - NR 132 kV Rihand - Nagar Untari	NR - ER 132 kV Rihand - Garhwa	ER-NR 400 kV Naubatpur - Ballia D/C	Total ER - NR	NER-NR HVDC Biswanath Chariali - Agra	Total NER-NR	
1-Jul-23	4.30	0.00	8.03	35.47	9.15	11.60	1.99	6.76	0.27	1.38	0.87	1.12	1.12		0.00	9.36	91.42	7.40	7.40	
2-Jul-23	4.65	1.15	9.12	34.12	8.53	12.71	4.58	6.95	1.45	1.44	0.88	1.20			0.00	9.01	95.79	7.44	7.44	
3-Jul-23	5.81	5.40	11.32	35.57	10.88	16.02	6.89	8.29	3.44	1.30	1.07	0.94			0.00	11.26	118.19	7.39	7.39	
4-Jul-23	5.05	4.61	11.43	35.02	11.61	16.54	4.74	8.78	1.64	1.30	1.18	1.51		0.00	0.00	12.03	115.44	7.00	7.00	
5-Jul-23	1.99	0.00	10.08	36.20	9.53	9.80	3.46	6.80	0.10	1.79	1.01	1.32			0.00	9.66	91.74	7.20	7.20	
6-Jul-23	0.00	0.00	10.75	35.05	8.62	10.01	3.24	5.65	0.00	2.05	1.33	0.95			0.00	8.60	86.25	7.32	7.32	
7-Jul-23	0.00	0.00	11.56	35.07	9.16	10.16	3.00	5.81	0.00	2.13	1.34	0.98			0.00	9.46	88.67	11.41	11.41	
8-Jul-23	0.96	0.00	9.13	35.19	7.57	8.62	2.44	4.35	0.00	2.70	0.00	0.85			0.00	7.40	79.21	11.73	11.73	
9-Jul-23	0.00	0.00	9.63	37.67	7.57	9.85	2.52	4.93	0.00	2.29	1.25	1.20	0.00		0.00	7.41	84.32	12.00	12.00	
10-Jul-23	1.56	0.00	11.61	43.10	9.64	9.52	3.61	6.18	0.00	2.38	1.12	1.30			0.00	9.75	99.77	12.17	12.17	
11-Jul-23	2.25	0.00	11.56	37.02	9.87	10.82	3.29	7.04	0.00	2.50	0.86	1.60			0.00	9.99	96.80	13.78	13.78	
12-Jul-23	3.14	0.00	9.89	29.38	9.61	11.51	3.61	6.99	0.00	2.52	0.96	1.86			0.00	9.61	89.08	12.57	12.57	
13-Jul-23	4.15	1.54	8.54	23.49	9.41	11.65	3.76	6.71	0.44	2.34	1.20	1.94			0.00	9.35	84.52	8.51	8.51	
14-Jul-23	1.60	0.34	8.61	23.45	7.80	12.72	2.84	6.80	0.26	2.15	1.44	1.82			0.00	7.62	77.45	12.01	12.01	
15-Jul-23	2.14	0.00	7.65	23.74	6.58	11.80	2.06	6.76	0.00	1.69	0.95	1.27			0.00	6.27	70.91	11.87	11.87	
16-Jul-23	2.33	0.00	8.53	23.50	6.20	11.60	2.14	7.15	0.34	0.65	0.38	1.60	0.00	0.00	0.00	5.81	70.23	11.64	11.64	
17-Jul-23	3.88	5.07	10.78	24.68	8.17	15.17	4.69	9.12	2.07	1.58	1.03	1.97	0.00	0.00		8.05	96.26	12.08	12.08	
18-Jul-23	1.95	1.92	11.01	24.10	6.82	12.92	2.92	8.31	0.78	1.62	0.99	1.68			0.00	6.46	81.48	12.10	12.10	
19-Jul-23	1.50	0.00	10.13	24.60	6.55	9.78	1.57	6.81	0.00	1.98	0.52	1.72			0.00	6.15	71.31	12.12	12.12	
20-Jul-23	2.57	0.69	11.71	23.95	7.47	11.62	2.33	8.28	0.31	1.71	0.84	1.75			0.00	7.08	80.31	12.11	12.11	
21-Jul-23	5.45	7.03	13.49	23.07	9.70	15.21	5.20	10.21	3.72	1.35	1.29	1.97	0.00		0.00	9.61	107.30	11.54	11.54	
22-Jul-23	3.90	2.20	11.32	23.63	7.78	13.90	3.96	9.43	1.98	1.41	1.20	1.28			0.00	7.54	89.53	11.28	11.28	
23-Jul-23	1.19	0.00	10.37	22.59	6.13	12.68	2.93	7.77	0.22	1.55	1.12	0.92	0.00		0.00	5.69	73.16	11.30	11.30	
24-Jul-23	3.68	6.65	6.79	11.60	8.65	14.42	3.65	10.64	2.35	1.45	1.17	1.49			0.00	8.32	80.86	11.60	11.60	
25-Jul-23	4.34	8.42	12.10	11.90	9.40	15.14	3.72	7.70	3.64	1.40	1.25	1.79	0.00		0.00	9.24	90.04	12.20	12.20	
26-Jul-23	1.86	2.18	9.57	12.05	7.20	12.99	2.29	8.82	1.39	1.51	1.09	1.57	0.00		0.00	6.73	69.25	12.12	12.12	
27-Jul-23	1.03	0.00	9.29	11.76	6.01	12.06	1.76	8.12	0.09	1.80	0.65	1.24		0.00		5.41	59.22	12.14	12.14	
28-Jul-23	0.00	0.00	7.94	12.40	3.57	10.22	0.57	5.96	0.00	2.23	0.36	0.99	0.00	0.00		2.65	46.89	12.11	12.11	
29-Jul-23	0.00	0.00	4.59	12.58	1.98	10.57	0.08	5.92	0.00	1.94	0.73	0.97	0.00	0.00		0.97	40.33	12.09	12.09	
30-Jul-23	1.68	0.00	5.79	11.49	3.65	13.38	0.50	7.85	0.16	1.84	0.69	1.34			0.00	2.93	51.30	11.50	11.50	
31-Jul-23	1.20	0.00	8.42	11.55	7.42	11.45	2.43	7.70	0.00	1.66		1.31			0.00	7.11	60.25	11.55	11.55	
Total	74.16	47.20	300.74	784.99	242.23	376.44	92.77	228.59	24.65	55.64	28.77	43.45	1.12	0.00	0.00	236.53	2537.28	339.28	339.28	

Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day

Date	15.2 Import-Export of NR with ER & NER during July 2023																		
	Export of NR to ER (NR- ER)																	Export of NR to NER (NR-NER)	
	NR - ER 765 kV Sasaram - Fatehpur	NR - ER 765 kV Gaya - Varanasi 2*S/C	NR - ER 765 kV Gaya Ballia	NR - ER HVDC Alipurduar - Agra	NR-ER 400 kV Patna - Ballia D/C	NR - ER 400 kV Muzaffarpur - Gorakhpur D/C	NR - ER 400 kV Biharshariff - Ballia D/C	NR - ER 400 kV Motihari - Gorakhpur D/C	NR - ER 400 kV Biharshariff - Varanasi D/C	NR-ER 400 kV Sasaram - Varanasi	NR-ER 400 kV Sasaram - Allahabad	NR - ER 220 kV Sahupuri- Karamnasa	NR - ER 132 kV Sahupuri - Karamnasa	NR - ER 132 kV Rihand -Nagar Untari	NR - ER 132 kV Rihand - Garhwa	NR-ER 400 kV Ballia- Naubatpur D/C	Total NR -ER	NR-NER HVDC Biswanath Chariali - Agra	Total NR-NER
1-Jul-23	0.00	1.17	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.52	0.00	1.69	0.00	0.00
2-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.57	0.00	0.57	0.00	0.00
3-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.68	0.00	0.68	0.00	0.00
4-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.72	0.00	0.00	0.72	0.00	0.00
5-Jul-23	0.00	1.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.71	0.00	1.89	0.00	0.00
6-Jul-23	0.08	4.07	0.00	0.00	0.00	0.00	0.00	0.00	1.11	0.00	0.00	0.00			0.70	0.00	5.96	0.00	0.00
7-Jul-23	0.10	2.88	0.00	0.00	0.00	0.00	0.00	0.00	0.94	0.00	0.00	0.00			0.67	0.00	4.59	0.00	0.00
8-Jul-23	0.00	4.80	0.00	0.00	0.00	0.00	0.00	0.00	2.28	0.00	0.82	0.00			0.70	0.00	8.60	0.00	0.00
9-Jul-23	0.20	5.25	0.00	0.00	0.00	0.00	0.00	0.00	1.53	0.00	0.00	0.00	0.00		0.74	0.00	7.72	0.00	0.00
10-Jul-23	0.00	3.50	0.00	0.00	0.00	0.00	0.00	0.00	0.84	0.00	0.00	0.00			0.73	0.00	5.07	0.00	0.00
11-Jul-23	0.00	2.33	0.00	0.00	0.00	0.00	0.00	0.00	0.27	0.00	0.00	0.00			0.70	0.00	3.30	0.00	0.00
12-Jul-23	0.00	0.40	0.00	0.00	0.00	0.00	0.00	0.00	0.18	0.00	0.00	0.00			0.63	0.00	1.21	0.00	0.00
13-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.75	0.00	0.75	0.00	0.00
14-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.73	0.00	0.73	0.00	0.00
15-Jul-23	0.00	1.41	0.00	0.00	0.00	0.00	0.00	0.00	0.15	Sasaram	0.00	0.00			0.71	0.00	2.27	0.00	0.00
16-Jul-23	0.00	0.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.48	0.00	0.68	0.00	1.46	0.00	0.00
17-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.48	0.65		0.00	1.13	0.00	0.00
18-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.74	0.00	0.74	0.00	0.00
19-Jul-23	0.00	0.90	0.00	0.00	0.00	0.00	0.00	0.00	0.35	0.00	0.00	0.00			0.70	0.00	1.95	0.00	0.00
20-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.75	0.00	0.75	0.00	0.00
21-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.77	0.00	0.77	0.00	0.00
22-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.68	0.00	0.68	0.00	0.00
23-Jul-23	0.00	2.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.75	0.00	2.98	0.00	0.00
24-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.76	0.00	0.76	0.00	0.00
25-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.80	0.00	0.80	0.00	0.00
26-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.75	0.00	0.75	0.00	0.00
27-Jul-23	0.00	0.88	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.84		0.00	1.72	0.00	0.00
28-Jul-23	2.40	6.46	0.00	0.00	0.00	0.00	0.00	0.00	1.27	0.00	0.00	0.00	0.00	0.72		0.00	10.85	0.00	0.00
29-Jul-23	0.67	5.37	0.00	0.00	0.00	0.00	0.00	0.00	1.28	0.00	0.00	0.00	0.00	0.44		0.00	7.76	0.00	0.00
30-Jul-23	0.00	0.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.69	0.00	0.84	0.00	0.00
31-Jul-23	0.00	2.66	0.00	0.00	0.00	0.00	0.00	0.00	0.19	0.00		0.00			0.74	0.00	3.59	0.00	0.00
Total	3.45	45.94	0.00	0.00	0.00	0.00	0.00	0.00	10.39	0.00	0.82	0.00	0.96	3.37	18.35	0.00	83.28	0.00	0.00
Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day																			

Date	15.3 Import and Export of ER with WR & NER during July 2023																							
	Export of WR to ER (WR-ER)								Import of WR from ER (ER - WR)								Export of ER to NER (ER-NER)				Import of ER from NER (NER-ER)			
	WR - ER 765 kV Dharamjaygarh - Ranchi 2x5/C	WR - ER 765 kV Dharamjaygarh - Jharsuguda Q/C	WR - ER 765 kV Durg - Jharsuguda D/C	WR - ER 400 kV Sipat - Ranchi D/C	WR - ER 400 kV Raigarh - Jharsuguda-2x5/D/C	WR - ER 220 kV Korba - Budhipadar D/C	WR - ER 220 kV Raigarh - Budhipadar	Total WR - ER	ER - WR 765 kV Dharamjaygarh - Ranchi 2x5/C	ER-WR 765 kV Dharamjaygarh - Jharsuguda D/C	ER-WR 765 kV Durg - Jharsuguda D/C	ER - WR 400 kV Sipat - Ranchi D/C	ER - WR 400 kV Raigarh - Jharsuguda-2x5/D/C	ER - WR 220 kV Korba - Budhipadar D/C	ER - WR 220 kV Raigarh - Budhipadar	Total ER - WR	ER - NER 400 kV Binaguri - Bongaigaon D/C	ER - NER 400 kV Alipurduar - Bongaigaon D/C	ER - NER 220 kV Birpara - Salakati D/C	Total ER - NER	NER - ER 400 kV Binaguri - Bongaigaon D/C	NER - ER 400 kV Alipurduar - Bongaigaon 2x5/D/C	NER - ER 220 kV Birpara - Salakati D/C	Total NER - ER
1-Jul-23	17.5	21.0	0.6	2.4	0.2	0.8	0.0	42.57	0.9	0.4	2.1	0.5	3.3	0.1	1.5	8.81	6.94	2.54	1.26	10.75	0.00	0.00	0.00	0.00
2-Jul-23	19.2	25.3	0.5	2.6	0.0	1.1	0.0	48.70	0.3	0.1	1.4	0.4	5.0	0.1	1.9	9.26	6.50	2.26	1.27	10.03	0.00	0.01	0.00	0.01
3-Jul-23	18.9	16.1	0.1	2.4	0.0	0.4	0.0	37.81	0.0	0.5	2.7	0.3	6.2	0.2	2.2	11.97	4.85	0.85	1.05	6.75	0.00	0.94	0.00	0.94
4-Jul-23	22.7	16.2	0.0	3.1	0.0	0.5	0.0	42.41	0.1	0.0	3.5	0.3	6.8	0.3	2.4	13.29	5.90	2.50	1.21	9.61	0.00	1.04	0.00	1.04
5-Jul-23	27.9	17.5	0.1	4.9	0.0	0.5	0.0	50.81	0.1	0.4	2.3	0.2	5.0	0.4	1.7	9.99	8.15	4.70	1.88	14.73	0.00	0.01	0.00	0.01
6-Jul-23	25.9	16.9	0.2	4.3	0.0	1.3	0.0	48.61	0.2	1.4	3.7	0.3	4.0	0.0	2.3	11.90	7.87	4.14	1.75	13.75	0.00	0.01	0.00	0.01
7-Jul-23	22.3	12.0	0.0	4.6	0.4	1.5	0.0	40.84	0.5	1.1	6.0	0.2	2.1	0.0	1.7	11.51	8.80	4.32	1.95	15.06	0.00	0.00	0.00	0.00
8-Jul-23	20.7	17.8	0.2	4.2	0.1	1.4	0.0	44.33	0.7	0.3	4.8	0.2	3.4	0.0	2.2	11.57	7.38	3.76	2.12	13.25	0.00	0.08	0.00	0.08
9-Jul-23	26.1	20.9	1.5	5.0	0.0	0.6	0.0	54.08	0.1	0.0	2.6	0.0	4.0	0.1	1.7	8.59	6.12	1.33	1.48	8.93	0.00	0.23	0.00	0.23
10-Jul-23	25.6	20.3	0.2	4.5	0.0	1.0	0.0	51.61	0.0	0.0	2.9	0.1	3.7	0.0	1.6	8.30	7.66	0.95	1.25	9.86	0.00	2.08	0.00	2.08
11-Jul-23	22.9	21.7	0.5	4.3	0.2	1.7	0.0	51.30	0.1	0.0	2.6	0.2	2.2	0.0	1.5	6.54	7.39	1.53	1.37	10.29	0.00	0.59	0.00	0.59
12-Jul-23	19.0	26.3	0.9	3.5	0.2	0.3	0.0	50.15	0.6	0.0	1.9	0.3	3.3	0.5	1.8	8.50	4.53	0.78	0.84	6.15	0.00	1.01	0.00	1.01
13-Jul-23	19.7	28.3	1.2	3.1	0.0	0.1	0.0	52.46	1.3	0.0	2.3	0.7	5.6	0.8	2.1	12.67	1.35	0.19	0.35	1.89	0.45	3.13	0.13	3.71
14-Jul-23	20.7	20.3	0.4	3.3	0.1	0.3	0.0	45.01	0.6	0.0	3.2	0.4	5.5	0.5	2.3	12.50	4.01	3.30	1.32	8.62	0.07	0.93	0.02	1.02
15-Jul-23	17.4	24.6	0.5	3.6	0.7	1.3	0.0	48.13	1.4	0.0	2.5	0.5	2.1	0.1	2.2	8.88	4.08	2.79	1.25	8.13	0.00	0.29	0.00	0.29
16-Jul-23	18.3	26.5	0.7	3.4	0.0	1.3	0.0	50.12	1.4	0.0	2.3	0.5	3.9	0.0	2.3	10.44	2.82	1.01	0.83	4.65	0.02	0.65	0.01	0.68
17-Jul-23	16.8	19.4	0.0	2.7	0.1	0.9	0.0	39.89	1.4	0.0	5.1	0.6	4.7	0.1	2.5	14.40	0.85	0.13	0.30	1.29	0.53	3.65	0.15	4.33
18-Jul-23	24.3	14.3	0.4	4.4	0.1	0.2	0.0	43.73	0.3	0.2	3.9	0.3	5.4	0.5	1.2	11.85	2.98	1.68	0.74	5.40	0.14	1.48	0.05	1.67
19-Jul-23	26.2	16.6	2.0	5.3	0.8	0.7	0.0	51.54	0.0	1.0	1.2	0.0	1.9	0.1	2.5	6.58	5.06	3.34	0.97	9.37	0.00	0.17	0.01	0.18
20-Jul-23	28.7	22.7	3.5	5.7	0.4	0.7	0.0	61.60	0.0	0.0	0.2	0.0	2.7	0.1	2.8	5.75	5.18	3.53	1.27	9.98	0.00	0.03	0.00	0.03
21-Jul-23	25.7	21.0	2.1	4.4	0.0	0.1	0.0	53.40	0.0	0.0	0.6	0.0	4.9	0.6	2.5	8.58	2.58	1.17	0.67	4.42	0.09	1.61	0.18	1.89
22-Jul-23	25.4	14.4	2.0	5.2	0.3	0.3	0.0	47.62	0.0	0.5	0.9	0.0	2.6	0.4	2.2	6.50	4.99	3.42	1.35	9.77	0.00	0.02	0.00	0.02
23-Jul-23	27.5	11.6	1.2	6.3	0.8	1.2	0.0	48.69	0.0	1.3	1.1	0.0	1.6	0.2	1.8	6.07	3.03	1.28	0.73	5.04	0.00	0.57	0.01	0.58
24-Jul-23	23.1	8.3	0.0	4.2	0.3	0.9	0.0	36.78	0.2	2.7	2.0	0.1	3.1	0.1	2.7	10.91	3.16	3.82	1.18	8.16	0.01	0.04	0.00	0.06
25-Jul-23	25.2	7.5	0.0	4.1	0.0	0.3	0.0	37.06	0.0	2.8	3.8	0.2	6.9	0.5	3.1	17.39	3.42	4.37	1.28	9.07	0.00	0.05	0.00	0.06
26-Jul-23	26.9	17.5	0.7	4.8	0.0	0.3	0.0	50.20	0.0	1.7	1.6	0.1	5.6	0.5	3.1	12.56	3.25	4.70	1.35	9.29	0.00	0.00	0.00	0.00
27-Jul-23	28.4	12.5	1.1	5.3	0.0	0.5	0.0	47.84	0.0	1.0	1.1	0.1	6.0	0.2	3.0	11.44	3.62	5.33	1.53	10.48	0.00	0.00	0.00	0.00
28-Jul-23	33.4	19.5	1.7	6.8	0.0	0.5	0.0	61.93	0.0	0.3	0.6	0.1	3.9	0.0	2.1	6.88	3.14	4.61	1.37	9.12	0.00	0.00	0.00	0.00
29-Jul-23	28.2	30.6	1.7	5.7	0.4	0.3	0.0	66.88	0.0	0.0	0.4	0.0	3.3	0.0	1.2	4.89	1.89	2.71	1.06	5.66	0.19	0.27	0.00	0.46
30-Jul-23	26.1	26.1	2.1	4.8	0.4	0.9	0.0	60.53	0.1	0.0	1.1	0.2	4.1	0.0	2.1	7.66	1.62	2.30	0.91	4.83	0.48	0.70	0.05	1.23
31-Jul-23	23.2	20.1	0.8	4.7	0.5	1.3	0.0	50.71	0.6	0.0	1.5	0.3	2.3	0.0	2.2	6.84	3.10	3.69	1.33	8.12	0.12	0.98	0.00	1.11
Total	734.00	593.61	26.78	133.75	6.26	22.96	0.00	1517.34	10.81	15.65	71.84	7.22	125.00	6.23	66.29	303.04	142.19	83.03	37.25	262.48	2.10	20.59	0.62	23.31

Date	15.4 Import and Export of SR with ER & WR during July 2023																							
	Export of ER to SR (ER - SR)				Import of ER from SR (SR - ER)				Export of WR to SR (WR-SR)								Import of WR from SR (SR - WR)							
	ER - SR HVDC Gazuwaka	ER - SR HVDC Talchar - Kolar	ER-SR 765 kV Angul- Srikakulam D/C	Total ER - SR	SR - ER HVDC Gazuwaka	SR - ER HVDC Talchar -	SR-ER 765 kV Angul- Srikakulam D/C	Total SR - ER	WR - SR HVDC Bhadrawati	WR - SR 765 kV Sholapur - Raichur 2x5/C	WR-SR 765 kV Wardha - Nizamabad D/C	WR-SR 400KV Kolhapur- Kudgi D/C	WR-SR HVDC Raigarh- Puglur	WR-SR 220kV Xeldem - Ambewadi S/C	WR-SR 220kV Ponda - Ambewadi S/C	Total WR - SR	SR - WR HVDC Bhadrawati	SR - WR 765 kV Raichur- Sholapur 2x5/C	SR-WR 765 kV Wardha - Nizamabad D/C	SR-WR 400KV Kolhapur-Kudgi D/C	SR-WR HVDC Raigarh- Puglur	WR-SR 220kV Xeldem - Ambewadi S/C	WR-SR 220kV Ponda - Ambewadi S/C	Total SR - WR
1-Jul-23	12.48	39.71	48.12	100.31	0.00	0.00	0.00	0.00	9.87	3.99	33.36	0.00	27.75	2.14	0.00	77.11	1.21	7.24	0.00	21.29	0.00	0.00	0.00	29.74
2-Jul-23	12.48	27.73	44.55	84.77	0.00	0.00	0.00	0.00	9.60	1.21	30.15	0.00	19.00	2.16	0.00	62.13	0.00	12.71	0.00	23.30	0.00	0.00	0.00	36.01
3-Jul-23	12.50	19.82	37.09	69.41	0.00	0.00	0.00	0.00	6.11	0.52	26.36	0.00	25.36	2.16	0.00	60.53	2.46	24.77	0.00	27.30	0.00	0.00	0.00	54.53
4-Jul-23	9.46	18.31	38.94	66.71	0.00	0.00	0.00	0.00	0.00	0.23	30.86	0.00	21.29	2.16	0.00	54.55	10.66	21.82	0.00	27.53	0.00	0.00	0.00	60.01
5-Jul-23	7.70	19.30	41.15	68.15	0.00	0.00	0.00	0.00	0.00	1.08	36.93	0.00	22.40	2.13	0.01	62.56	17.52	16.07	0.00	25.74	0.00	0.00	0.00	59.32
6-Jul-23	10.22	19.30	40.43	69.95	0.00	0.00	0.00	0.00	0.00	2.37	37.75	0.00	18.50	2.09	0.01	60.70	14.00	15.82	0.00	22.99	1.96	0.00	0.00	54.78
7-Jul-23	10.34	19.30	42.91	72.55	0.00	0.00	0.00	0.00	0.00	4.28	39.82	0.00	39.88	2.15	0.01	86.13	15.48	11.58	0.00	21.85	0.00	0.00	0.00	48.91
8-Jul-23	10.34	23.67	42.90	76.91	0.00	0.00	0.00	0.00	6.36	5.75	35.73	0.00	42.91	2.07	0.00	92.82	8.48	11.40	0.00	21.07	0.00	0.00	0.00	40.95
9-Jul-23	10.18	34.56	35.73	80.47	0.00	0.00	0.00	0.00	7.56	0.86	25.00	0.00	22.93	2.02	0.00	58.37	0.00	17.70	0.00	23.90	0.00	0.00	0.00	41.60
10-Jul-23	7.44	39.22	44.73	91.40	0.00	0.00	0.00	0.00	7.32	6.70	39.03	0.00	33.57	1.99	0.00	88.61	0.00	5.49	0.00	19.80	0.00	0.00	0.00	25.29
11-Jul-23	10.08	39.14	41.13	90.35	0.00	0.00	0.00	0.00	7.25	3.99	34.53	0.00	41.73	2.13	0.00	89.63	0.00	8.84	0.00	21.63	0.00	0.00	0.00	30.47
12-Jul-23	10.10	43.43	38.56	92.09	0.00	0.00	0.00	0.00	16.93	3.40	31.05	0.00	56.66	2.14	0.00	110.18	0.00	11.54	0.00	27.76	0.00	0.00	0.00	39.30
13-Jul-23	10.07	35.89	38.59	84.55	0.00	0.00	0.00	0.00	14.87	1.14	29.83	0.00	52.29	2.15	0.00	100.29	0.00	11.05	0.00	25.47	0.00	0.00	0.00	36.52
14-Jul-23	10.06	33.71	41.93	85.70	0.00	0.00	0.00	0.00	7.80	3.38	36.67	0.00	60.96	2.06	0.00	110.89	3.20	8.81	0.00	24.29	0.00	0.00	0.00	36.31
15-Jul-23	9.99	38.94	43.77	92.71	0.00	0.00	0.00	0.00	5.07	5.79	42.62	0.00	91.24	2.23	0.00	146.95	2.08	3.02	0.00	21.54	0.00	0.00	0.00	26.64
16-Jul-23	10.00	39.67	41.81	91.48	0.00	0.00	0.00	0.00	18.53	7.39	36.86	0.00	42.10	2.17	0.00	107.04	0.00	4.95	0.00	19.62	0.00	0.00	0.00	24.57
17-Jul-23	10.07	34.19	43.22	87.48	0.00	0.00	0.00	0.00	10.16	4.77	37.26	0.00	27.91	2.22	0.00	82.32	0.00	6.04	0.00	21.28	0.00	0.00	0.00	27.32
18-Jul-23	4.91	27.77	38.37	71.05	2.20	0.00	0.00	2.20	7.30	1.99	26.28	0.00	18.68	2.10	0.01	56.35	9.41	13.36	0.10	23.20	0.00	0.00	0.00	46.08
19-Jul-23	0.00	32.27	30.37	62.64	4.86	0.00	0.00	4.86	0.00	0.40	16.72	0.00	22.50	1.80	0.01	41.43	16.95	18.43	0.94	27.83	0.00	0.00	0.00	64.15
20-Jul-23	0.00	29.71	27.00	56.71	4.90	0.00	0.00	4.90	0.00	0.32	12.49	0.00	17.17	1.85	0.01	31.83	21.34	24.63	1.10	29.96	0.00	0.00	0.00	77.03
21-Jul-23	0.00	28.52	32.42	60.95	4.89	0.00	0.00	4.89	0.00	0.93	16.39	0.00	15.39	1.79	0.01	34.51	18.05	22.03	1.70	28.81	0.00	0.00	0.00	70.61
22-Jul-23	0.00	32.87	30.08	62.96	4.89	0.00	0.00	4.89	0.00	0.15	12.72	0.00	24.99	1.69	0.01	39.56	11.01	24.66	0.31	29.85	0.00	0.00	0.00	65.82
23-Jul-23	0.00	27.76	27.09	54.85	5.70	0.00	0.00	5.70	0.00	0.15	10.79	0.00	15.36	1.66	0.01	27.97	15.45	30.47	1.27	31.89	0.00	0.00	0.00	79.09
24-Jul-23	0.00	28.65	32.84	61.49	9.71	0.00	0.00	9.71	0.00	0.89	16.78	0.00	15.72	1.66	0.01	35.06	23.79	24.89	2.67	31.50	0.00	0.00	0.00	82.84
25-Jul-23	0.00	34.46	33.71	68.17	13.72	0.00	0.00	13.72	0.00	1.17	13.83	0.00	11.30	1.65	0.01	27.96	19.17	28.95	2.34	31.37	3.43	0.00	0.00	85.26
26-Jul-23	0.00	35.99	34.45	70.43	16.72	0.00	0.00	16.72	0.00	1.19	15.97	0.00	6.58	1.56	0.01	25.31	16.66	23.72	4.36	31.98	5.79	0.00	0.00	82.51
27-Jul-23	0.00	27.60	34.45	57.58	9.58	0.00	0.00	9.58	0.00	0.97	11.22	0.00	3.11	1.21	0.00	16.52	23.78	25.54	3.18	30.15	5.46	0.00	0.00	88.13
28-Jul-23	0.00	35.53	36.08	71.62	9.47	0.00	0.00	9.47	0.00	3.11	20.34	0.00	13.76	2.01	0.00	39.22	20.22	14.96	0.36	26.71	0.00	0.00	0.00	62.25
29-Jul-23	0.00	38.78	42.98	81.76	10.66	0.00	0.00	10.66	0.00	5.21	32.22	0.00	19.48	2.34	0.00	59.26	18.81	9.58	0.00	25.71	0.00	0.00	0.00	54.11
30-Jul-23	0.00	35.18	41.24	76.42	10.12	0.00	0.00	10.12	0.00	4.34	30.79	0.00	16.66	2.13	0.00	53.91	14.06	11.14	0.00	25.43	0.00	0.00	0.00	50.63
31-Jul-23	0.00	36.64	42.18	78.82	6.68	0.00	0.00	6.68	4.89	6.27	31.72	0.00	38.42	2.13	0.00	83.43	4.79	7.27	0.00	25.26	0.00	0.00	0.00	37.32
TOTAL	178.44	977.61	1184.37	2340.42	114.11	0.00	0.00	114.11	139.64	83.93	852.05	0.00	885.62	61.74	0.16	2023.14	308.60	478.51	18.33	796.02	16.65	0.00	0.00	1618.10

अंतरराष्ट्रीय विद्युत विनिमय [भारत 'से' निर्यात]/ 'को' आयात]
Transnational Exchange ('Export from'/'Import to' India)

* Based on daily operational data

Date	16.1 Import from neighbouring countries during July 2023 (All figures in MU)													
	Import from Bhutan								Import from Bangladesh		Import from Nepal			Import from Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurdwār D/C*	400 kV Jigmeling-Alipurdwār D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-Jul-23	18.18	5.31	2.14	0.37	5.46	8.26	1.12	0.56	0.00	0.00	0.00	9.34	0.00	0.00
2-Jul-23	18.29	5.40	2.73	0.46	5.75	8.70	0.93	0.66	0.00	0.00	0.00	7.62	0.00	0.00
3-Jul-23	18.43	5.48	2.91	0.55	6.08	9.20	0.93	0.68	0.00	0.00	0.00	8.80	0.00	0.00
4-Jul-23	18.35	5.42	2.83	0.52	5.99	9.06	0.88	0.52	0.00	0.00	0.00	8.54	0.00	0.00
5-Jul-23	18.28	5.37	2.90	0.70	5.83	8.81	0.95	0.67	0.00	0.00	0.00	8.87	0.00	0.00
6-Jul-23	18.23	5.34	3.00	0.68	5.79	8.76	0.95	0.64	0.00	0.00	0.00	8.01	0.00	0.00
7-Jul-23	18.32	5.23	3.14	0.80	5.92	8.98	0.69	0.76	0.00	0.00	0.00	8.56	0.00	0.00
8-Jul-23	18.10	5.35	2.90	0.74	5.99	9.07	0.81	0.80	0.00	0.00	0.00	8.92	0.00	0.00
9-Jul-23	18.38	5.44	2.78	0.69	5.92	8.95	0.74	0.81	0.00	0.00	0.00	7.10	0.00	0.00
10-Jul-23	17.86	5.21	2.71	0.73	5.53	8.38	0.51	0.56	0.00	0.00	0.00	7.79	0.00	0.00
11-Jul-23	16.64	4.82	2.66	0.54	5.99	9.07	0.58	0.61	0.00	0.00	0.00	8.28	0.00	0.00
12-Jul-23	16.05	4.67	2.69	0.20	5.86	8.87	0.94	0.57	0.00	0.00	0.00	9.01	0.00	0.00
13-Jul-23	18.55	5.25	2.95	0.61	6.04	9.15	1.07	0.10	0.00	0.00	0.00	10.65	0.00	0.00
14-Jul-23	18.53	5.55	3.32	0.74	5.95	8.99	0.60	0.00	0.00	0.00	0.00	10.29	0.00	0.00
15-Jul-23	18.52	5.52	2.96	0.67	6.08	9.18	1.31	0.00	0.00	0.00	0.00	9.86	0.00	0.00
16-Jul-23	18.57	5.55	2.78	0.50	5.92	8.94	1.05	0.00	0.00	0.00	0.00	9.62	0.00	0.00
17-Jul-23	18.63	5.58	2.83	0.48	6.00	9.06	0.87	0.00	0.00	0.00	0.00	8.78	0.00	0.00
18-Jul-23	16.95	6.51	3.13	0.85	6.01	9.07	1.20	0.18	0.00	0.00	0.00	7.87	0.00	0.00
19-Jul-23	17.32	7.44	2.62	0.07	5.65	8.51	1.25	0.66	0.00	0.00	0.00	7.21	0.00	0.00
20-Jul-23	18.23	5.34	2.91	0.68	5.61	8.49	1.32	0.51	0.00	0.00	0.00	6.17	0.00	0.00
21-Jul-23	19.60	4.19	2.89	0.54	5.69	8.57	0.98	0.50	0.00	0.00	0.00	6.96	0.00	0.00
22-Jul-23	18.48	5.49	3.10	0.76	5.60	8.50	1.15	0.57	0.00	0.00	0.00	7.44	0.00	0.00
23-Jul-23	18.48	5.49	2.72	0.61	5.69	8.64	1.11	0.51	0.00	0.00	0.00	6.91	0.00	0.00
24-Jul-23	18.23	5.34	4.28	0.00	4.52	6.82	1.17	0.45	0.00	0.00	0.00	6.75	0.00	0.00
25-Jul-23	18.09	5.24	4.68	0.00	4.87	7.36	1.25	0.42	0.00	0.00	0.00	6.47	0.00	0.00
26-Jul-23	18.13	5.27	4.81	0.00	5.56	8.40	1.22	0.56	0.00	0.00	0.00	6.93	0.00	0.00
27-Jul-23	18.11	5.26	5.24	0.00	5.60	8.46	1.23	0.55	0.00	0.00	0.00	6.96	0.00	0.00
28-Jul-23	18.53	5.53	3.08	0.51	5.56	8.42	1.25	0.58	0.00	0.00	0.00	7.07	0.00	0.00
29-Jul-23	18.66	5.61	3.63	0.29	5.62	8.48	1.31	0.55	0.00	0.00	0.11	8.53	0.00	0.00
30-Jul-23	18.77	5.68	3.21	0.70	5.67	8.55	1.16	0.57	0.00	0.00	0.00	7.91	0.00	0.00
31-Jul-23	18.71	5.63	3.47	0.87	3.69	5.56	1.21	0.37	0.00	0.00	0.00	6.88	0.00	0.00
Total	564.22	168.51	97.98	15.84	175.45	265.26	31.74	14.92	0.00	0.00	0.11	250.10	0.00	0.00
Based on SEM/Energy meter data for links where available (*Mangdechū generation receipt at APD through a bypassed arrangement at 400kV Punatsanchu station)														

Date	16.2 Export to neighbouring countries during July 2023													
	(All figures in MU)													
	Export to Bhutan								Export to Bangladesh		Export to Nepal			Export to Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurwar D/C*	400 kV Jigmeling-Alipurwar D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.06	3.04	0.22	0.00	0.00	0.04
2-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.11	3.37	0.33	0.00	0.00	0.03
3-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.76	3.27	0.50	0.00	0.00	0.04
4-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.65	3.46	0.29	0.00	0.00	0.03
5-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.07	3.41	0.08	0.00	0.00	0.03
6-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.06	3.30	0.63	0.00	0.00	0.04
7-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.18	2.98	0.75	0.00	0.00	0.03
8-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.29	3.63	0.52	0.00	0.00	0.03
9-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.11	3.78	0.93	0.00	0.00	0.03
10-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.63	3.78	1.09	0.00	0.00	0.04
11-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.65	3.45	0.63	0.00	0.00	0.03
12-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.62	3.61	0.70	0.00	0.00	0.03
13-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.24	3.85	0.38	0.00	0.00	0.03
14-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.17	3.41	0.58	0.00	0.00	0.03
15-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.47	3.60	0.24	0.00	0.00	0.03
16-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.22	3.20	0.34	0.00	0.00	0.03
17-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.63	3.62	0.07	0.00	0.00	0.03
18-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.52	3.75	0.54	0.00	0.00	0.03
19-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.39	3.50	1.16	0.00	0.00	0.04
20-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.63	3.19	0.82	0.00	0.00	0.03
21-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.05	3.12	0.75	0.00	0.00	0.03
22-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.96	3.13	0.59	0.00	0.00	0.03
23-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.99	3.25	1.14	0.00	0.00	0.03
24-Jul-23	0.00	0.00	0.00	0.65	0.00	0.00	0.00	0.00	21.82	3.39	1.03	0.00	0.00	0.03
25-Jul-23	0.00	0.00	0.00	1.05	0.00	0.00	0.00	0.00	21.64	3.12	0.62	0.00	0.00	0.04
26-Jul-23	0.00	0.00	0.00	1.02	0.00	0.00	0.00	0.00	21.39	2.95	0.19	0.00	0.00	0.03
27-Jul-23	0.00	0.00	0.00	1.15	0.00	0.00	0.00	0.00	21.81	2.94	0.19	0.00	0.00	0.02
28-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.22	3.08	0.09	0.00	0.00	0.03
29-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.29	2.87	0.00	0.00	0.00	0.04
30-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.03	2.84	0.46	0.00	0.00	0.01
31-Jul-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.66	3.25	0.25	0.00	0.00	0.04
Total	0.00	0.00	0.00	3.87	0.00	0.00	0.00	0.00	676.32	103.13	16.11	0.00	0.00	1.02

Based on SEM/Energy meter data for links where available (*Mangdechu generation receipt at APD through a bypassed arrangement at 400kV Punatsanchu station)

17. एसटीओए (द्विपक्षीय एवं सामूहिक) एवं डी एस एम बिलिंग का ब्योरा – जुलाई 2023

STOA (BILATERAL AND COLLECTIVE) & DSM BILLING DETAILS - JUL 2023

द्विपक्षीय एसटीओए BILATERAL SHORT TERM OPEN ACCESS

जुलाई 2023 Jul 2023	Jul 2023		Apr'23- Jul'23	
नोडल क्षे.भा.प्रे.के. का नाम Name of Nodal RLDC	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Energy Approved(MU)	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Energy Approved (MU)
ऊ. क्षे. NR	1181	6638	4136	16191
प.क्षे. WR	941	1699	3842	9318
द.क्षे. SR	818	1355	3238	9935
पू.क्षे. ER	788	907	2945	3796
पूर्वोत्तर क्षे. NER	144	181	427	254
कुल TOTAL	3872	10780	14588	39494

एसटीओए SHORT TERM OPEN ACCESS

	सामूहिक एसटीओए Collective STOA		द्विपक्षीय एसटीओए Bilateral STOA	
माह MONTH	क्रेताओं / विक्रेताओं की सं. NO. of Buyers/Sellers	अनुमोदित ऊर्जा (मि.यु.) Approved Energy (MU)	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Approved Energy (MU)
अप्रैल 2023 Apr'23	5580	6941	3322	9714
मई 2023 May'23	5798	6626	3469	8418
जून 2023 Jun'23	5783	6966	3925	10583
जुलाई 2023 Jul'23	6428	6719	3872	10780
कुल TOTAL	23589	27252	14588	39494

* Approved Energy (MU) includes the Real Time Market (RTM) transactions (started from 1st Jun-20) as well, while No. of Buyers/Sellers are excluding RTM transactions.

** Approved Energy (MU) includes the Real Time Market (RTM) transactions (started from 1st Jun-20) as well, while No. of Buyers/Sellers are excluding RTM transactions

मासिक डी एस एम बिलिंग का ब्योरा* 2021-22
MONTHLY DSM BILLING DETAILS* 2021-22

अंतिम आँकड़े
Provisional data
subject to change
करोड़ रु. में (RS. IN CRORES)

क्षेत्र REGION →	उत्तरी क्षेत्र NORTH	पश्चिमी क्षेत्र WEST	दक्षिणी क्षेत्र SOUTH	पूर्वी क्षेत्र EAST	पूर्वोत्तर क्षेत्र NORTH EAST
सप्ताह WEEK ↓					
03.07.23 to 09.07.23	141.30	252.23	193.19	169.27	---
10.07.23 to 16.07.23	162.39	204.96	226.00	174.84	---
17.07.23 to 23.07.23	114.49	192.93	159.05	119.75	---
24.07.23 to 30.07.23	118.52	151.81	175.06	127.71	---

* Amount shown is Payable to DSM pool ^Provisional Data

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.) POWER MARKET INFORMATION (Source IEX & PXIL) पावर एक्सचेंज के माध्यम से विनिमय – माह:- जुलाई 2023 EXCHANGES THROUGH POWER EXCHANGES - JULY 2023								
क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्सचेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU					
			Integrated Day Ahead Market (I-DAM)		Real Time Market (RTM)		Total (I-DAM+RTM)	
			विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase
1	पंजाब Punjab	उत्तरी क्षेत्र NR	43.04	141.62	131.31	112.91	174.35	254.53
2	हरियाणा Haryana		269.33	23.89	100.61	18.52	369.94	42.41
3	राजस्थान Rajasthan		87.67	290.09	78.75	236.79	166.42	526.89
4	दिल्ली Delhi		150.40	21.12	92.28	56.79	242.69	77.91
5	उत्तर प्रदेश Uttar Pradesh		369.15	276.59	42.63	134.60	411.78	411.18
6	उत्तराखंड Uttarakhand		18.15	38.09	40.64	81.18	58.79	119.27
7	हिमाचल प्रदेश Himachal Pradesh		236.45	11.64	117.82	39.77	354.27	51.42
8	जम्मू और कश्मीर J & K		27.65	0.00	159.57	20.84	187.22	20.84
9	केंद्र शासित चंडीगढ़ UT Chandigarh		38.04	0.23	2.43	0.16	40.47	0.39
10	ए डी हाइड्रो AD Hydro		63.51	0.00	8.83	0.00	72.34	0.00
11	करचम वांगतू Karcham Wangtoo		84.61	0.00	26.36	2.54	110.97	2.54
12	श्री सीमेंट Shree Cement		5.25	0.00	2.75	0.00	8.00	0.00
13	नाथपा झाकरी Nathpa Jhakri		0.00	0.00	0.00	0.00	0.00	0.00
14	लांको बुधिल LANCO Budhil		5.13	0.00	0.31	0.00	5.44	0.00
15	सिंगरौली Singrauli		0.00	0.00	10.51	0.00	10.51	0.00
16	दादरी-I Dadri TPS		0.00	0.00	4.00	0.00	4.00	0.00
17	दादरी-II Dadri stg-II		0.00	0.00	2.81	0.00	2.81	0.00
18	ऊंचाहार-I Unchahar-I TPS		0.00	0.00	0.72	0.00	0.72	0.00
19	ऊंचाहार-II Unchahar-II TPS		0.00	0.00	1.86	0.00	1.86	0.00
20	ऊंचाहार-III Unchahar-III TPS		0.00	0.00	0.60	0.00	0.60	0.00
21	रिहंद-I Rihand STPS-I		0.00	0.00	9.29	0.00	9.29	0.00
22	रिहंद-II Rihand STPS-II		0.00	0.00	6.95	0.00	6.95	0.00
23	रिहंद-III Rihand STPS-III		0.00	0.00	7.26	0.00	7.26	0.00
24	एपीसीपीएल झज्जर APCPL Jhajjar		0.00	0.00	0.00	0.00	0.00	0.00
25	सैज Sainj HEP		17.39	0.00	2.13	2.60	19.51	2.60
26	दादरी गैस Dadri GPP		0.00	0.00	0.00	0.00	0.00	0.00
27	औरैया गैस NTPC Auraiya		0.00	0.00	0.00	0.00	0.00	0.00
28	अंता गैस NTPC Anta		0.00	0.00	0.02	0.00	0.02	0.00
29	रामपुर एचईपी RAMPUR HEP		0.00	0.00	0.00	0.00	0.00	0.00
30	ऊंचाहार-IV UNCHAHAR-IV		0.00	0.00	1.89	0.00	1.89	0.00
31	एनटीपीसी कोलडाम NTPC KOLDAM		0.00	0.00	0.00	0.00	0.00	0.00
32	कोटेश्वर Koteswar		0.00	0.00	0.00	0.00	0.00	0.00
33	टांडा स्टेज-२ Tanda Stg-2		0.00	0.00	3.20	0.00	3.20	0.00
34	अदानी सौर एनर्जी टू लिमिटेड Adani Solar Energy Jodhpur Two Limited		10.46	0.00	0.00	0.00	10.46	0.00
35	सिंगोली भटवारी ह्येप Singoli Bhatwari HEP		28.56	0.00	4.39	0.15	32.95	0.15
36	रेलवे Railways		0.00	26.68	0.00	4.79	0.00	31.47
37	टिहरी Tehri		0.00	0.00	0.00	0.00	0.00	0.00
38	Renew Solar Energy (Jharkhand Three) Private Limited (RSEJTPL)		0.00	0.00	0.00	0.00	0.00	0.00
39	सोरांग एच.ई.पी. Sorang HEP, (Himachal Sorang Power Private Ltd)		1.24	0.00	0.41	0.00	1.65	0.00
40	अदानी संकर ऊर्जा जैसलमेर वन लिमिटेड (सौर)Adani Hybrid Energy Jaisalmer One Limited (Solar)		0.00	0.00	0.00	0.00	0.00	0.00
41	अदानी संकर ऊर्जा जैसलमेर टु लिमिटेड (सौर) Adani Hybrid Energy Jaisalmer Two Limited (Solar)		0.00	0.00	0.00	0.00	0.00	0.00

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.) POWER MARKET INFORMATION (Source IEX & PXIL) पावर एक्सचेंज के माध्यम से विनिमय – माह:- जुलाई 2023 EXCHANGES THROUGH POWER EXCHANGES - JULY 2023								
क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्सचेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU					
			Integrated Day Ahead Market (I-DAM)		Real Time Market (RTM)		Total (I-DAM+RTM)	
			विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase
42	अदानी संकर ऊर्जा जैसलमेर थ्री लिमिटेड (सौर) Adani Hybrid Energy Jaisalmer Three Limited (Solar)	उत्तरी क्षेत्र NR	0.00	0.00	0.00	0.00	0.00	0.00
43	अयाना रिनूवेबल पावर वन लिमिटेड Ayana Renewable Power One Private Limited		0.00	0.00	0.00	0.00	0.00	0.00
44	Adani Hybrid Energy Jaisalmer Four Limited (Solar)अदानी संकर ऊर्जा जैसलमेर चार लिमिटेड (सौर)		0.00	0.00	0.00	0.00	0.00	0.00
45	रिणीव सोलर ऊर्जा प्राइवेट लिमिटेड Renew Solar Urja private limited		0.00	0.00	0.00	0.00	0.00	0.00
46	NEA-NR एन ई ए-एन आर		0.00	6.62	0.00	0.00	0.00	6.62
47	रेनेव सूर्या रवि प्राइवेट लिमिटेड (सौर) Renew Surya Ravi Private Limited (Solar)		45.38	0.00	0.82	0.00	46.20	0.00
48	अजूरै पावर मपले प्राइवेट लिमिटेड (सौर) Azure Power Maple Pvt Ltd (Solar)		0.00	0.00	0.00	0.00	0.00	0.00
49	अदानी ग्रीन हाइब्रिड सौर जैसलमेर Adani Green Hybrid Solar Jaisalmer		0.00	0.00	0.00	0.00	0.00	0.00
50	अदानी ग्रीन हाइब्रिड पवन जैसलमेरAdani Green Hybrid Wind Jaisalmer		0.00	0.00	0.00	0.00	0.00	0.00
51	अदानी ग्रीन सौर जैसलमेर Adani Green Solar Jaisalmer		0.00	0.00	0.00	0.00	0.00	0.00
52	ABC Renewable Energy (RJ-01) Private Limited एबीसी नवीकरणीय ऊर्जा (आर-01) प्राइवेट लिमिटेड		0.00	0.00	0.00	0.00	0.00	0.00
53	SBSR Power Cleantech Eleven Private Limited एसबीएसआर पावर क्लीनटेक लिमिटेड		7.24	0.00	0.09	0.00	7.32	0.00
54	टनकपुर पावर स्टेशन Tanakpur Power Station		0.00	0.00	0.00	0.00	0.00	0.00
55	उरी पावर स्टेशन Uri I Power Station		0.00	0.00	0.00	0.00	0.00	0.00
56	उरी पावर स्टेशन Uri II Power Station		0.00	0.00	0.00	0.00	0.00	0.00
57	चमेरा पावर स्टेशन Chamera I Power Station		0.00	0.00	0.00	0.00	0.00	0.00
58	चमेरा पावर स्टेशन Chamera II Power Station		0.00	0.00	0.00	0.00	0.00	0.00
59	चमेरा पावर स्टेशन Chamera III Power Station		0.00	0.00	0.00	0.00	0.00	0.00
60	धौलीगंगा पावर स्टेशन Dhauliganga Power Station		0.00	0.00	0.00	0.00	0.00	0.00
61	दुलहसती पावर स्टेशन Dulhashti Power Station		0.00	0.00	0.00	0.00	0.00	0.00
62	किसनगंगा पावर स्टेशन Kishanganga Power Station		0.00	0.00	0.00	0.00	0.00	0.00
63	पार्वती पावर स्टेशन Parbati III Power Station		0.00	0.00	0.00	0.00	0.00	0.00
64	सेवा पावर स्टेशन Sewa II Power Station		0.00	0.00	0.00	0.00	0.00	0.00
65	सलाल पावर स्टेशन Salal Power Station		0.00	0.00	0.00	0.00	0.00	0.00
66	ADANI SOLAR ENERGY JAISALMER ONE PRIVATE LIMITED AHS अदानी सौर ऊर्जा जैसलमेर वन प्राइवेट लिमिटेड ए एच अस		0.00	0.00	0.00	0.00	0.00	0.00
67	पार्वती ह. ई. प्रोजेक्ट Parbati II HE Project		0.00	0.00	0.00	0.00	0.00	0.00
68	एनटीपीसी अंटा जीपीएस NTPC ANTA GPS RLNG		0.00	0.00	0.00	0.00	0.00	0.00
69	एनटीपीसी औरैया जीपीएस NTPC AURAIYA GPS RLNG		0.00	0.00	0.00	0.00	0.00	0.00
70	एनटीपीसी दादरी जीपीएस NTPC DADRI GPS RLNG		0.00	0.00	0.00	0.00	0.00	0.00
71	अदानी संकर ऊर्जा जैसलमेर चार लिमिटेड ADANI HYBRID ENERGY JAISALMER FOUR LIMITED AHW		0.00	0.00	0.00	0.00	0.00	0.00
72	मध्य प्रदेश Madhya Pradesh	पश्चिमी क्षेत्र WR	145.05	61.50	404.36	37.21	549.41	98.70
73	महाराष्ट्र Maharashtra		97.91	272.57	56.70	133.49	154.61	406.06
74	गुजरात Gujarat		45.95	493.73	18.48	76.42	64.43	570.15
75	छत्तीसगढ़ Chhattisgarh		174.65	6.64	66.27	18.45	240.92	25.09
76	गोवा डब्लू आर Goa WR		11.69	11.40	0.39	0.75	12.08	12.14
77	जिंदल पावर JINDAL POWER		45.19	0.00	6.19	0.00	51.38	0.00
78	लान्को Lanco		0.00	0.00	0.00	0.00	0.00	0.00
79	ए सी बी आई एल ACBIL		66.85	0.00	9.14	0.01	75.99	0.01
80	जिंदल स्टील JINDAL STEEL		41.46	0.00	3.90	1.44	45.35	1.44
81	बालको 2 BALCO 2		0.00	0.00	0.00	0.00	0.00	0.00
82	जीएमआर वरोरा GMR Warora		7.39	0.00	2.93	0.00	10.32	0.00

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.) POWER MARKET INFORMATION (Source IEX & PXIL) पावर एक्सचेंज के माध्यम से विनिमय – माह:- जुलाई 2023 EXCHANGES THROUGH POWER EXCHANGES - JULY 2023								
क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्सचेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU					
			Integrated Day Ahead Market (I-DAM)		Real Time Market (RTM)		Total (I-DAM+RTM)	
			विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase
83	एस्सार पावर एम पी Essar Power MP Ltd	पश्चिमी क्षेत्र WR	85.96	0.00	23.29	0.60	109.25	0.60
84	के एस के महानदी KSK Mahanadi		0.00	0.00	0.00	0.00	0.00	0.00
85	जिंदल पावर लिमिटेड Jindal Power Ltd. Stg-II		44.52	0.00	7.44	0.39	51.96	0.39
86	रायगढ़ एनर्जी लिमिटेड Raigarh Energy Ltd.		0.00	0.00	0.00	0.00	0.00	0.00
87	डी बी पावर DB Power		64.02	0.00	8.00	0.00	72.03	0.00
88	धारीवाल Dhariwal		0.00	0.00	0.00	0.00	0.00	0.00
89	जेपी निगरी JAYPEE NIGRIE		181.57	0.00	2.60	0.00	184.17	0.00
90	डीजेन मेगा पावर DGEN Mega power		0.00	0.00	0.00	0.00	0.00	0.00
91	एम बी पावर MB Power		89.52	0.00	24.23	0.00	113.75	0.00
92	अदानी पावर स्टेज 3 Adani Power STG3		0.00	0.00	0.00	0.00	0.00	0.00
93	टॉरेंट पावर Torrent Power		0.00	0.00	0.00	0.00	0.00	0.00
94	मारुति क्लीन कोल पावर Maruti Clean Coal Power		12.13	0.00	3.64	0.28	15.78	0.28
95	झाबुआ पावर Jhabua Power		6.71	0.00	6.46	0.00	13.18	0.00
96	विंध्याचल स्टेज-1 Vindhyachal stg-1		0.00	0.00	2.37	0.00	2.37	0.00
97	विंध्याचल स्टेज-2 Vindhyachal stg-2		0.00	0.00	1.51	0.00	1.51	0.00
98	विंध्याचल स्टेज-3 Vindhyachal stg-3		0.00	0.00	1.60	0.00	1.60	0.00
99	विंध्याचल स्टेज-4 Vindhyachal stg-4		0.00	0.00	1.01	0.00	1.01	0.00
100	विंध्याचल स्टेज-5 Vindhyachal stg-5		0.00	0.00	1.43	0.00	1.43	0.00
101	मौदा स्टेज-1 Mouda Stg-1		0.00	0.00	1.85	0.00	1.85	0.00
102	सिपत स्टेज-1 Sipat Stg-1		0.00	0.00	1.53	0.00	1.53	0.00
103	सिपत स्टेज-2 Sipat Stg-2		0.00	0.00	1.28	0.00	1.28	0.00
104	सासन Sasan UMPP		0.00	0.00	29.05	0.00	29.05	0.00
105	भिलाई एनएसपीसीएल NSPCL Bhilai		0.00	0.00	0.00	0.00	0.00	0.00
106	कोरबा स्टेज-3 Korba Stg 3		0.00	0.00	0.98	0.00	0.98	0.00
107	टीआरएन एनर्जी TRN Energy		0.00	0.00	8.28	0.00	8.28	0.00
108	गांधार Gandhar		0.00	0.00	0.00	0.90	0.00	0.90
109	कोरबा Korba		0.00	0.00	2.56	0.00	2.56	0.00
110	आर के एम पावर RKM Power		27.00	0.00	13.38	0.00	40.38	0.00
111	कावस Kawas		0.00	0.00	0.00	0.54	0.00	0.54
112	मौदा स्टेज-2 Mouda Stg-2		0.00	0.00	3.56	0.00	3.56	0.00
113	एस के एस पावर SKS Power		18.81	0.00	2.67	0.27	21.48	0.27
114	एन टी पी सी सोलापुर NTPC Sholapur		0.00	0.00	2.98	0.00	2.98	0.00
115	एन टी पी सी गदरवारा NTPC Gadawara		0.00	0.00	5.55	0.00	5.55	0.00
116	एन टी पी सी लारा NTPC Lara		0.00	0.00	1.03	0.00	1.03	0.00
117	रायपुर एनेर्जेंस Raipur Energen		0.00	0.00	0.00	0.00	0.00	0.00
118	आर्सेलर मितल ArcelorMittal Nippon steel (ESIL)		0.00	19.23	0.00	18.67	0.00	37.90
119	खरगोन Khargone STPP		0.00	0.00	2.15	0.00	2.15	0.00
120	कोस्टल गुजरात पावर लिमिटेड Coastal Gujrat Power Limited		0.00	0.00	0.00	0.00	0.00	0.00
121	कंटीनम पावर ट्रेडिंग Continuum Power Trading (TN) Private Limited		5.98	0.00	0.00	0.00	5.99	0.00
122	ऑस्ट्रो एनर्जी प्राइवेट लिमिटेड Ostro Energy Private Limited (OEPL)		0.00	0.00	0.00	0.00	0.00	0.00
123	दमन डीयू/दादरा और नगर हवेली Daman & Diu/Dadar and Nagar Haveli		0.00	147.48	4.19	23.18	4.19	170.66

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.) POWER MARKET INFORMATION (Source IEX & PXIL) पावर एक्सचेंज के माध्यम से विनिमय – माह:- जुलाई 2023 EXCHANGES THROUGH POWER EXCHANGES - JULY 2023								
क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्सचेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU					
			Integrated Day Ahead Market (I-DAM)		Real Time Market (RTM)		Total (I-DAM+RTM)	
			विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase
124	अदानी विंड एनेर्जी Adani Wind Energy Kutchh Five Limited	पश्चिमी क्षेत्र WR	0.00	0.00	0.00	0.00	0.00	0.00
125	आर जी पि पि एल RGPL		0.00	0.00	0.00	1.07	0.00	1.07
126	Kakrapar Atomic Power Station काकरपर अटॉमिक पावर स्टेशन		0.00	0.00	0.00	0.00	0.00	0.00
127	BALCO बालको		0.00	8.71	0.00	12.64	0.00	21.35
128	अदानी विंड एनर्जी कछ फोर लिमिटेड नखटरना ADANI WIND ENERGY KUTCHH FOUR Ltd. Nakhatrana		28.77	0.00	0.00	0.00	28.77	0.00
129	एनटीपीसी झनोर गांधार जीपीएस NTPC Jhanor Gandhar GPS RLNG		0.00	0.00	0.00	0.00	0.00	0.00
130	एनटीपीसी कावस गपद NTPC Kawas GPS RLNG		0.00	0.00	0.00	0.00	0.00	0.00
131	रत्नागिरी गैस और पावर प्राइवेट लिमिटेड Ratnagiri Gas and Power Private Limited-LTRLNG		0.00	0.00	0.00	0.00	0.00	0.00
132	विंड फाइव एनर्जी लिमिटेड WIND FIVE RENERGY LIMITED		9.43	0.00	0.00	0.00	9.43	0.00
133	एनटीपीसी झनोर गांधार जीपीएस कॉम गैस NTPC Jhanor Gandhar GPS_COMGAS		0.00	0.00	0.00	0.00	0.00	0.00
134	धारीवाल एस टी यू DHARIWAL STU OTH		2.21	0.00	0.00	0.00	2.21	0.00
135	धारीवाल आइ एस टी एस Dhariwal ISTS		0.37	0.00	0.00	0.00	0.37	0.00
136	इनोक्स ग्रीन एनर्जी सर्विसेस लिमिटेड INOX GREEN ENERGY SERVICES LIMITED		1.24	0.00	0.04	0.00	1.28	0.00
137	अदानी पावर लिमिटेड रायपुर टीपीपी Adani Power Limited Raipur TPP		86.23	0.00	13.43	0.00	99.66	0.00
138	टाटा पावर कंपनी लिमिटेड (MTPS) The Tata Power Co Ltd (MTPS)		0.00	0.00	2.39	0.00	2.39	0.00
139	आंध्र प्रदेश Andhra Pradesh	दक्षिणी क्षेत्र SR	85.45	237.14	57.49	195.90	142.94	433.03
140	कर्नाटक Karnataka		160.02	45.24	66.81	28.86	226.82	74.11
141	केरल Kerala		2.72	178.79	3.89	156.24	6.61	335.03
142	तमिलनाडु Tamilnadu		47.21	197.11	107.40	69.18	154.60	266.29
143	पांडिचेरी Pondicherry		22.53	0.00	8.39	0.10	30.93	0.10
144	गोवा एस आर Goa SR		0.00	6.90	0.00	1.43	0.00	8.33
145	तेलंगाना TELANGANA		35.92	348.05	75.03	501.74	110.95	849.79
146	कोस्टल एनर्जेन Costal Energen		0.00	0.00	0.00	0.00	0.00	0.00
147	आई एल और एफ एस IL&FS		2.30	0.00	0.35	0.00	2.65	0.00
148	सिंहद्री स्टेशन -2 Simhadri Stg-2		0.00	0.00	0.78	0.00	0.78	0.00
149	एनएलसी इंडिया स्टेशन 1 विस्तार NLC India Station I Expansion		0.00	0.00	1.11	0.00	1.11	0.00
150	एन एल सी स्टेशन-1 NLC TPS-II Stg-1		0.00	0.00	0.00	0.00	0.00	0.00
151	एन एल सी स्टेशन-2 NLC TPS-II Stg-2		0.00	0.00	0.00	0.00	0.00	0.00
152	रामागुंडम स्टेशन-1 Ramagundam Stg-1		0.00	0.00	2.11	0.00	2.11	0.00
153	तलचर स्टेशन -2 Talcher Stg-2		0.00	0.00	3.16	0.00	3.16	0.00
154	रामागुंडम स्टेशन-3 Ramagundam Stg-3		0.00	0.00	0.88	0.00	0.88	0.00
155	सेम्कोर्प गायत्री Sembcorp Gayatri		50.63	0.00	11.36	0.00	61.99	0.00
156	वल्लूर Vallur		0.00	0.00	0.00	0.00	0.00	0.00
157	कुड्गी NTPC Kudgi		0.00	0.00	2.07	0.00	2.07	0.00
158	सिंहद्री स्टेशन -1 Simhadri Stg-1		0.00	0.00	0.98	0.00	0.98	0.00
159	एन एल सी टी पी एस -11 एक्सपन NLC TPS-II Expn		0.00	0.00	0.00	0.00	0.00	0.00
160	एनटीपीएल NTPL		0.00	0.00	3.74	0.00	3.74	0.00
161	नयवेली न्यू थर्मल पावर प्रोजेक्ट Neyveli New Thermal Power Project		0.00	0.00	4.36	0.00	4.36	0.00
162	रामगुंडम फ्लोटिंग सोलर पी वी स्टेशन Ramagundam Floating Solar PV Station		5.00	0.00	0.00	0.00	5.00	0.00
163	सिंहद्री फ्लोटिंग सोलर Simhadri Floating solar		1.75	0.00	0.00	0.00	1.75	0.00
164	सिंहद्री एफएसपी 15 एमवी प्रोजेक्ट Simhadri FSP 15 MW Project		2.58	0.00	0.00	0.00	2.58	0.00
165	सेम्कोर्प एनर्जी इंडिया लिमिटेड Sembcorp energy india limited		12.84	0.00	4.06	0.00	16.91	0.00

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.) POWER MARKET INFORMATION (Source IEX & PXIL) पावर एक्सचेंज के माध्यम से विनिमय – माह:- जुलाई 2023 EXCHANGES THROUGH POWER EXCHANGES - JULY 2023								
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			Integrated Day Ahead Market (I-DAM)		Real Time Market (RTM)		Total (I-DAM+RTM)	
			विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase
166	जे एस डब्ल्यू रिनोव एनर्जी टू लिमिटेड JSW RENEW ENERGY TWO LIMITED	दक्षिणी क्षेत्र SR	0.00	0.00	0.00	0.00	0.00	0.00
167	ओस्ट्रो कन्नड पावर प्राइवेट लिमिटेड Ostro Kannada Power Private Limited		30.44	0.00	4.80	0.00	35.24	0.00
168	सिमहापुरी एनर्जी लिमिटेड Simhapuri Energy Limited		1.09	0.00	4.14	2.26	5.23	2.26
169	परमपूज्य सोलर एनर्जी प्राइवेट लिमिटेड PARAMPUJYA SOLAR ENERGY PRIVATE LIMITED		0.00	0.00	0.00	0.00	0.00	0.00
170	पश्चिम बंगाल West Bengal	पूर्वी क्षेत्र ER	2.32	1044.63	21.57	278.18	23.90	1322.81
171	ओड़िशा Odisha		34.20	63.04	104.03	83.09	138.23	146.14
172	बिहार Bihar		165.55	15.64	153.64	28.30	319.19	43.94
173	झारखंड Jharkhand		12.12	30.28	40.47	23.76	52.59	54.04
174	सिक्किम Sikkim		12.64	0.00	8.68	1.93	21.32	1.93
175	दामोदर घाटी परियोजना DVC		14.11	101.27	3.31	57.34	17.41	158.62
176	मैथन पावर लिमिटेड Maithon P. Ltd		0.00	0.00	0.00	0.00	0.00	0.00
177	आधुनिक पा. लिमिटेड Adhunik P.Ltd		6.06	0.00	6.35	0.00	12.40	0.00
178	चुजाचेन Chuzachen HEP		11.98	0.00	0.00	0.00	11.98	0.00
179	जीएमआर कमलंगा GMR Kamalanga		0.32	0.00	0.07	0.00	0.39	0.00
180	जिंदल आई पी टी एल JIPTL		126.42	0.00	0.06	0.00	126.48	0.00
181	जोरथांग Jorethang		0.00	0.00	3.32	0.00	3.32	0.00
182	फरक्का-III Farakka-III		0.00	0.00	1.39	0.00	1.39	0.00
183	बाढ़-II Barh Stage-II		0.00	0.00	5.06	0.00	5.06	0.00
184	कहलगाँव-II Kahalgaon Stage-II		0.00	0.00	9.02	0.00	9.02	0.00
185	फरक्का Farakka		0.00	0.00	3.51	0.00	3.51	0.00
186	कहलगाँव Kahalgaon		0.00	0.00	4.15	0.00	4.15	0.00
187	तालचर Talcher		0.00	0.00	0.90	0.00	0.90	0.00
188	तीस्ता-III Teesta-III		0.00	0.00	0.00	0.00	0.00	0.00
189	दिक्चु Dikchu HEP		27.04	0.00	0.21	0.00	27.25	0.00
190	ताशिदिंग Tashiding HEP		0.00	0.00	4.93	0.00	4.93	0.00
191	केबीयूएनएल KBUNL		0.00	0.00	0.88	0.00	0.88	0.00
192	दर्लीपल्ली Darlipalli		0.00	0.00	0.33	0.00	0.33	0.00
193	नबीनगर NPGC Nabinagar		0.00	0.00	13.84	0.00	13.84	0.00
194	नेपाल विद्युत प्राधिकरण Nepal Electricity Authority		251.12	0.00	0.00	0.00	251.12	0.00
195	Barh STPS Stage-I बाढ़ एसटीपीएस स्टेज-1		0.00	0.00	3.85	0.00	3.85	0.00
196	NTPC North Karanpura STPS नॉर्थ करनपुरा		0.00	0.00	0.00	0.00	0.00	0.00
197	Druk Green Power Corporation Limited-Bhutan द्रुक ग्रीन पावर कॉर्पोरेशन लिमिटेड-भूटान		0.00	0.00	0.00	0.00	0.00	0.00
198	NTPC BARH1 Unit 2 एनटीपीसी बाढ़ 1		0.00	0.00	99.35	0.00	99.35	0.00
199	सिक्किम ऊर्जा लिमिटेड Sikkim Urja Limited		185.14	0.00	0.03	0.00	185.17	0.00
200	अरुणाचल प्रदेश Arunachal Pradesh	पूर्वोत्तर क्षेत्र NER	22.66	0.00	3.14	1.36	25.80	1.36
201	असम Assam		10.02	37.49	30.74	47.45	40.76	84.94
202	मणिपुर Manipur		1.59	0.00	14.11	3.41	15.71	3.41
203	मेघालय Meghalaya		6.35	13.78	0.07	10.58	6.42	24.35
204	मिज़ोरम Mizoram		4.24	0.00	0.00	0.00	4.24	0.00
205	नागालैंड Nagaland		4.24	0.00	1.69	0.00	5.93	0.00
206	त्रिपुरा Tripura		0.00	1.46	3.47	11.36	3.47	12.83
207	अगरतला जीटीपीपी Agartala GTPP_AG0		0.00	0.00	0.06	0.00	0.06	0.00

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.)
POWER MARKET INFORMATION (Source IEX & PXIL)
पावर एक्स्चेंज के माध्यम से विनिमय – माह:- जुलाई 2023
EXCHANGES THROUGH POWER EXCHANGES - JULY 2023

क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU					
			Integrated Day Ahead Market (I-DAM)		Real Time Market (RTM)		Total (I-DAM+RTM)	
			विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase
208	असम जीबीपीपी Assam GBPP_KT0	पूर्वोत्तर क्षेत्र NER	0.00	0.00	0.18	0.00	0.18	0.00
209	पलाताना Palatana		8.65	0.00	12.79	0.00	21.44	0.00
210	Doyang HEP दोयांग हेप		0.00	0.00	0.00	0.00	0.00	0.00
211	बोंगाईगाँव Bongaigaon TPP		1.06	0.00	4.80	0.00	5.86	0.00
212	कामेंग Kameng HEP		1.05	0.00	7.43	0.00	8.48	0.00
213	रंगानदी हाइड्रो पावर स्टेशन Ranganadi Hydro Power Station		0.00	0.00	0.00	0.00	0.00	0.00
214	लोकटक पावर स्टेशन LOKTAK POWER STATION		0.00	0.00	0.00	0.00	0.00	0.00
			4179	4179	2540	2540	6719	6719

19. INFORMATION ABOUT RENEWABLE ENERGY
CERTIFICATE MECHANISM

MONTH : JULY 2023

RE Source & Unit wise break up (01.07.2023-31.07.2023)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	14	4	322	5
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	4	5	0	0
5	Small Hydro	0	0	0	0
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	0	0	5	1
9	Bio-fuel cogeneration	0	0	0	0
	Total	18	9	327	6

RECs Issued (01.07.2023-31.07.2023)

Sr.No.	Non Solar	Solar	Total
1	468719	82728	551447

Redemption of REC (01.07.2023-31.07.2023)

Sr.No.	Non Solar	Solar	Total
1	602269	166141	768410

19. INFORMATION ABOUT RENEWABLE ENERGY
CERTIFICATE MECHANISM

MONTH : JULY 2023

RE Source & Unit wise break up (Apr'23-Jul'23)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	339	11	425	8
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	58	10	65	3
5	Small Hydro	42	7	36	6
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	5	1	5	1
9	Bio-fuel cogeneration	4	1	19	4
	Total	448	30	550	22

RECs Issued (Apr'23-Jul'23)

Sr.No.	Non Solar	Solar	Total
1	1317053	415618	1732671

Redemption of REC (Apr'23-Jul'23)

Sr.No.	Non Solar	Solar	Total
1	1736155	298528	2034683

19. INFORMATION ABOUT RENEWABLE ENERGY
CERTIFICATE MECHANISM

MONTH : JULY 2023

RE Source & Unit wise break up Since Inception to Jul'23

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	3064	518	2894	515
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	1327	502	1250	468
5	Small Hydro	686	40	692	41
6	Others	4	2	3	1
7	Geothermal	0	0	0	0
8	Biomass	402	37	378	35
9	Bio-fuel cogeneration	813	89	379	54
	Total	6296	1188	5596	1114

RECs Issued since Inception to Jul'23

Sr.No.	Non Solar	Solar	Total
1	89197193	12826356	102023549

Redemption of REC since Inception to Jul'23

Sr.No.	Non Solar	Solar	Total
1	69264340	11926916	81191256

REC Closing balance as on 31.07.2023

Sr.No.	Non Solar	Solar	Total
1	19932853	899440	20832293

20.Details of Grid Events during the Month of July 2023
Northern Region



Sl.No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (H:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GI-2	Uttar Pradesh	01-Jul-2023 06:39	01-Jul-2023 10:34	03:55	0	0	0.000	0.000	45312	52864	i) 400/220KV Bareilly(UP) has double main transfer bus scheme at both 400KV & 220KV level. During antecedent condition, 400/220 kv 315 MVA ICT 3 at Bareilly(UP), 400 KV Bareilly-Umao (UP) Ckt-1, 400 KV Bareilly(UP)-Bareilly(PG) (PG) Ckt-1 and 80 MVAR Bus Reactor No 1 at 400KV Bareilly(UP) were connected to 400KV Bus 1 at Bareilly(UP) and rest of the elements were connected to Bus 2. ii) As reported, at 06:39 hrs on 01st July, 2023, 400KV Bus 1 at Bareilly(UP) tripped due to bus bar protection mal-operation and hence all the elements connected to Bus 1 also tripped and Bus 1 became dead. iii) As per PMU at 400KV Bareilly(PG), no fault is observed in the system. iv) As per SCADA, no change in demand is observed in UP control area. After tripping load shifted to remaining 400/220/33 kv 315 MVA ICT 1 & 2 at Bareilly(UP).	1) 400KV Bus 1 at Bareilly(UP) 2) 400/220 kv 315 MVA ICT 3 at Bareilly(UP) 3) 400 KV Bareilly-Umao (UP) Ckt-1 4) 400 KV Bareilly(UP)-Bareilly(PG) (PG) Ckt-1 5) 80 MVAR Bus Reactor No 1 at 400KV Bareilly(UP)
2	GI-1	Himachal Pradesh	01-Jul-2023 16:08	01-Jul-2023 16:14	00:06	0	150	0.000	0.236	54276	63555	i) 220/66KV Uperanangal (HP) has double main bus scheme at both 220KV & 66KV level. During antecedent condition, power was flowing through 220/66KV 80/100MVA ICT-1 & 2 at Uperanangal(HP) to the feeders connected at 66KV level of Uperanangal (HP). ii) As reported, at 16:08 hrs on 01.07.2023, 220/66KV 80/100MVA ICT-1 at Uperanangal(HP) tripped on R-Phase overcurrent earth fault (Ir=1.12KA) on fault in 66KV feeder. Due to this, 220/66KV 80/100MVA ICT-2 at Uperanangal(HP) also tripped due to overloading. This resulted in further unloading of feeders connected at 66KV level and load loss at Uperanangal (HP). iii) As per PMU at 400KV Nallagahri(PG), no fault is observed in the system. HP has been communicated to share the DR/EL for confirmation of fault nature and location. iv) As per SCADA, change in demand of approx. 150MW is observed in HP control area. v) As reported by SLDC-HP, similar events are observed at 23:08hrs on 24.06.2023, 23:46hrs on 26.06.2023, 07:08hrs on 27.06.2023, 04:26hrs and 09:29hrs on 28.06.2023 where 66KV Ambuja and Sanerh feeder relay operated and as fault did not clear by opening of these two feeders, further tripping of 220/66KV 80/100MVA ICT-1 & 2 at Uperanangal(HP) occurred. P&T team visited at site on 28.06.2023. All settings were checked and found okay, but reason for tripping was not found. P & T Team again visited at site on 02.07.2023 after tripping on 01.07.2023 and testing and assessment of fault is in progress.	1) 220/66KV 80/100MVA ICT-1 at Uperanangal(HP) 2) 220/66KV 80/100MVA ICT-2 at Uperanangal(HP)
3	GI-1	Delhi	03-Jul-2023 12:24	03-Jul-2023 12:32	00:08	0	245	0.000	0.349	62300	70281	i) During antecedent condition, 220KV Bawana –Shalimarbagh Ckt -I, 220KV Shalimarbagh – Wazirpur Ckt. I & II, 220KV Shalimarbagh – DMRC Ckt. -I, 220KV Shalimarbagh –SGTH Ckt. -II and 220KV Shalimarbagh – Rohini-I Ckt -I were connected to 220KV Bus-1 at Shalimarbagh(DTL) and 220KV Bawana –Shalimarbagh Ckt. -II, 220KV Shalimarbagh – DMRC Ckt. -II, 220/66KV 100MVA Pt. Tr. –II, 220/66KV 100MVA Pt. Tr. –II, 220/33KV 100MVA Pt. Tr. –II, 220KV Shalimarbagh – DMRC Ckt. -II were connected to 220KV Bus-2 at Shalimarbagh(DTL). 220KV Bus Coupler at Shalimar Bagh was in off position. ii) As reported, at 12:24 hrs, 220 KV Bawana – Shalimarbagh (DTL) Ckt-1 tripped on R-B-N double phase to ground fault. PLCC communication was not healthy during the incident and hence line tripped from Bawana end only. There are two lines installed at Bawana end for protection of 220 KV Bawana – Shalimarbagh (DTL) Ckt-1: distance relay (Main-II) and differential relay (Main-II). Distance relay operated during this event at Bawana end, but DR of the same is not available because of memory overwrite. iii) As per DR of differential relay installed at Bawana end of 220 KV Bawana – Shalimarbagh (DTL) Ckt-1, line tripped on R-B-N double phase to ground fault (fault sensed in zone-2) with fault current of 9.96kA and 1.35kA in R and B phase respectively and fault clearing time of 432ms. iv) As per PMU at Bawana(DTL), R-B-N double phase to earth fault is observed with delayed fault clearing time of 440 ms. v) As per SCADA, load loss of approx. 245MW is observed in Delhi control area is observed. vi) As reported, load on Shalimarbagh was restored by charging 220 KV Shalimarbagh-Rohini Ckt-1 at 12:32 hrs.	1) 220 KV Bawana – Shalimarbagh (DTL) Ckt-1
4	GI-2	Himachal Pradesh	09-Jul-2023 10:28	09-Jul-2023 16:35	06:07	0	0	0.000	0.000	43993	47878	i) During antecedent condition, 77MW unit-1,2&3 all were under shutdown due to high silt (~9000PPM) condition since around 23-33 hrs of 08th July 2023. ii) As reported, due to heavy discharge in the river, land slide occurred and led to the collapsing of tower no. 1 from Chamera-II end at 10:28 hrs. Collapsing of tower no.1 further resulted into collapsing of gantry of 220 KV Chamera_3(NH)-Chamba(PG) (PG) Ckt-1 & 2 at Chamera_3 end. Collapsing of gantry led to the damage of 02 nos CVT, 03 nos of lightning arrestors & 01 no Wave trap at Chamera_3(NH) end. iii) As reported, at the time, bus bar protection of 220KV Bus-1&2 at Chamba(PG) operated which led to the tripping of all the elements connected at 220KV side of Chamba(PG). Details of faults at Chamba(PG) end are yet to be received. iv) As per PMU, R-N fault with delayed clearance in 560msec followed by Y-N fault which cleared within 100msec is observed. v) As per SCADA, no change in demand and load is observed.	1) 220KV Bus 1 at Chamba(PG) 2) 220KV Bus 2 at Chamba(PG) 3) 220 KV Chamera_3(NH)-Chamba(PG) (PG) Ckt-1 4) 220 KV Chamera_3(NH)-Chamba(PG) (PG) Ckt-2 5) 400/220 kv 315 MVA ICT 1 at Chamba(PG) 6) 400/220 kv 315 MVA ICT 2 at Chamba(PG) 7) 220 KV Chamba(PG)-Karian(HP) (HPSEB) Ckt-1
5	GD-1	Uttarakhand	14-Jul-2023 02:48	14-Jul-2023 05:16	02:28	108	35	0.232	0.058	46478	60431	i) During antecedent condition, 33MW Unit-1, 2 and 3 at Singoli Bhatwari HEP were generating approx. 36MW each respectively. ii) As reported, at 12:24 hrs, station transformer (33KV/0.4KV) which is connected on tertiary winding of 400/220 kv 315 MVA ICT 2 at Srinagar(UK) damaged with heavy blast and hence ICT 2 tripped on differential protection operation. iii) The station T/F was placed under the 220KV Bus 1 at Srinagar(UK). Due to blast in station T/F, heavy arc developed and T/F oil spread on 220 KV Bus 1. Therefore, bus fault created and bus bar protection operated at 220KV Bus 1 at Srinagar(UK). iv) During the same time, 220 KV Singoli Bhatwari(Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-1 & 2 and 400/220 kv 315 MVA ICT 1 at Srinagar(UK) also tripped. v) The power generated by 33MW Unit-1, 2 and 3 at Singoli Bhatwari HEP were evacuating through 220 KV Singoli Bhatwari(Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-1 & 2. Hence, due to tripping of both 220 KV Singoli Bhatwari(Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-1 & 2, 33MW Unit-1, 2 and 3 at Singoli Bhatwari HEP tripped due to loss of evacuation path and blackout occurred at 220KV Singoli Bhatwari HEP. vi) As per DR of 400/220 kv 315 MVA ICT 2 at Srinagar(UK), ICT tripped on differential protection operation (3 phase to ground fault is observed). vii) As per DR of 220 KV Singoli Bhatwari(Singoli(LTUHP))-Srinagar(UK) (end) (PTCUL) Ckt-1 & 2, lines tripped on Y-N phase to ground fault (fault sensed in zone-4 indicating bus fault) with fault current of 510 A in Y phase. viii) As per DR of 400/220 kv 315 MVA ICT 1 at Srinagar(UK), Y-N phase to ground fault is observed with fault current of 2.33kA and 4.62kA in respectively HV and LV side of Y-phase of the ICT. ix) As per PMU at Roorkee(PG), 3-phase to ground fault followed by Y-N phase to ground fault is observed with fault clearing time of 80 ms. x) As per SCADA, load loss of approx. 35MW is observed in Uttarakhand control area and generation loss of approx. 108MW at Singoli Bhatwari HEP are observed.	1) 220 KV Singoli Bhatwari(Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-1 2) 220 KV Singoli Bhatwari(Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-2 3) 400/220 kv 315 MVA ICT 1 at Srinagar(UK) 4) 400/220 kv 315 MVA ICT 2 at Srinagar(UK) 5) 33MW Unit-1 at Singoli Bhatwari HEP 6) 33MW Unit-2 at Singoli Bhatwari HEP 7) 33MW Unit-3 at Singoli Bhatwari HEP
6	GI-2	Rajasthan	18-Jul-2023 19:56	18-Jul-2023 22:37	02:41	0	0	0.000	0.000	58065	69404	i) 400/220KV Bhadla(Raj) has one and half breaker scheme at 400KV side and double main & transfer bus scheme at 220KV side. ii) During antecedent condition, 400/220 kv 500 MVA ICT 1&3 at Bhadla(RS) were carrying ~141MW each and 400/220 kv 500 MVA ICT 2 at Bhadla(RS) was not in service. iii) As reported, at 19:56hrs, R-N phase to earth fault occurred on 400 KV Bhadla(RS)-Bhadla(PG) (PG) Ckt-2. Fault was in Z-1 from Bhadla(RS) end and in Z-2 from Bhadla(PG) end. iv) On this fault, Main-1 relay (distance protection) at Bhadla(RS) of 400 KV Bhadla(RS)-Bhadla(PG) (PG) Ckt-2 didn't operate due to issue in DC supply and Main-2 relay (differential protection) at Bhadla(RS) also didn't operate due to issue in signal communication. (Rajasthan has been communicated to resolve the issues in Main protection of the line at the earliest) v) Distance protection at Bhadla(RS) end of 400 KV Bhadla(RS)-Bhadla(PG) (PG) Ckt-2 sensed the fault in Z-2. However line tripped in ~250msec on directional earth fault O/C protection operation. As reported, TMS setting of the O/C relay was sensitive and same has been corrected to ensure its proper operation. vi) Distance protection of adjacent 400KV lines at Bhadla(RS) sensed the fault in Z-4 from Bhadla(RS) end. 400 KV Bhadla(RS)-Bhadla(PG) (PG) Ckt-1 tripped from Bhadla(RS) end in Z-4 with ~160msec delay and 400 KV Bhadla-Ramgarh (RS) Ckt-1 & 2 tripped from Ramgarh end in Z-2 with ~300msec delay (Z-4 time delay of these lines at Bhadla(RS) was ~450msec). Rajasthan has been communicated to review the time delay setting of distance protection relay and keep them in line with the NR protection philosophy. vii) As fault was also fed by 220KV side through ICTs, it cleared with the tripping of 220KV Bus sectionalizer on O/C E/F with ~200msec delay. viii) 220KV Bus-1(B) was remain intact. ix) As per PMU at Bhadla(PG), R-N fault with the delayed clearance in ~220msec is observed. x) As per SCADA, no change in load in Rajasthan control area is observed.	1) 400 KV Bhadla(RS)-Bhadla(PG) (PG) Ckt-1 2) 400 KV Bhadla(RS)-Bhadla(PG) (PG) Ckt-2 3) 400 KV Bhadla-Ramgarh (RS) Ckt-1 4) 400 KV Bhadla-Ramgarh (RS) Ckt-2

20.Details of Grid Events during the Month of July 2023
Northern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (H:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
7	GD-1	Rajasthan	20-Jul-2023 13:53	20-Jul-2023 15:29	01:36	2526	0	4.140	0.000	61013	74741	i) On 20th July 2023 at 13:53 hrs, as reported, R-N phase to earth fault occurred on 220kV Bhadla(PG)- MRPL ckt, R-ph jumper snapped at tower location no. 56 ii) On this fault, CB at Bhadla(PG) end of the MRPL line (connected at 220kV Bus-18) didn't open and therefore, LBB of MRPL bay at Bhadla(PG) operated. iii) LBB operation led to the tripping of 400/220kV 500 MVA ICT 5 at Bhadla(PG) and 220kV lines to RE stations i.e., MRPL, CSPJL, ACME & MAHOBA connected at 220kV Bus-18 at Bhadla(PG). iv) As per PMU at Bhadla(PG), R-N phase to earth fault with delayed clearance in 320msec is observed. v) At the same time, drop in RE generation at RE stations connected at other ISTS pooling station in Rajasthan RE complex also occurred on LVRT. vi) As per PMU, total drop in RE generation was approx. 2526MW. Out of total ~2526MW approx. 770MW lost due to tripping of RE stations on LBB protection operation as mentioned above and rest ~1750MW RE generation was due to partial/delayed recovery of RE generation after LVRT. Out of this ~1750MW RE generation, ~850MW RE generation recovered within 02 min of the event.	1) 220 KV Bhadla(PG) - Bus 3 2) 220 KV Bhadla(PG)-ACME Solar(ACM) (ACME) Ckt-1 3) 220 KV Bhadla(PG)-Mahoba Solar(Adani) (Adani) Ckt-1 4) 220 KV Bhadla(PG)-CS_Jodhpur SL_BHD_PG (Cleansolar_Jodhpur) (Cleansolar_Jodhpur) Ckt-1 5) 220 KV Bhadla(PG)-Mahindra SL_BHD_PG (MAHINDRA) (MAHINDRA) Ckt-1 6) 400/220 KV 500 MVA ICT 5 at Bhadla(PG)
8	GD-1	Uttarakhand	21-Jul-2023 00:44	21-Jul-2023 01:50	01:06	108	30	0.200	0.040	54019	74608	i) During antecedent condition, 33MW Unit-1, 2 and 3 at Singoli Bhatwari HEP were generating approx. 36MW each respectively. ii) As reported, at 00:44 hrs, 220 KV Singoli Bhatwari(Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-1 & 2 tripped from both ends on Y-N phase to earth fault. Both the circuits are on same tower. iii) As per DR of Singoli Bhatwari end, fault occurred on both the lines. Fault was in Z-1 (~54km) from Singoli Bhatwari end. A/R started in both the lines however, in ckt-1 it didn't complete due to DT received from Srinagar after ~50msec end and in ckt-2 line tripped after ~300msec on under voltage stage-2 operation. iv) As per DR of Srinagar end, fault occurred on both the lines. Fault was in Z-2 (~67km) from Srinagar(UK) end and fault current was around ~1.5KA. Carrier received from Singoli Bhatwari end however, A/R didn't operate and three phase tripping occurred instantaneously. v) Due to tripping of both 220 KV Singoli Bhatwari(Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-1 & 2, 33MW Unit-1, 2 and 3 at Singoli Bhatwari HEP tripped due to loss of evacuation path. vi) As per PMU at Roorkee(PG), Y-N phase to ground fault with no A/R operation and cleared within 100 ms is observed vii) As per SCADA, load loss of approx. 30MW is observed in Uttarakhand control area and generation loss of approx. 108MW at Singoli Bhatwari HEP are observed.	1) 220 KV Singoli Bhatwari(Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-1 2) 220 KV Singoli Bhatwari(Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-2 3) 33MW Unit-1 at Singoli Bhatwari HEP 4) 33MW Unit-2 at Singoli Bhatwari HEP 5) 33MW Unit-3 at Singoli Bhatwari HEP
9	GD-1	Delhi	23-Jul-2023 20:21	23-Jul-2023 20:25	00:04	0	314	0.000	0.443	51817	70881	i) 220kV side of 400/220kV Mandola(PG) has double main & transfer bus scheme and 220/66/33kV Gopalpur(DTL) & 220/66kV Narela has double main bus scheme. ii) During antecedent condition, 400/220kV 500MVA ICT-4, 220kV feeders i.e., Gopalpur ckt-1 & Narela ckt-2 were connected at 220kV Bus-1 at Mandola(PG) and 400/220kV 500MVA ICT-2, 125MVA reactor-1, 220kV feeders i.e., Gopalpur ckt-2 & Narela ckt-1 were connected at 220kV Bus-2 at Mandola(PG). ICT-2&4 were carrying approx. 215MW each, 220kV Narela D/C were carrying approx. 72MW each and 220kV Gopalpur ckt-1 & 2 were carrying approx. 87MW & 153MW respectively as per SCADA data. iii) Load of 220kV Gopalpur 5/s and 220kV Sabji Mandi 5/s was connected through 220kV Mandola-Gopalpur Ckt. I&II and load of 220kV Narela 5/s was connected through 220kV Mandola - Narela Ckt. I&II. iv) As reported, at 20:21 hrs, dropper jumper of CT at 220kV side of 400/220 KV 500 MVA ICT-4 at Mandola(PG) snapped which led to the differential current in ICT-4. v) As per PMU at Mandola(PG) & DR of 220kV feeders at Mandola(PG), there was no fault in system. However, bus bar protection at 220kV side operated and elements connected to 220kV Bus-1&2 tripped. 220kV Bus-3&4 and elements connected to them were remained intact. vi) As there was no fault in system, operation of bus bar protection was due to maloperation of bus bar relay. As reported, bus bar relay at 220kV side is ALSTOM FAC&A (conventional electromechanical relay) and thus DR of the same is also not available. vii) As reported, new bus bar panel has been procured and commissioning work of new bus bar protection is in process. viii) Due to tripping of 220 KV Mandola(PG)-Gopalpur(DTL) (DTL) Ckt-1 & ckt-2 supply to 220/66/33kV Gopalpur(DTL) 5/s lost and load of Gopalpur 5/s (~157MW) and Sabji Mandi 5/s (~37MW) affected. And due to tripping of 220 KV Mandola(PG)-Narela(DV) (DTL) Ckt-1&2 load of 220kV Narela (~120MW) affected. ix) As reported, load of 220kV Narela 5/s was restored at 20:24 hrs through 220kV DSIDC Bawana-Narela ckt-1&2 and load of 220kV Gopalpur & 220kV Sabji mandi was restored at 20:26 hrs through 220kV Wazirabad-Gopalpur ckt-1&2.	1) 220kV Bus 1 at Mandola(PG) 2) 220kV Bus 2 at Mandola(PG) 3) 220 KV Mandola(PG)-Gopalpur(DTL) (DTL) Ckt-1 4) 220 KV Mandola(PG)-Gopalpur(DTL) (DTL) Ckt-2 5) 220 KV Mandola(PG)-Narela(DV) (DTL) Ckt-1 6) 220 KV Mandola(PG)-Narela(DV) (DTL) Ckt-2 7) 400/220 KV 500 MVA ICT 2 at Mandola(PG) 8) 400/220 KV 500 MVA ICT 4 at Mandola(PG) 9) 125MVA reactor-1 at 220kV side
10	GD-1	Rajasthan	24-Jul-2023 03:29	24-Jul-2023 09:48	06:19	0	205	0.000	0.305	49195	67222	i) 220/132kV Bhiwadi(RS) has double main bus scheme at 220kV side. ii) As reported, at 03:29hrs, 220/132kV 160MVA ICT-1 at Bhiwadi(RS) caught fire which created internal fault in the ICT and due to this LBB operated. Rajasthan has been asked to share the reason of LBB operation. iii) During this time, 220 KV Bhiwadi(PG)-Bhiwadi(RS) (RS) Ckt-1 & 2 and 220/132kV 160MVA ICT-2 & 3 at Bhiwadi(RS) tripped which resulted in complete blackout of 220/132kV Bhiwadi(RS) 5/s. Rajasthan has been asked to share the reason of tripping of all the elements at Bhiwadi(RS). iv) As reported by SLDC Rajasthan, 220 KV Bhiwadi(PG)-Bhiwadi(RS) (RS) Ckt-1 tripped from 220kV Bhiwadi(RS) end only. Line was manually tripped from PGCL end. 220 KV Bhiwadi(PG)-Bhiwadi(RS) (RS) Ckt-2 tripped from both ends. v) As per DR of Bhiwadi(PG) of 220 KV Bhiwadi(PG)-Bhiwadi(RS) (RS) Ckt-2, Y-N fault at distance of ~7.7km (Z-2) & fault current of 13.68kA is observed. vi) As per PMU at Bhiwadi(PG), two consecutive Y-N phase to earth faults with the delayed clearance of 360msec are observed. vii) As per SCADA, change in demand of approx. 205MW in Rajasthan control area is observed.	1) 220 KV Bhiwadi(PG)-Bhiwadi(RS) (RS) Ckt-1 2) 220 KV Bhiwadi(PG)-Bhiwadi(RS) (RS) Ckt-2 3) 220/132kV 160MVA ICT-1 at Bhiwadi(RS) 4) 220/132kV 160MVA ICT-2 at Bhiwadi(RS) 5) 220/132kV 160MVA ICT-3 at Bhiwadi(RS)
11	GI-2	Haryana	25-Jul-2023 14:44	25-Jul-2023 18:00	03:16	0	190	0.000	0.258	54660	73753	i) 220kV side of 400/220kV Bahadurgarh(PG) has double main & transfer bus scheme. ii) As reported, at 14:22 hrs, sparking was observed in Y-phase isolator of 400/220 KV 315 MVA ICT 2 at Bahadurgarh(PG) and ICT 2 tripped. POWERGRID has been communicated to share the exact detail of protection operation. iii) Further at 14:44 hrs, bus bar protection operated at 220kV Bus 1 & 2 at Bahadurgarh(PG) and 400/220 KV 315 MVA ICT 1 & 2 at Bahadurgarh(PG) along with 220 KV Bahadurgarh(PG)-Nuna Majra(HV) (HVPNL) Ckt-1 & 2 tripped. POWERGRID has been communicated to share the exact nature and location of fault. iv) As per DR of 220 KV Bahadurgarh(PG) Nuna Majra(HV) (HVPNL) Ckt-2 of POWERGRID end, line tripped on B-N phase to earth fault in Z-2. v) As per PMU at Bahadurgarh(PG), B-N phase to earth fault with delayed fault clearance time of 240 ms is observed. vi) As per SCADA, change in demand of approx. 190MW in Haryana control area.	1) 400/220 KV 500 MVA ICT 2 at Bahadurgarh(PG) 2) 220kV Bus 1 at Bahadurgarh(PG) 3) 220kV Bus 2 at Bahadurgarh(PG) 4) 220 KV Bahadurgarh(PG)-Nuna Majra(HV) (HVPNL) Ckt-1 5) 220 KV Bahadurgarh(PG)-Nuna Majra(HV) (HVPNL) Ckt-2 6) 400/220 KV 315 MVA ICT 1 at Bahadurgarh(PG)
12	GD-1	Rajasthan	26-Jul-2023 09:23	26-Jul-2023 11:58	02:35	450	0	0.866	0.000	51973	59547	i) Total MW generation of Aavaada Surce, Aavaada RHN and Aavaada Sustainable are pooled at 400kV Aavaada pooling and total generation is evacuated through 400 KV Aavaada Pooling SL_BKN_PG (AEPJ)-Bikaner(PG) (AEPJ) Ckt. ii) As reported, at 09:23hrs, 400 KV Aavaada Pooling SL_BKN_PG (AEPJ)-Bikaner(PG) (AEPJ) Ckt tripped on Z-4 distance protection operation at Bikaner end. iii) As per PMU at Aavaada(P), no fault is observed in the system. iv) As reported, relay co-ordination and PT/CVT wiring at both the ends were checked and revalidated. CVT wiring mis-contact was found at Aava line main-1 relay at Bikaner end and the same was rectified during shutdown 31st July 2023 night hours. v) As per PMU, generation loss of approx. 450MW at Aavaada Pooling(P) is observed.	1) 400 KV Aavaada Pooling SL_BKN_PG (AEPJ)-Bikaner(PG) (AEPJ) Ckt

20.Details of Grid Events during the Month of July 2023
Northern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (H:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
13	GD-1	Uttar Pradesh	26-Jul-2023 04:46	26-Jul-2023 06:32	01:46	0	430	0.000	0.712	46861	60382	i) 400kV side of 400/220/132kV Gr.Noida(UP) has double main & transfer bus scheme. ii) During antecedent condition, power was flowing from 400kV side to 220kV side through 400/220 kV 315 MVA ICT 1 & 2 and 400/220 kV 500 MVA ICT 5 & 6 at Gr.Noida(UPC). iii) As reported, at 04:46 hrs, B-ph conductor of 400kV Bus 1 at Gr.Noida(UPC) broke and fell on Y-ph bay of 400 KV Dadri(NT)-Gr.Noida(UPC) (PG) Ckt-1 which created Y-B phase to phase fault at Bus 1. Due to this bus bar protection operated and elements on Bus 1 tripped. iv) It is further reported that due to delayed operation of bus coupler CB, elements on Bus 2 also tripped during the same time. Eventually 400/220/132kV Gr.Noida(UP) S/s became dead. v) As per DR, 400 KV Dadri(NT)-Gr.Noida(UPC) (DR end) (PG) Ckt-1 tripped on Y-B phase to phase fault with fault currents of 16.57kA and 15.56kA respectively in Y and B phase; fault sensed in zone-4. vi) As per PMU at Agra(PG), Y-B phase to phase fault with delayed fault clearance time of 440 ms is observed. vii) As per SCADA, change in demand of approx. 430MW in UP control area.	1) 400/220 kV 315 MVA ICT 1 at Gr.Noida(UPC) 2) 400/220 kV 315 MVA ICT 2 at Gr.Noida(UPC) 3) 400kV Bus 1 at Gr.Noida(UPC) 4) 400 KV Dadri(NT)-Gr.Noida(UPC) (PG) Ckt-1 5) 400 KV Gr.Noida(UPC)-Nawada(HV) (PG) Ckt-1 6) 400 KV Gr.Noida_2(UPC)-Gr.Noida(UPC) (UP) Ckt-1 7) 400/220 kV 500 MVA ICT 5 at Gr.Noida(UPC) 8) 400 KV Gr.Noida_2(UPC)-Gr.Noida(UPC) (UP) Ckt-2 9) 400/220 kV 500 MVA ICT 6 at Gr.Noida(UPC) 10) 400kV Bus 2 at Gr.Noida(UPC)
14	GI-1	Himachal Pradesh	26-Jul-2023 07:07	26-Jul-2023 09:11	02:04	0	125	0.000	0.229	47116	54598	i) 220/132kV Majri (HP) has double main bus scheme at both 220kV & 132kV level. During antecedent condition, active power flowing through 220 KV Khodri(LUK)-Majri(HP) (LUK) Ckt-1 & 2 was 72MW each respectively. ii) As reported, at 07:07 hrs, 220 KV Khodri(LUK)-Majri(HP) (UK) Ckt-1 tripped on Y-B phase to phase fault. As per DR of Khodri end, Y-B phase to phase fault in Z-2 with fault current of 3.42kA each respectively in Y and B phase and fault clearance time of 384ms is observed. iii) During the same time, 220 KV Khodri(LUK)-Majri(HP) (UK) Ckt-2 also tripped due to over-current protection operation (I _Y =1.959kA and I _B =1.785kA) from Majri end. Line tripped from Majri end only and remained charged from Khodri end. iv) As per PMU at Shahrampur(PG), Y-B phase to phase fault with delayed fault clearance time of 400ms is observed in the system. v) As per SCADA, change in demand of approx. 125MW is observed in HP control area.	1) 220 KV Khodri(LUK)-Majri(HP) (UK) Ckt-1 2) 220 KV Khodri(LUK)-Majri(HP) (UK) Ckt-2
15	GD-1	Uttar Pradesh	26-Jul-2023 05:30	26-Jul-2023 06:40	01:10	0	105	0.000	0.179	47234	58819	i) 220kV side of 220/132kV Muzaffarnagar2(Nara)(UP) has main and transfer bus scheme. ii) As reported, at 05:30 hrs, 220kV Nara(UP)-Roorkee(UK) (UP) Ckt tripped from Roorkee end only on R-N fault. iii) This fault was sensed by Nara(UP) end in Z-1 with fault current of approx. 4.8 kA and fault distance of 31.9 km. However, relay at Nara(UP) failed to initiate tripping command. iv) Due to this, 220 KV Meerut(PG)-Nara(UP) (PG) Ckt (fault current 3.517kA as per DR), 220kV Nara(UP)-Jansath(UP) Ckt (fault distance 82.9km from Nara) and 220kV Nara(UP)-Muzaffarnagar(UP) Ckt (fault current 2.63kA from Nara) tripped from remote end. Fault was sensed in zone-3 (as per DR). v) During this time, 220/132kV 160MVA ICT-1 at Nara(UP) also tripped due to over-current earth fault protection operation. vi) With the tripping of all the aforementioned elements, feeding source to Nara(UP) S/s lost and 220/132kV Muzaffarnagar2(Nara)(UP) S/s became dead. vii) As per SCADA, SCOT, 220kV Muzaffarnagar(UP)-Jansath(UP) Ckt also tripped during the same time (reason yet to be shared). viii) As per PMU at Roorkee(PG), R-N phase to earth fault with delayed fault clearance time of 1400 ms is observed. ix) As reported by SLDC-UP, change in demand of approx. 105MW occurred in UP control area.	1) 220kV Nara(UP)-Roorkee(UK) (UP) Ckt 2) 220 KV Meerut(PG)-Nara(UP) (PG) Ckt 3) 220kV Nara(UP)-Jansath(UP) Ckt 4) 220kV Nara(UP)-Muzaffarnagar(UP) Ckt 5) 220/132kV 160MVA ICT-1 at Nara(UP)
16	GI-1	Jammu and Kashmir	26-Jul-2023 14:19	26-Jul-2023 15:00	00:41	0	135	0.000	0.208	55367	64808	i) As reported, at 14:19 hrs, 220 KV Wagoora(PG)-Pampore(PDD) (PG) Ckt-1 tripped on R-N phase to earth fault with fault current of 1.18kA and fault distance of 57.4km from Wagoora(PG) end, fault was sensed in Zone-3 from Wagoora(PG). ii) At the same time, 220 KV Wagoora(PG)-Pampore(PDD) (PG) Ckt-2 also tripped from Pampore end only on over current protection operation. iii) As per PMU, R-N phase to phase fault with delayed fault clearance time of 960ms is observed in system. iv) As per SCADA, load loss of approx. 135MW occurred in J&K control area.	1) 220 KV Wagoora(PG)-Pampore(PDD) (PG) Ckt-1 2) 220 KV Wagoora(PG)-Pampore(PDD) (PG) Ckt-2
17	GI-2	Delhi	29-Jul-2023 12:26	29-Jul-2023 12:50	00:24	0	70	0.000	0.116	52986	60558	i) 400/220kV Maharanibagh(DTL) has double main bus scheme at both 400kV and 220kV level. 220 kV Maharanibagh-Lodhi road (DTL) Ckt-1&2 were feeding the load of 220kV Lodhi Road S/s. ii) As reported, at 12:26 hrs, 220 kV Maharanibagh-Lodhi road (DTL) Ckt-2 tripped due to Y-ph LA burst at Maharanibagh end. Fault sensed in zone-1 at Maharanibagh end. iii) Further, 220 kV Maharanibagh-Lodhi (DTL) Ckt-1 also tripped during the same time (reason yet to be shared) (866 operated at Lodhi road end). iv) As per PMU at Bulahgarh(PG), Y-N phase to earth fault is observed with fault clearing time of 80 ms. v) As per SCADA, load loss of approx. 200MW is observed in Delhi control area.	1) 220 kV Maharanibagh-Lodhi road (DTL) Ckt-1 2) 220 kV Maharanibagh-Lodhi Road (DTL) Ckt-2
18	GD-1	Rajasthan	31-Jul-2023 10:36	31-Jul-2023 22:39	12:03	1620	0	2.639	0.000	61385	68548	i) On 31st July, 2023, at 10:36:11:200 hrs, B-ph jumper at RSUPL end of 220kV Fatehgarh2-RSUPL ckt snapped at tower location no. 10 at distance ~18km from RSUPL end. ii) On this fault, As per DR of both the ends, line successfully auto reclosed from both ends. However after approx. 750msec of A/R operation, line tripped from RSUPL end on under voltage protection. RSUPL has been communicated to disable the under voltage protection in line at the earliest. iii) At 10:36:11:280 hrs, R-N fault occurred on 220kV Fatehgarh2-RJPL ckt. As per DR of both the ends, line successfully autoreclosed from both the ends. iv) As per PMU at Bikaner(PG), B-N phase to earth fault followed by R-N phase to earth fault which cleared within 80msec is observed. v) At the same time, drop in RE generation at RE stations connected at other ISTS pooling station in Rajasthan RE complex also occurred on LVRT (triggered due to B-N & R-N fault in the system). vi) During same time, 220kV Bhadai2(PG)-Nokhra ckt also tripped on SOTF over current protection at Nokhra end(as per DR). It was maloperation of protection due to incorrect SORF protection logic. Nokhra has been communicated to correct the logic of SOTF protection. vii) As per PMU, total loss in RE generation was approx. 1620MW. Out of total ~1620MW approx. 500MW lost due to tripping of RSUPL(260MW) & Nokhra(240MW) RE station as mentioned above and rest ~1120MW RE generation was due to partial/delayed recovery of RE generation after LVRT. Out of this 1120MW RE generation, ~400MW RE generation recovered within 01 min of the event.	1) 220 KV Renew Solar(Urja SL_FGARH_PG_RSUPL)-Fatehgarh_2(PG) (Renew Solar Urja (RSUPL)) Ckt-1

20.Details of Grid Events during the Month of July 2023
Western Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	WR	01-Jul-23 01:19	01-Jul-23 03:49	02:30	145.2	-	0.002	-	58756	48315	At 01:19 Hrs / 01-07-2023, 220 kV Bachau-Ostro 1 & 2 tripped due to blast of Yph LA of 33kV feeder 1. Generation loss of 145.2 MW occurred at 220 kV Ostro (Renew Power) wind power plant due to loss of evacuation path.	Tripping of 1. 220 kV Bachau-Ostro 1 & 2
2	GD-1	WR	04-Jul-23 20:09	05-Jul-23 12:33	16:24	20	-	0.000	-	69871	57616	At 20:09 Hrs / 04-07 -2023, 220 kV Bhuj-Kotda Madh tripped on B phase to earth fault. During patrolling, a rope was found hanging on B phase conductor which came in induction zone and line tripped. Generation loss of 20 MW occurred at 220 kV Kotda Madh (Alfanar) wind power plant due to the event.	Tripping of 1. 220 kV Bhuj-Kotda Madh
3	GD-1	WR	07-Jul-23 19:22	07-Jul-23 20:55	01:33	53	-	0.001	-	67942	55110	At 19:22 Hrs / 07-07-2023, 220 kV Bhuj-Vadva tripped on Y phase to earth fault at Vadva end only and A/R successfully at Bhuj end. Generation loss of 53 MW occurred at 220 kV Vadva (GIWEL) wind power plant due to the event.	Tripping of 1. 220 kV Bhuj- Vadva
4	GD-1	WR	09-Jul-23 20:54	09-Jul-23 22:10	01:16	-	197	-	0.004	66624	51197	At 20:54 Hrs/09-07-2023, 220 kV Magarwada Buses 1&2 tripped on Y-B phase fault on BB protection operation due to earth wire snapped and came in induction zone of 220 kV Magarwada Buses 1&2. Load loss of 197 MW occurred due to the event.	Tripping of 1. 220 kV Magarwada- Magarwada(PG) 1&2 2. 220 kV Magarwada- Vapi(PG) 1&2 3. 220 kV /66 kV Magarwada ICTs 1,2,3&4
5	GD-1	WR	12-Jul-23 05:03	13-Jul-23 16:22	11:19	31	-	0.000	-	63296	50634	At 05:03 Hrs/12-07-2023, 220 kV Jamkhambaliya- Sidhpur tripped on B-E fault. Generation loss of 31 MW occurred at 220 kV Sidhpur (Torrent) wind power plant due to loss of evacuation path. During patrolling, insulator flashover was found at location 31/04 and after replacement of insulator also line did not hold. On more insulator flashover found at location 34/00. Line charged after replacement of the insulator.	Tripping of 1. 220 kV Jamkhambaliya- Sidhpur
6	GD-1	WR	13-Jul-23 01:49	13-Jul-23 02:50	01:01	960	-	0.015	-	65842	52759	At 01:49 Hrs/13-07-2023, While test charging 400 kV MEL- Bilaspur 2 from MEL end, 400 kV Bilaspur-MEL 1 tripped on Over Voltage protection operation from Bilaspur end. Prior to the event, 400 kV MEL- Bilaspur 2 tripped on B-E fault at 00:05 Hrs/13-07-2023. Generation loss of 960 MW at 400 kV MEL due to loss of evacuation path.	Tripping of 1. 400 kV MEL- Bilaspur 1&2

20.Details of Grid Events during the Month of July 2023



Western Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
7	GD-1	WR	14-Jul-23 20:07	15-Jul-23 01:25	05:18	47	-	0.001	-	72228	56810	At 20:07 Hrs / 14-07 -2023, 220 kV Bhuj-Kotda Madh tripped on B-E fault. During patrolling, foreign material was found hanging on B phase conductor. Generation loss of 47 MW occurred at 220 kV Kotda Madh (Alfanar) wind power plant due to loss of evacuation path.	Tripping of 1. 220 kV Bhuj-Kotda Madh
8	GD-1	WR	15-Jul-23 16:47	15-Jul-23 22:05	05:18	205	-	0.003	-	68878	54677	At 16:47 Hrs / 15-07 -2023, 220 kV Bhuj- Dayapar 2 tripped on R-Y phase fault. Durring patrolling, R phase jumper open at Tower No. 154. Generation loss of 205 MW occurred at 220 kV Dayapar (INOX) wind power plant due to loss of evacuation path.	Tripping of 1. 220 kV Bhuj- Dayapar 2
9	GD-1	WR	18-Jul-23 17:10	18-Jul-23 21:40	04:30	54	-	0.001	-	63848	55106	At 17:10 Hrs/18-07-2023, 220 kV Bhuj-Kotda Madh tripped on B-E fault. Generation loss of 54 MW occurred at 220 kV Kotda Madh (Alfanar) wind power plant due to loss of evacuation path.	Tripping of 1. 220 kV Bhuj-Kotda Madh
10	GD-1	WR	20-Jul-23 02:04	20-Jul-23 02:55	00:51	720	-	0.012	-	61747	50037	At 02:04 Hrs / 20-07-2023, 400 kV Hadala-Vadinar-1 tripped on R-E fault. At the same time 400 kV Hadala-Vadinar-2 tripped on R-Y phase to phase fault on Zone-1 protection operation at Vadinar end only due to over-reach (As fault was on Line -1). Due to these trippings Vadinar (EPGL-Essar Power Gujarat Limited) units 1 & 2 (600 MW each) tripped due to loss of evacuation path, leading to blackout of Vadinar substation. Generation loss of 720 MW occurred at Vadinar (EPGL) power station.	Tripping of 1. 400 kV Hadala-Vadinar-1 & 2 2. Vadinar-Unit-1 & 2 (600 MW) 3. 400 kV Vadinar-Bus-1 & 2
11	GD-1	WR	20-Jul-23 20:15	20-Jul-23 20:55	00:40	615	-	0.009	-	71890	55647	At 20:15 Hrs / 20-07-2023, while charging of 400 kV Hadala-Vadinar-1, 400 kV Hadala-Vadinar-2 tripped on R-Y phase to phase fault, Zone-1 operation. Due to this tripping Vadinar (EPGL-Essar Power Gujarat Limited) units 1 & 2 (600 MW each) tripped due to loss of evacuation path, leading to blackout of Vadinar substation. Generation loss of 615 MW occurred at Vadinar (EPGL) power station.	Tripping of 1. 400 kV Hadala-Vadinar-2 2. Vadinar-Unit-1 & 2 (600 MW) 3. 400 kV Vadinar-Bus-1 & 2
12	GI-1	WR	22-Jul-23 15:43	22-Jul-23 20:04	04:21	3	-	0.000	-	61445	52186	At 15:43 Hrs / 22-07-2023, 220/33 kV Ramnagar Pahad ICT-1, 2 & 3 tripped from LV side on REF trip. At the time of these trippings heavy rains and lightning were reported by site. Generation loss of 3 MW occurred at 220 kV Ramnagar Pahad (Mahindra) solar plant due to the event.	Tripping of 1. 220/33 kV Ramnagar Pahad ICT-1, 2 & 3
13	GD-1	WR	26-Jul-23 19:38	26-Jul-23 20:50	01:12	70	-	0.001	-	69489	57913	At 19:38 Hrs./26-07-2023, 220 kV Jamkhambaliya- Khakarda line tripped from Khakarda end only on df/dt protection operation. As informed by AEPL, df/dt protection disabled at Khakarda end. Generation loss of 70 MW occurred at 220 kV Khakarda (AEPL) wind power plant due to loss of evacuation path.	Tripping of 1. 220 kV Jamkhambaliya- Khakarda
14	GD-1	WR	27-Jul-23 19:05	27-Jul-23 20:24	01:19	180	-	0.003	-	67375	56097	At 19:05 Hrs/27-07-2023, While FTC of 220/33 kV Dayapar ICT 4 from 220 kV side, 220 kV Bhuj- Dayapar-2 tripped from Dayapar end only on under voltage protection operation. Generation loss of 180 MW occurred at 220 kV Dayapar(INOX) wind power plant due to loss of evacuation path.	Tripping of 1. 220 kV Bhuj- Dayapar
15	GD-1	WR	28-Jul-23 09:44	28-Jul-23 11:18	01:34	250	-	0.004	-	64590	55795	At 09:44 Hrs/28-07-2023, 220 kV Bhuj- Naranpar tripped on B-E fault. Generation loss of 250 MW occurred at 220 kV Naranpar(GIWEL) wind power plant due to loss of evacuation path.	Tripping of 1. 220 kV Bhuj- Naranpar

20.Details of Grid Events during the Month of July 2023



Eastern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1 or 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GI-2	Farakka (FSTPP)	01.07.2023 02:17	01.07.2023 05:28	03:11	1420	0	4.61%	0.00%	30802	22584	At 02:17 Hrs. FSTPP Unit-4 (500 MW) tripped on Low Vacuum and other remaining four units of FSTPP St I & II went off Bar on low vacuum within next 45 Mins one by one. Around 1420 MW generation loss occurred.Unit Tripping time as below: 500 MW U#4: 02:17 Hrs 200 MW U#2: 02:31 Hrs 200 MW U#3: 02:31 Hrs 200 MW U#1: 02:35 Hrs 500 MW U#5: 03:00 Hrs	3*200 MW Unit at FSTPP 2*500 MW Unit at FSTPP
2	GI-2	GMR (Odisha)	31.07.2023 16:29	01.08.2023 16:40	24:11	262	0	0.93%	0.00%	28268	26624	At 16:29 Hrs, 400 kV GMR-Meramundali B tripped due to B_N fault. Consequently, 350 MW U#3 of GMR tripped due to loss of evacuation path. 262 MW generation loss occurred.	350 MW U#3 at GMR (STU) 400 kV GMR-Meramundali B

20.Details of Grid Events during the Month of July 2023
Southern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t. Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre fault and post fault system conditions)	Name of Elements (Tripped/Manually opened)
	(GI for 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Tamil Nadu	02-Jul-23 13:07	02-Jul-23 14:04	57mins	504	0	1.05%	0.00%	48020	49300	Complete Outage of 230kV NCTPS Stage-1 Generating Station of TANGEDCO: As per the reports submitted, the triggering incident was tripping of 230kV NCTPS ETPS line and 230kV NCTPS Kilapuk line on faults at 13:07hrs and 13:09hrs respectively. Tripping of these lines led to tripping of NCTPS Unit-1,2 & 3 on over-frequency protection. This led to a complete outage of the 230kV NCTPS Stage-1 Generating Station.	1. 230kV NCTPS Unit-1,2&3 2. 230kV NCTPS ETPS line 3. 230kV NCTPS Kilapuk line
2	GD-1	Karnataka	27-Jul-23 09:41	27-Jul-23 10:09	28mins	0	98	0.00%	0.24%	42616	41205	Complete Outage of 220kV/110kV Ambewadi SS of KPTCL: During antecedent conditions, all 220kV elements were connected to 220kV Bus-1 at 220kV/110kV Ambewadi SS. As per the reports submitted, the triggering incident was B-N fault in 220kV Nagheri Ambewadi line-2 and the line tripped. Because of this, 220kV Ambewadi Nagheri line-1 got overloaded and caused the B-phase bus jumper connected to the line to fail. This resulted in a B-phase bus fault causing BNP to operate and all the elements connected to the 220kV Bus tripped. This led to a complete outage of 220kV/110kV Ambewadi SS.	1. 220kV/110kV Ambewadi Transformer-1,2&3 2. 220kV Nagheri Ambewadi Line-1&2 3. 220kV Ambewadi Narendra Line-1&2 4. 220kV Ambewadi Ponda Line-1&2
3	GD-1	Telangana	27-Jul-23 16:10	27-Jul-23 21:27	5hrs 17mins	0	0	0.00%	0.00%	42297	40176	Complete Outage of 220kV/11kV BG Kothur LIS SS of TSTRANSCO: As per the reports submitted, the triggering incident was tripping of all 220kV connected lines at 220kV/11kV BG Kothur LIS SS on CTD operation due to DC supply failure. This resulted in complete outage of 220kV/11kV BG Kothur LIS SS.	1. 220kV BG Kothur Manuguru 2. 220kV BG Kothur Chintur 3. 220kV BG Kothur KTPS Stage-V 4. 220kV BG Kothur KTPS Stage-III
4	GI-2	Andhra Pradesh	03-Jul-23 17:37	03-Jul-23 18:33	54 mins	0	0	0.00%	0.00%	43575	45632	Tripping of 400kV Bus-1 of 400kV Hinduja Generating station: As per the reports submitted, the triggering incident was B-N fault in 400kV Guddigudem Hinduja line-1. At Hinduja end, the main breaker failed to open causing LBB to operate and all the main breakers connected to 400kV Bus-1 tripped.	1. 400kV Guddigudem Hinduja Line-1
5	GI-1	Telangana	06-Jul-23 00:34	06-Jul-23 01:27	53 mins	0	0	0.00%	0.00%	40442	39752	Tripping of 220kV Bus-2 of 220kV Upper Jurala PH of TSGENCO: During antecedent conditions, there was no generation at 220kV Upper Jurala PH. As per the reports submitted, the triggering incident was R-N fault in 220kV Jurala Raichur_KA Line-2. At the same time, the Bus Coupler tripped on over current protection. Tripping of only connected line and bus coupler resulted in de-energization of 220kV Bus-2 at 220kV Upper Jurala PH.	1. 220kV Jurala Raichur_KA Line-2 2. 220kV Bus coupler at 220kV Jurala PH
6	GI-2	Andhra Pradesh	11-Jul-23 19:54	11-Jul-23 23:00	3hrs 6mins	0	0	0.00%	0.00%	43527	44793	Tripping of 400kV Bus-3 and Bus-4 of 400kV/220kV Kalpakka SS of APTRANSCO: As per the reports submitted, the triggering incident was R-phase CT failure in Bus coupler-2 bay of 400kV/220kV Kalpakka SS. Immediately, 400kV Bus-3 and Bus-4 BNP operated and all the elements connected to the Bus-3 and Bus-4 tripped.	1. 400kV Kalpakka Gajuwaka Line-1&2 2. 400kV Kalpakka Simhari Line-2 3. 400kV kalpakka Vemagiri Line-1 4. 400kV/220kV Kalpakka ICT-3
7	GI-2	Tamil Nadu	27-Jul-23 07:12	27-Jul-23 09:46	2hr 34mins	0	0	0.00%	0.00%	37555	38621	Tripping of 400kV Bus-1 of 400kV/230kV/110kV Alamythy SS of TANTRANSCO: 400kV/230kV/110kV Alamythy SS is operating with one and half breaker scheme at 400kV level. As per the reports submitted, the triggering incident was bus-side B-phase CT failure in 400kV/230kV ICT-5 at 400kV/230kV/110kV Alamythy SS. Immediately, 400kV Bus-1 BNP operated and the main breakers connected to the 400kV Bus-1 tripped. This resulted in de-energisation of 400kV Bus-1 of 400kV/230kV/110kV Alamythy SS.	1. 400kV/230kV ICT-5 at Alamythy

20.Details of Grid Events during the Month of July 2023

North-Eastern Region



Sl No.	Category of Grid Event (G1 for 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM:SS)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD 1	New Umtru Generating Station of Meghalaya Power System	03-Jul-23 14:46	03-Jul-23 14:53	00:07:00	20	0	0.69%	0.00%	2898	2447	New Umtru Generating Station of Meghalaya Power System was connected with the rest of NER Grid through 132 kV EPIP II - New Umtru and 132 kV Umtru - New Umtru lines. At 14:46 Hrs on 03.07.2023, 132 kV EPIP II - New Umtru and 132 kV Umtru - New Umtru lines tripped. Due to tripping of these elements, New Umtru Generating Station of Meghalaya Power System got separated from NER Grid and subsequently collapsed due to loss of evacuation path. Power supply was extended to New Umtru Generating Station of Meghalaya Power System by charging 132 kV EPIP II - New Umtru line at 14:53 Hrs on 03.07.2023.	132 kV EPIP II - New Umtru and 132 kV Umtru - New Umtru lines.
2	GD 1	Sarupathar and Bokajan areas of Assam Power System	04-Jul-23 10:00	04-Jul-23 10:20	00:20:00	0	24	0.00%	0.90%	2404	2661	Sarupathar and Bokajan areas of Assam Power System were connected with rest of NER grid through 132 kV Sarupathar -Golaghat line. 132 kV Bokajan - Dimapur(PG) line was under planned shutdown prior to event. At 10:00 Hrs on 04.07.2023, 132 kV Sarupathar - Golaghat line tripped. Due to tripping of this element, Sarupathar and Bokajan areas of Assam Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extended to Sarupathar and Bokajan areas of Assam Power System by charging 132 kV Sarupathar - Golaghat line at 10:20 Hrs on 04.07.2023.	132 kV Sarupathar - Golaghat line
3	GD 1	Sarupathar and Bokajan areas of Assam Power System	04-Jul-23 10:48	04-Jul-23 11:17	00:29:00	0	22	0.00%	0.84%	2340	2608	Sarupathar and Bokajan areas of Assam Power System were connected with rest of NER grid through 132 kV Sarupathar -Golaghat line. 132 kV Bokajan - Dimapur(PG) line was under planned shutdown prior to event. At 10:48 Hrs on 04.07.2023, 132 kV Sarupathar - Golaghat line tripped. Due to tripping of this element, Sarupathar and Bokajan areas of Assam Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extended to Sarupathar and Bokajan areas of Assam Power System by charging 132 kV Dimapur (PG) - Bokajan line at 11:17 Hrs on 04.07.2023.	132 kV Sarupathar - Golaghat line
4	GD 1	Margherita(Ledo), Rupai, Chapakhowa areas of Assam Power System and Roing, Tezu, Namsai, Pasighat, Along, Daporijo and Ziro areas of Arunachal Pradesh Power System	04-Jul-23 18:32	04-Jul-23 20:47	02:15:00	0	72	0.00%	2.41%	2781	2987	Margherita(Ledo), Rupai, Chapakhowa areas of Assam Power System and Roing, Tezu, Namsai, Pasighat, Along, Daporijo and Ziro areas of Arunachal Pradesh Power System were connected with rest of NER grid through 132 kV Tinsukia - Margherita(Ledo) line. 132kV Ranganadi - Ziro and 132 kV Tinsukia - Rupai lines were under outage prior to event. At 18:32 Hrs on 04.07.2023, 132 kV Tinsukia - Margherita(Ledo) line tripped. Due to tripping of this element, Margherita(Ledo), Rupai, Chapakhowa areas of Assam Power System and Roing, Tezu, Namsai, Pasighat, Along, Daporijo and Ziro areas of Arunachal Pradesh Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extended to Roing, Tezu, Namsai, Pasighat, Along, Daporijo and Ziro by charging 132 kV Ziro - Ranganadi line at 20:47 Hrs on 04.07.2023. Power supply was extended to Margherita(Ledo), Rupai, Chapakhowa areas of Assam Power System a 132 kV Tinsukia - Margherita(Ledo) line at 23:16 Hrs on 04.07.2023.	132 kV Tinsukia - Margherita(Ledo) line
5	GD 1	Tuirial Generating Station of Mizoram Power System	05-Jul-23 18:05	05-Jul-23 18:17	00:12:00	18	0	0.68%	0.00%	2639	3043	Tuirial Generating Station of Mizoram Power System was connected with the rest of NER Grid through 132 kV Tuirial-Kolasib line. At 18:05 Hrs on 05.07.2023, 132 kV Tuirial-Kolasib line tripped. Due to tripping of this element, Tuirial Generating Station of Mizoram Power System got separated from rest of NER Grid and subsequently collapsed due to loss of evacuation path. Power supply was extended to Tuirial Generating Station of Mizoram Power System by charging 132 kV Tuirial-Kolasib line at 18:17 Hrs on 05.07.2023	132 kV Tuirial-Kolasib line
6	GD 1	Tuirial Generating Station of Mizoram Power System	06-Jul-23 16:10	06-Jul-23 16:20	00:10	18	0	1%	0%	2203	2794	Tuirial Generating Station of Mizoram Power System was connected with the rest of NER Grid through 132 kV Tuirial-Kolasib line. At 16:10 Hrs on 06.07.2023, 132 kV Tuirial-Kolasib line tripped. Due to tripping of this element,Tuirial Generating Station of Mizoram Power System got separated from rest of NER Grid and subsequently collapsed due to loss of evacuation path. Power supply was extended to Tuirial Generating Station of Mizoram Power System by charging 132 kV Tuirial-Kolasib line at 16:20 Hrs on 06.07.2023	132 kV Tuirial-Kolasib line

20.Details of Grid Events during the Month of July 2023

North-Eastern Region



Sl No.	Category of Grid Event (G1 1or 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM:SS)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	Generation Loss(MW)					Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)			
7	GD 1	Pasighat, Roing, Tezu & Namsai areas of Arunchal Pradesh Power System and Chapakhowa area of Assam Power System	14-Jul-23 13:20	14-Jul-23 13:29	00:09	0	19	0%	1%	2613	2607	Pasighat, Roing, Tezu & Namsai areas of Arunchal Pradesh Power System and Chapakhowa area of Assam Power System were connected with rest of NER grid through 132 kV Along-Pasighat and 132 kV Rupai-Chapakhowa lines. At 13:20 Hrs on 14.07.2023, 132 kV Along-Pasighat and 132 kV Rupai-Chapakhowa lines tripped. Due to tripping of these elements, Pasighat, Roing, Tezu & Namsai areas of Arunchal Pradesh Power System and Chapakhowa area of Assam Power System got separated from the rest of NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extended to Pasighat, Roing, Tezu & Namsai areas of Arunchal Pradesh Power System and Chapakhowa area of Assam Power System by charging 132 kV Rupai-Chapakhowa Line at 13:29 Hrs on 14.07.2023.	132 kV Along-Pasighat and 132 kV Rupai-Chapakhowa lines
8	GD 1	Karbi Langpi Generating Station of Assam Power System	15-Jul-23 10:33	15-Jul-23 10:47	00:14:00	50	0	1.74%	0.00%	2875	2764	Karbi Langpi Generating Station of Assam Power System was connected with the rest of NER Grid through 220 kV Sarusajai-Karbi Langpi 1 line. 220 kV Sarusajai-Karbi Langpi 2 line was under planned shutdown since 08:52 Hrs on 15.07.2023 for maintenance of Isolator. At 10:46 Hrs on 15.07.2023, 220 kV Sarusajai-Karbi Langpi 1 line tripped. Due to tripping of this element, Karbi Langpi Generating Station of Assam Power System got separated from rest of NER Grid and subsequently collapsed due to loss of evacuation path. Power supply was extended to Karbi Langpi Generating Station of Assam Power System by charging 220 kV Sarusajai-Karbi Langpi 1 line at 10:47 Hrs on 15.07.2023	220 kV Sarusajai-Karbi Langpi 1 line
9	GD 1	Meluri & Kiphire areas of Nagaland Power System	18-Jul-23 17:13	18-Jul-23 17:54	00:41:00	16	9	0.47%	0.32%	3381	2808	Meluri & Kiphire areas of Nagaland Power System were connected with rest of grid through 132 kV Kohima-Meluri line. At 17:13 Hrs on 18.07.2023, 132 kV Kohima-Meluri line tripped. Due to tripping of this element, Meluri & Kiphire areas of Nagaland Power System got separated from rest of NER Grid and subsequently collapsed due to load generation mismatch in these areas. 132 kV Kohima-Meluri line was declared faulty at 17:54 Hrs on 18.07.2023. Power supply was extended to Meluri & Kiphire areas of Nagaland Power System by charging 132 kV Kohima-Meluri line at 11:40 Hrs on 19.07.2023.	132 kV Kohima-Meluri line
10	GD 1	Karbi Langpi Generating Station of Assam Power System	18-Jul-23 10:46	18-Jul-23 11:17	00:31:00	50	0	1.75%	0.00%	2850	2478	Karbi Langpi Generating Station of Assam Power System was connected with the rest of NER Grid through 220 kV Sarusajai-Karbi Langpi 1 line. 220 kV Sarusajai-Karbi Langpi 2 line was under state approved shutdown from 06:58 Hrs on 18.07.2023. At 10:46 Hrs on 18.07.2023, 220 kV Sarusajai-Karbi Langpi 1 line tripped. Due to tripping of this element, Karbi Langpi Generating Station of Assam Power System got separated from rest of NER Grid and subsequently collapsed due to loss of evacuation path. Power supply was extended to Karbi Langpi Generating Station of Assam Power System by charging 220 kV Sarusajai-Karbi Langpi 1 line at 11:17 Hrs on 18.07.2023	220 kV Sarusajai-Karbi Langpi 1 line
11	GD 1	Nirjuli area of Arunachal Pradesh Power System	20-Jul-23 14:03	20-Jul-23 14:57	00:54:00	0	23	0.00%	0.75%	2867	3055	Nirjuli area of Arunachal Pradesh Power System was connected with rest of NER Grid through 132 kV Lekhi - Nirjuli line. 132 kV Gohpur - Nirjuli line was under shutdown to avoid over loading of 132 kV Itanagar - Lekhi line At 14:03 Hrs on 20.07.2023, 132 kV Lekhi - Nirjuli line tripped. Due to tripping of this element, Nirjuli area of Arunachal Pradesh Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in this area. Power supply was extended to Nirjuli area of Arunachal Pradesh Power System by charging 132 kV Lekhi - Nirjuli line at 14:57 Hrs on 20.07.2023.	132kV Lekhi - Nirjuli line
12	GD 1	Karong area of Manipur Power System	22-Jul-23 09:31	22-Jul-23 10:03	00:32	0	12	0%	0%	2798	2627	Karong area of Manipur Power System was connected with the rest of NER Grid through 132 kV Imphal (MSPCL) - Karong & 132 kV Karong - Kohima lines. At 09:31 Hrs on 22.07.2023, 132 kV Imphal (MSPCL) - Karong & 132 kV Karong - Kohima lines tripped. Due to tripping of these elements, Karong area of Manipur Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in this area. Power supply was extended to Karong area of Manipur Power System by charging 132 kV Imphal (MSPCL) - Karong line at 10:03 Hrs on 22.07.2023	132 kV Imphal (MSPCL) - Karong & 132 kV Karong - Kohima lines

20.Details of Grid Events during the Month of July 2023

North-Eastern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM:SS)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(G1 1or 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
13	GD 1	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System	22-Jul-23 19:07	22-Jul-23 19:46	00:39	27	36	1%	1%	3273	3182	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System were connected with rest of NER grid through 132 kV Balipara- Tenga line. At 19:07 Hrs on 22.07.2023, 132 kV Balipara- Tenga line tripped. Due to tripping of this element, Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System got separated from rest of NER Grid and subsequently collapsed due to load generation mismatch in these areas. 132 kV Balipara- Tenga line was declared faulty at 19:46 Hrs on 22.07.2023. Power supply was extended to Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System by charging 132 kV Balipara- Tenga line at 13:51 Hrs on 23.07.2023.	132 kV Balipara- Tenga line
14	GD 1	Lunshnong area of Meghalaya Power System	23-Jul-23 10:40	23-Jul-23 12:10	01:30	0	30	0%	1%	2799	2227	Lunshnong area of Meghalaya Power System was connected with rest of NER grid through 132 kV Khleihriat-Lumshnong line. 132 kV Lumshnong-Panchgram line tripped at 08:36 Hrs on 23.07.2023 and was declared faulty. At 10:40 Hrs on 23.07.2023, 132 kV Khleihriat-Lumshnong line tripped. Due to tripping of this element, Lumshnong area of Meghalaya Power System got separated from rest of the NER Grid and subsequently collapsed due to no source available in this area. Power supply was extended to Lunshnong area of Meghalaya Power System by charging 132 kV Khleihriat-Lumshnong line at 12:10 Hrs on 23.07.2023.	132 kV Khleihriat-Lumshnong line
15	GD 1	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System	23-Jul-23 18:04	23-Jul-23 18:35	00:31	17	22	1%	1%	3178	2618	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System were connected with rest of NER grid through 132 kV Balipara- Tenga line. At 18:04 Hrs on 23.07.2023, 132 kV Balipara- Tenga line tripped. Due to tripping of this element, Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System got separated from rest of NER Grid and subsequently collapsed due to load generation mismatch in these areas. Power supply was extended to Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System by charging 132 kV Balipara- Tenga line at 18:35 Hrs on 23.07.2023.	132 kV Balipara- Tenga line
16	GD 1	Umtru and New Umtru Power Stations of Meghalaya Power System	23-Jul-23 20:38	23-Jul-23 20:46	00:08	36	0	1%	0%	3586	3142	Umtru and New Umtru Power Stations of Meghalaya Power System were connected with rest of NER grid through 132 kV Killing - EPIP II D/C, 132 kV EPIP II - Umtru D/C, 132 kV EPIP II - New Umtru, 132 kV Umtru - Umiam IV D/C, 132 kV Umtru - Umiam III D/C, 132 kV Umtru - New Umtru lines. 132 kV Umtru - Sarusajai D/C lines and 132 kV Umtru - Kahilpara D/C lines were idle charged to avoid overloading of 132 kV EPIP-II - Umtru D/C lines. At 20:38 Hrs on 23.07.2023, 132 kV Killing - EPIP II D/C, 132 kV EPIP II - Umtru D/C, 132 kV EPIP II - New Umtru, 132 kV Umtru - Umiam IV D/C, 132 kV Umtru - Umiam III D/C, 132 kV Umtru - New Umtru lines tripped. Due to tripping of the elements, Umtru and New Umtru Power Stations of Meghalaya Power System got separated from rest of the NER Grid and subsequently collapsed due to loss of evacuation path. Power supply was extended to Umtru and New Umtru Power Stations of Meghalaya Power System by charging 132 kV Umtru - New Umtru line at 20:46 Hrs on 23.07.2023.	132 kV Killing - EPIP II D/C, 132 kV EPIP II - Umtru D/C, 132 kV EPIP II - New Umtru, 132 kV Umtru - Umiam IV D/C, 132 kV Umtru - Umiam III D/C, 132 kV Umtru - New Umtru lines
17	GD 1	Dhaligaon,Barpeta, Kokrajhar, Gosaigaon, Bilasipara, Gauripur and Bornagar areas of Assam Power System	25-Jul-23 11:45	25-Jul-23 11:47	00:02	0	130	0%	5%	2785	2760	Dhaligaon, Barpeta, Kokrajhar, Gosaigaon, Bilasipara, Gauripur and Bornagar areas of Assam Power System were connected with rest of NER grid through 160 MVA 220/132 kV ICT-I & 100 MVA 220/132 kV ICT-II at BTPS. 132 kV BTPS(AS) - Dhaligaon 1 line tripped at 06:30 Hrs on 25.07.2023 and was under ESD due to isolator issue at BTPS. 132 kV Nalbari-Barpeta line was under shutdown to avoid overloading of 132 kV Rangia-Nalbari line and 132 kV Gossaigaon-Gauripur line was under shutdown to avoid overloading of 132 kV BTPS-Kokrajhar D/C lines. At 11:45 Hrs on 25.07.2023, 160 MVA 220/132 kV ICT-I & 100 MVA 220/132 kV ICT-II at BTPS tripped. Due to tripping of these elements, Dhaligaon, Barpeta, Kokrajhar, Gosaigaon, Bilasipara, Gauripur and Bornagar areas of Assam Power System got separated from rest of the NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extended to Dhaligaon, Barpeta, Kokrajhar, Gosaigaon, Bilasipara, Gauripur and Bornagar areas of Assam Power System by charging 100 MVA 220/132kV ICT-II at BTPS at 11:47 Hrs on 25.07.2023.	160 MVA 220/132 kV ICT-I & 100 MVA 220/132 kV ICT-II at BTPS.

20.Details of Grid Events during the Month of July 2023

North-Eastern Region



Sl No.	Category of Grid Event (G1 1or 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM:SS)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	Generation Loss(MW)					Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)			
18	GD 1	Zuangtui, Saitual areas and Vankal Solar Power Station of Mizoram Power System	25-Jul-23 13:13	25-Jul-23 13:42	00:29	18	20	1%	1%	2735	2797	Zuangtui, Saitual areas and Vankal Solar Power Station of Mizoram Power System were connected with rest of NER grid through 132 kV Melriat(PG) - Zuangtui line. 132 kv Serchip-Zuangtui line tripped at 13:08 Hrs on 25.07.2023. At 13:13 Hrs on 25.07.2023, 132 kv Melriat(PG) - Zuangtui line tripped. Due to tripping of this element, Zuangtui, Saitual areas and Vankal Solar Power Station of Mizoram Power System got separated from rest of the NER Grid and subsequently collapsed due to load generation mismatch in these areas. Power supply was extended to Zuangtui, Saitual areas and Vankal Solar Power Station of Mizoram Power System by charging 132 kV Melriat-Zuangtui line at 13:42 Hrs on 25.07.2023.	132 kv Melriat(PG) - Zuangtui line
19	GD 1	Pare HEP of Arunachal Pradesh Power System	25-Jul-23 14:11	25-Jul-23 14:34	00:23	120	0	4%	0%	2749	2715	Pare HEP of Arunachal Pradesh Power System were connected with rest of NER grid through 132 kv Pare - Itanagar D/C lines. 132 kv Ranganadi -Pare-1 and 132kv Pare - Lekhi lines were under shutdown for straightening of 132 kv Ranganadi-Lekhi/Nirjuli line and commissioning of 132 kv Pare-North Lakhimpur transmission line and L/LO at Nirjuli. At 14:11 Hrs on 25.07.2023, 132 kv Pare - Itanagar D/C lines tripped. Due to tripping of these elements, Pare HEP of Arunachal Pradesh Power System got separated from rest of the NER Grid and subsequently collapsed due to loss of evacuation path. 132 kv Pare - Itanagar D/C lines were declared faulty at 14:34 Hrs on 25.07.2023. Power supply was extended to Pare HEP of Arunachal Pradesh Power System by charging 132 kv Pare - Itanagar 2 line at 18:41 Hrs on 25.07.2023.	132 kv Pare - Itanagar D/C lines
20	GD 1	Itanagar area and Pare HEP of Arunachal Pradesh Power System	27-Jul-23 22:15	27-Jul-23 22:48	00:33	119	18	4%	1%	3400	3241	Itanagar area and Pare HEP of Arunachal Pradesh Power System were connected with rest of NER grid through 132 kv BNC-Itanagar D/C and 132 kv Lekhi-Itanagar lines. At 22:15 Hrs on 27.07.2023, 132 kv BNC-Itanagar D/C and 132 kv Lekhi-Itanagar lines tripped. Due to tripping of these elements, Itanagar area and Pare HEP of Arunachal Pradesh Power System got separated from rest of the NER Grid and subsequently collapsed due to load generation mismatch in these areas. Power supply was extended to Itanagar area and Pare HEP of Arunachal Pradesh Power System by charging 132 kv Lekhi-Itanagar line at 22:48 Hrs on 27.07.2023.	132 kv BNC-Itanagar D/C and 132 kv Lekhi-Itanagar lines
21	GD 1	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System	28-Jul-23 09:04	28-Jul-23 09:44	00:40	11	21	0%	1%	2918	2544	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System were connected with rest of NER grid through 132 kv Balipara- Tenga line. At 09:04 Hrs on 28.07.2023, 132 kv Balipara- Tenga line tripped. Due to tripping of this element, Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System got separated from rest of NER Grid and subsequently collapsed due to load generation mismatch in these areas. 132 kv Balipara- Tenga line was declared faulty at 09:44 Hrs on 28.07.2023. Power supply was extended to Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System by charging 132 kv Balipara- Tenga line at 15:31 Hrs on 28.07.2023.	132 kv Balipara- Tenga line
22	GD 1	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System	29-Jul-23 01:14	29-Jul-23 01:43	00:29	5	18	0%	1%	2955	2826	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System were connected with rest of NER grid through 132 kv Balipara- Tenga line. At 01:14 Hrs on 29.07.2023, 132 kv Balipara- Tenga line tripped. Due to tripping of this element, Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System got separated from rest of NER Grid and subsequently collapsed due to load generation mismatch in these areas. Power supply was extended to Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System by charging 132 kv Balipara- Tenga line at 01:43 Hrs on 29.07.2023.	132 kv Balipara- Tenga line
23	GD 1	Dharmanagar area of Tripura Power System	29-Jul-23 11:05	29-Jul-23 11:25	00:20	0	18	0%	1%	2971	2465	Dharmanagar area of Tripura Power System was connected with the rest of NER Grid through 132 kv Dharmanagar - Dullavchera line. 132 kv Dharmanagar - PK Bari Line was declared faulty since 10:43 Hrs of 29.07.2023. At 11:05 Hrs on 29.07.2023, 132 kv Dharmanagar - Dullavchera line tripped. Due to tripping of this element, Dharmanagar area of Tripura Power System was separated from the rest of NER Grid and subsequently collapsed due to no source available in this area. Power supply was extended to Dharmanagar area of Tripura Power System by charging 132 kv Dharmanagar - Dullavchera line at 11:25 Hrs on 29.07.2023.	132 kv Dharmanagar - Dullavchera line

20.Details of Grid Events during the Month of July 2023

North-Eastern Region



Sl No.	Category of Grid Event (G1 for 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM:SS)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	Generation Loss(MW)					Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)			
24	GD 1	Kohima, Meluri and Kiphire areas of Nagaland Power System	29-Jul-23 13:26	29-Jul-23 14:48	01:22	23	13	1%	0%	3067	2828	Kohima, Meluri and Kiphire areas of Nagaland Power System were connected with rest of NER grid through 132 kV Dimapur-Kohima, 132 kV Kohima-Chiephobozou, 132 kV Kohima-Karong &132 kV Meluri-Kiphire lines. At 13:26 Hrs on 29.07.2023, 132 kV Dimapur-Kohima, 132 kV Kohima-Chiephobozou, 132 kV Kohima-Karong & 132 kV Meluri-Kiphire lines tripped. Due to tripping of these elements, Kohima, Meluri and Kiphire areas of Nagaland Power System got separated from rest of NER Grid and subsequently collapsed due to load generation mismatch in these areas. Power supply was extended to Kohima, Meluri and Kiphire areas of Nagaland Power System by charging 132 kV Dimapur-Kohima line at 14:48 Hrs on 29.07.2023.	132 kV Dimapur-Kohima, 132 kV Kohima-Chiephobozou, 132 kV Kohima-Karong &132 kV Meluri-Kiphire lines
25	GD 1	Bokajan area of Assam Power System	30-Jul-23 21:24	30-Jul-23 23:18	01:54	0	10	0%	0%	3466	3223	Bokajan area of Assam Power System was connected with rest of NER grid through 132 kV Bokajan - Sarupathar line. 132 kV Bokajan - Dimapur(PG) line was under outage from 21:23 Hrs on 30.07.2023 At 21:24 Hrs on 30.07.2023, 132 kV Bokajan - Sarupathar line tripped. Due to tripping of this element, Bokajan area of Assam Power System got separated from rest of the NER Grid and subsequently collapsed due to no source available in this area. Power supply was extended to Bokajan area of Assam Power System by charging 132 kV Bokajan - Sarupathar line at 23:18 Hrs on 30.07.2023.	132 kV Bokajan - Sarupathar line
26	GD 1	Tuirial Generating Station of Mizoram Power System	30-Jul-23 17:41	30-Jul-23 17:58	00:17	22	0	1%	0%	3027	2884	Tuirial Generating Station of Mizoram Power System was connected with the rest of NER Grid through 132 kV Tuirial-Kolasib line. At 17:14 Hrs on 30.07.2023, 132 kV Tuirial-Kolasib line tripped. Due to tripping of this element, Tuirial Generating Station of Mizoram Power System got separated from rest of NER Grid and subsequently collapsed due to loss of evacuation path. Power supply was extended to Tuirial Generating Station of Mizoram Power System by charging 132 kV Tuirial-Kolasib line at 17:58 Hrs on 30.07.2023	132 kV Tuirial-Kolasib line
27	GI-II	Assam	18-Jul-23 10:31	18-Jul-23 12:00	01:29	20	0	1%	0%	2855	2635	AGBPP Unit 7 tripped at 10:31 Hrs on 18.07.2023 due to Leakage of high capacity NRV. Revision done from Block No. 49 on 18.07.2023.	AGBPP Unit 7
28	GI-I	Tripura	24-Jul-23 06:02	24-Jul-23 08:00	01:58	10	0	0%	0%	2797	2362	AGTCCPP Unit 6 tripped at 06:02 Hrs on 24.07.2023 due to Turbine bearing temperature high. Revision done from Block No. 33 on 24.07.2023.	AGTCCPP Unit 6
29	GI-II	Tripura	28-Jul-23 12:57	28-Jul-23 14:30	01:33	526	0	19%	0%	2803	2591	Palatana Module 1 and Module 2 tripped at 12:57 Hrs and 12:59 Hrs respectively on 28.07.2023 due to loss of Flame in GTG-1 and generator protection operated in GTG-2. Revision done from Block No. 59 on 28.07.2023.	Palatana Module 1 and Module 2
30	GI-I	Tripura	30-Jul-23 19:29	30-Jul-23 20:45	01:16	18	0	1%	0%	3414	3165	AGTCCPP Unit 5 tripped at 19:29 Hrs on 30.07.2023 due to Exhaust Steam Pressure High Trip. Revision done from Block No. 84 on 30.07.2023.	AGTCCPP Unit 5

21. NEW ELEMENTS COMMISSIONED* IN JULY 2023

COMMISSIONING OF NEW GENERATING UNITS

REGION	S.NO.	ENTITY	UNIT NAME / LOCATION	ADD. CAPACITY (MW)	UNIT NO.	INS. CAPACITY (MW)	DATE®
NR	1	SBSR POWER CLEANTECH ELEVEN PRIVATE LIMITED	SBSR Power Cleantech Eleven Private Limited (Solar)/Rajasthan	25	-	300	11-07-2023
WR	2	ALFANAR NETRA VAYU	Alfanar Netra Vayu(Wind)/Gujarat	41.8	I	96.8	14-07-2023
	3	ADANI WIND ENERGY KUTCHHH FOUR LTD	Adani Wind Energy Kutchh Four Ltd(Wind)/Gujarat	33.6	I	134.4	21-07-2023
	4	TORRENT SOLARGEN LIMITED	Torrent Solargen Limited (Wind)/Gujarat	13.5	I	70.2	04-07-2023
	5	TORRENT SOLARGEN LIMITED	Torrent Solargen Limited (Wind)/Gujarat	16.2	I	86.4	06-07-2023
	6	TORRENT SOLARGEN LIMITED	Torrent Solargen Limited(Wind)/ Gujarat	24.3	I	110.7	12-07-2023
	7	TORRENT SOLARGEN LIMITED	Torrent Solargen Limited(Wind)/ Gujarat	4.3	I	115	15-07-2023
SR	8	JSW RENEW TWO ENERGY LIMITED	JSW Renew Two(Wind)/Tamilnadu	21.6	-	170.1	29-07-2023

®RE (Solar & Wind generation added capacity FTC is in phases. Date mentioned is the date by which complete (Added) capacity

COMMISSIONING OF NEW ICTs

REGION	S.NO.	ENTITY	SUB-STATION	VOLTAGE (kV)	ICT NO.	CAPACITY (MVA)	DATE
NR	1	PGCIL	Patiala(PG)	400/220/33	IV	500	09-07-2023
SR	2	PGCIL	KOCHI	400/220	III	500	13-07-2023

COMMISSIONING OF NEW TRANSMISSION LINES

REGION	S.NO.	ENTITY	LINE NAME	CKT NO.	VOLTAGE (kV)	DATE
NR	1	PGCIL	400kV Bikaner_2 (PBTS)-Bikaner(PG)	I	400	01-07-2023
	2	PGCIL	400kV Bikaner_2 (PBTS)-Bikaner(PG)	II	400	08-07-2023
WR	3	INOX	220KV Bhuj-I- Dayapur Line	I	220	28-07-2023
SR	4	KPTCL	400kV YTPS - KALBURGI	II	400	05-07-2023
	5	WKTL	765 KV Kurnool-Maheshwaram	I	765	12-07-2023
	6	WKTL	766 KV Kurnool-Maheshwaram	II	765	13-07-2023
ER	7	BSPTCL	220 kV Muzaffarpur (PG) – Amnour (BSPTCL)	II	220	10-07-2023
NER	8	PGCIL	132 kV Roing-Chapakhowa	I	132	04-07-2023
	9	PGCIL	132 kV Roing-Chapakhowa	II	132	04-07-2023
	10	MUML	132 kV North Lakhimpur-Nirjuli	I	132	31-07-2023
	11	MUML & NEEPCO	132 kV Nirjuli -Pare	I	132	31-07-023

LILO OF TRANSMISSION LINES

REGION	S.NO.	ENTITY	LINE NAME (LILOed LINE)	LILO AT	VOLTAGE (kV)	DATE

COMMISSIONING OF BUS/LINE REACTORS

REGION	S.NO.	ENTITY	SUB-STATION / LINE (@SUB-STATION)	VOLTAGE (kV)	CAPACITY (MVAR)	DATE

COMMISSIONING OF HVDC/FACTS DEVICE/SYSTEM

REGION	S.NO.	NAME	SUB-STATION / CONSTITUENT	RATINGS	VOLTAGE (kV)	DATE

*Includes first time charging (FTC) as well

विद्युत विभव रूपरेखा माह - जुलाई 2023

VOLTAGE PROFILE - JULY 2023

Date of reporting: 04-Aug-23

क्र.सं. No.	क्षेत्र REGION	उपकेंद्र SUBSTATION	VOLTAGE < V(lower)* (V=380,728 kV)	V(lower) < VOLTAGE < V(upper)*	VOLTAGE > V(upper)* (V=420,800 kV)	Voltage Deviation Index (%age of time voltage is outside range)	Number of hours voltage was outside IEGC band during month	उच्चतम (कि.वॉ.) Maximum(kV)	निम्नतम(कि.वॉ.) Minimum(kV)	औसत (कि.वॉ.) Average(kV)
1	पूर्वी क्षेत्र ER	अंगुल ANGUL	0%	100%	0%	0%	0	794	750	775
2		दालिपली DARLIPALI	0%	100%	0%	0%	0	788	765	779
3		गया GAYA	0%	100%	0%	0%	0	792	742	766
4		जौरत JERAT	0%	100%	0%	0%	0	796	738	765
5		झारसुगुडा JHARSUGUDA	0%	100%	0%	0%	0	795	773	787
6		मिदनापुर MEDINIPUR	0%	100%	0%	0%	0	793	746	769
7		रांची RANCHI	0%	100%	0%	0%	0	794	762	779
8		सासाराम SASARAM	0%	100%	0%	0%	0	781	743	764
1		आगरा AGRA	0%	100%	0%	0%	0	796	742	777
2		आगरा (फतेहगढ़) AGRA (FATEHABAD)	0%	100%	0%	0%	0	794	744	770
3	उत्तरी क्षेत्र NR	अजमेर AJMER	0%	100%	0%	0%	0	800	745	778
4		अलीगढ़ ALIGARH	0%	100%	0%	0%	0	798	740	772
5		अनूपरा सी ANPARA-C	0%	100%	0%	0%	0	780	754	765
6		अनूपरा डी ANPARA-D	0%	100%	0%	0%	0	777	751	763
7		अन्ता ANTA	0%	98%	2%	2%	17	807	761	788
8		बलिया BALLIA	0%	100%	0%	0%	0	793	737	769
9		बारा BARA	0%	100%	0%	0%	0	785	754	770
10		भादला BHADLA	0%	100%	0%	0%	0	793	717	771
11		भादला-2 BHADLA-2	0%	100%	0%	0%	0	799	704	773
12		भिवानी BHIWANI	0%	100%	0%	0%	0	797	745	775
13	पश्चिमी क्षेत्र WR	बीकानेर BIKANER	0%	100%	0%	0%	0	798	714	772
14		बरेली BAREILLY	0%	100%	0%	0%	1	803	739	773
15		चित्तौड़गढ़ CHITTORGARH	0%	100%	0%	0%	3	803	759	781
16		फतेहगढ़-2 FATEHGARH-2	0%	100%	0%	0%	1	798	708	772
17		फतेहपुर FATEHPUR	0%	100%	0%	0%	0	794	730	770
18		ग्रेटर नोडा GREATER NODA	0%	100%	0%	0%	0	800	743	774
19		हावड़ा HAPUR	0%	100%	0%	0%	1	801	737	769
20		जवाहरपुर JAWAHARPUR	0%	0%	0%	0%	0	0	0	0
21		झटिकरा JHATIKARA	0%	100%	0%	0%	0	792	741	770
22		कानपुर जीआइएस KANPUR GIS	0%	100%	0%	0%	0	790	740	769
23	पश्चिमी क्षेत्र WR	खेतीरि KHETIRI	0%	100%	0%	0%	0	797	737	773
24		कोटेश्वर KOTESHWAR	0%	100%	0%	0%	0	795	739	769
25		ललितपुर LALITPUR	0%	100%	0%	0%	0	777	741	758
26		लखनऊ LUCKNOW	0%	100%	0%	0%	0	794	737	770
27		मैनपुरी MAINPURI	0%	100%	0%	0%	0	802	750	777
28		मेरठ MEERUT	0%	100%	0%	0%	0	800	740	775
29		मोगा MOGA	0%	100%	0%	0%	0	799	734	768
30		ओबरा सी OBRA C	0%	100%	0%	0%	0	778	745	759
31		उरई ORAI	0%	100%	0%	0%	2	804	749	782
32		फागी PHAGI	0%	100%	0%	0%	0	800	748	779
33	पश्चिमी क्षेत्र WR	उन्नाव UNNAO	0%	100%	0%	0%	0	789	737	762
34		वाराणसी VARANASI	0%	100%	0%	0%	0	794	749	773
1		अकोला AKOLA	0%	100%	0%	0%	0	795	751	776
2		औरंगाबाद AURANGABAD	0%	100%	0%	0%	0	798	749	778
3		भोपाल (बीडीटीसीएल) BHOPAL (BDTCL)	0%	100%	0%	0%	0	792	754	771
4		भुज BHUJ	0%	100%	0%	0%	0	794	747	774
5		भुज-2 BHUJ-II	0%	100%	0%	0%	0	789	742	769
6		बिलासपुर BILASPUR	0%	100%	0%	0%	0	790	759	774
7		बीना BINA	0%	100%	0%	0%	0	800	756	779
8		बनसकांठा BANASKANTHA	0%	100%	0%	0%	2	804	761	782
9	पश्चिमी क्षेत्र WR	चाम्पा CHAMPA	0%	92%	8%	8%	58	809	766	789
10		धुले (बीडीटीसीएल) DHULE (BDTCL)	0%	100%	0%	0%	0	794	758	777
11		धर्मजयगढ़ DHARAMJAIGARH	0%	100%	0%	0%	0	789	759	773
12		रायपुर पूलिंग RAIPUR POOLING	0%	100%	0%	0%	0	796	767	784
13		एकतुनी EKTUNI	0%	100%	0%	0%	0	797	749	775
14		गदरवाड़ा GADARWARA	0%	100%	0%	0%	2	802	768	783
15		ग्वालियर GWALIOR	0%	100%	0%	0%	0	800	747	780
16		इंदौर INDORE	0%	100%	0%	0%	0	788	754	771
17		जबलपुर JABALPUR	0%	99%	1%	1%	8	804	762	782
18		खडवा KHANDWA	0%	100%	0%	0%	0	791	756	774
19	दक्षिणी क्षेत्र SR	कोरडी KORADI	0%	100%	0%	0%	0	791	747	767
20		लकाडिया LAKADIYA	0%	100%	0%	0%	0	799	755	779
21		रायगढ़ पूलिंग RAIGARH POOLING	0%	100%	0%	0%	0	798	773	782
22		पाटने PADGHE	0%	99%	1%	1%	8	808	751	782
23		पारुली PARULI	0%	98%	2%	2%	15	807	756	787
24		पुणे PUNE	0%	98%	2%	2%	13	810	747	780
25		राजनन्दगाँव RAJNANDGAON	0%	96%	4%	4%	26	806	774	782
26		सासन SASAN	0%	100%	0%	0%	0	791	761	776
27		सतना SATNA	0%	100%	0%	0%	2	801	762	783
28		सिवनी SEONI	0%	100%	0%	0%	0	798	762	781
29	पूर्वोत्तर क्षेत्र NER	सीपत SIPAT	0%	100%	0%	0%	0	788	761	776
30		सोलापुर SOLAPUR	0%	99%	1%	1%	10	805	759	787
31		तिरोडा TIRODA	0%	100%	0%	0%	0	781	744	764
32		तामनार TAMNAR	0%	100%	0%	0%	0	796	773	784
33		वडोदरा VADODARA	0%	100%	0%	0%	0	794	751	776
34		विंध्याचल पूलिंग VINDHYACHAL PS	0%	100%	0%	0%	0	795	764	780
35		वर्धा WARDHA	0%	100%	0%	0%	3	805	753	784
36		वरौरा WARORA	0%	100%	0%	0%	1	802	760	786
1		कडपा CUDDAPAH	0%	98%	2%	2%	18	807	751	783
2		चिल्कलरीपेट CHILAKALURIPETA	0%	100%	0%	0%	0	800	735	769
3	पूर्वोत्तर क्षेत्र NER	कूर्नूल KURNOOL	0%	100%	0%	0%	0	799	745	776
4		महेश्वरम MAHESHWARAM	0%	96%	4%	4%	29	808	736	782
5		निजामाबाद NIZAMABAD	0%	88%	12%	12%	88	813	744	789
6		नेल्लूर पूलिंग NELLORE PS	0%	100%	0%	0%	0	794	748	772
7		रायचूर RAICHUR	0%	100%	0%	0%	0	800	750	780
8		श्रीकाकुलम SRIKAKULAM	0%	99%	1%	1%	8	806	745	782
9		थिरुवल्लम THIRUVALEM	0%	100%	0%	0%	1	801	743	769
10		वेंमगिरी VEMAGIRI	0%	99%	1%	1%	8	806	739	777
1		बालिपारा BALIPARA (400 kV)	0%	0%	0%	0%	0	0	0	0
2		बिस्वनाथ चरिआलि BISWANATH CHARIALI (400 kV)	0%	100%	0%	0%	0	418	396	407
3	पूर्वोत्तर क्षेत्र NER	बोंगाईगाव BONGAIGAOON (400 kV)	0%	100%	0%	0%	0	411	396	405
4		बोंगाईगाव टीपीएस BONGAIGAOON TPS (400 kV)	0%	100%	0%	0%	0	415	401	408
5		इम्फाल IMPHAL (400 kV)	0%	100%	0%	0%	0	417	394	404
6		बर्निहाट BYRNIHAT (400 kV)	0%	100%	0%	0%	0	417	402	410
7		कामेंग KAMENG (400 kV)	0%	100%	0%	0%	0	410	395	402
8		अज़रा AZARA (400 kV)	0%	100%	0%	0%	0	400	386	395
9		मिसा MISA (400 kV)	0%	100%	0%	0%	0	412	392	402
10		न्यू मरियानी NEW MARIANI (400 kV)	0%	100%	0%	0%	0	414	395	405
11		न्यू कोहिमा NEW KOHIMA (400 kV)	0%	100%	0%	0%	0	417	393	406
12		पालाटाना PALATANA (400 kV)	0%	100%	0%	0%	0	415	400	408
13	पूर्वोत्तर क्षेत्र NER	पूबकचनबारी PK BARI (400 kV)	0%	100%	0%	0%	0	414	396	404
14		रंगानादी RANGANADI (400 kV)	0%	100%	0%	0%	1	421	401	410
15		सिलचर SILCHAR (400 kV)	0%	100%	0%	0%	0	416	390	406
16		सुबान्सिरी SUBANSIRI (400 kV)	0%	0%	0%	0%	0	0	0	0
17		सूर्ययामनिनगर SURJYAMANINAGAR (400 kV)	0%	100%	0%	0%	0	413	395	404
18		थोबाल THOUBAL (400 kV)	0%	100%	0%	0%	0	410	384	397

All listed stations are 765 kV stations unless otherwise mentioned

*Percentage is calculated w.r.t. Time of one month.

23.List of Outages of Elements monitored by NLDC for the month of July-2023

S. No.	Name of Transmission Element	Type of the element	Region Involved	Owner/ Agency	Catagory	Tripping Date and Time	Brief Reason/Relay Indication	Restoration Date and Time	Outage Duration	Number of Cumulative tripping in the month
Inter Regional Elements										
1	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-I	HVDC	WR, NR	Adani Trans Ind Ltd	MO	05-Jul-2023 23:23	Voltage Regulation	06-Jul-2023 06:42	07:19:00	0
2	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-II	HVDC	WR, NR	Adani Trans Ind Ltd	MO	05-Jul-2023 23:31	Voltage Regulation	06-Jul-2023 06:37	07:06:00	0
3	+/- 800kV HVDC Champa - Kurukshetra Pole-I	HVDC	WR, NR	POWERGRID	MO	02-Jul-2023 14:20	Voltage Regulation	02-Jul-2023 20:36	06:16:00	0
4	+/- 800kV HVDC Champa - Kurukshetra Pole-II	HVDC	WR, NR	POWERGRID	MO	02-Jul-2023 14:20	Voltage Regulation	02-Jul-2023 21:01	6:41	0
5	+/- 500kV HVDC Talcher - Kolar Pole-I	HVDC	ER, SR	POWERGRID	TL	03-Jul-2023 03:28	B-Phase convertor transformer Bushing blasted at HVDC Kolar	08-Jul-2023 00:15	116:47:00	1
6	+/- 205kV HVDC GAJIWAKA Pole-I	HVDC	SR, ER	POWERGRID	SE	04-Jul-2023 16:32	for attending the Thyristor redundancy lost alarm	04-Jul-2023 21:23	04:51:00	0
7	+/- 205kV HVDC GAJIWAKA Pole-II	HVDC	SR, ER	POWERGRID	TL	04-Jul-2023 17:36	Communication fail in between Pole control (PCP) to AC Control & protection (ACP) Servers.	04-Jul-2023 21:09	03:33:00	1
8	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-II	HVDC	WR, NR	Adani Trans Ind Ltd	MO	06-Jul-2023 21:21	To control NR Import	07-Jul-2023 07:00	09:39:00	0
9	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-I	HVDC	WR, NR	Adani Trans Ind Ltd	MO	06-Jul-2023 21:27	To contro NR Import	07-Jul-2023 07:25	09:58:00	0
10	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-I	HVDC	WR, NR	Adani Trans Ind Ltd	MO	11-Jul-2023 19:44	to change the power flow direction	12-Jul-2023 01:28	05:44:00	0
11	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-II	HVDC	WR, NR	Adani Trans Ind Ltd	MO	11-Jul-2023 20:04	to change the power flow direction	12-Jul-2023 01:33	05:29:00	0
12	800kV HVDC AGRA-BNC Pole-I	HVDC	NR, NER	POWERGRID	SO	13-Jul-2023 16:49	broken jumper at location no-3615	13-Jul-2023 21:49	05:00:00	0
13	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-II	HVDC	WR, NR	Adani Trans Ind Ltd	TL	13-Jul-2023 16:50	Tripped on DC line fault	13-Jul-2023 21:19	04:29:00	1
14	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-II	HVDC	WR, NR	Adani Trans Ind Ltd	SO	14-Jul-2023 01:06	For preparing poles for reversal due to high NR Import in morning hours	14-Jul-2023 06:40	05:34:00	1
15	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-I	HVDC	WR, NR	Adani Trans Ind Ltd	SO	14-Jul-2023 01:14	For preparing poles for reversal due to high NR Import in morning hours	14-Jul-2023 06:48	05:34:00	0
16	+/- 800kV HVDC Champa - Kurukshetra Pole-II	HVDC	WR, NR	POWERGRID	TS	14-Jul-2023 19:10	Mal function of Control Card which in turn triggered Pole-2 DC Differential Protection	14-Jul-2023 22:11	03:01:00	1
17	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-II	HVDC	WR, NR	Adani Trans Ind Ltd	TL	15-Jul-2023 07:37	Tripped on DC line fault	15-Jul-2023 19:44	12:07:00	2
18	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-I	HVDC	WR, NR	Adani Trans Ind Ltd	MO	16-Jul-2023 11:28	To control loading of 400kV Dehgam-Jhanor	17-Jul-2023 14:51	27:23:00	0
19	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-II	HVDC	WR, NR	Adani Trans Ind Ltd	MO	16-Jul-2023 11:33	To control loading of 400kV Dehgam-Jhanor	17-Jul-2023 14:56	27:23:00	2
20	+/- 205kV HVDC Bhadravati Pole-I	HVDC	WR, SR	POWERGRID	TS	19-Jul-2023 12:31	Thyristor redundancy exceeded"alarm	19-Jul-2023 19:10	06:39:00	1
21	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-II	HVDC	WR, NR	Adani Trans Ind Ltd	TL	19-Jul-2023 16:54	DC line fault	19-Jul-2023 22:03	05:09:00	3
22	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-II	HVDC	WR, NR	Adani Trans Ind Ltd	MO	21-Jul-2023 18:27	Power Regulation	24-Jul-2023 16:45	70:18:00	3
23	+/- 500kV HVDC APL Mundra - Mohindargarh Pole-I	HVDC	WR, NR	Adani Trans Ind Ltd	MO	21-Jul-2023 18:32	Power Regulation	24-Jul-2023 16:35	70:03:00	0
24	+/- 205kV HVDC Bhadravati Pole-I	HVDC	WR, SR	POWERGRID	SE	22-Jul-2023 10:41	ESD To replace 2 Nos. of thyristor/gate card	22-Jul-2023 21:28	10:47:00	1
25	+/- 800kV HVDC Raigarh - Pugalur Pole-I	HVDC	WR, SR	POWERGRID	MO	25-Jul-2023 11:03	power regulation (to control loading of 400kv Kolhapur-Kolhapur)	25-Jul-2023 19:15	08:12:00	0
26	+/- 800kV HVDC Raigarh - Pugalur Pole-II	HVDC	WR, SR	POWERGRID	MO	25-Jul-2023 11:13	power regulation (to control loading of 400kv Kolhapur-Kolhapur)	25-Jul-2023 19:34	08:21:00	0
27	+/- 800kV HVDC Raigarh - Pugalur Pole-III	HVDC	WR, SR	POWERGRID	MO	25-Jul-2023 11:15	Power Regulation	25-Jul-2023 12:26	01:11:00	0
28	+/- 800kV HVDC Raigarh - Pugalur Pole-IV	HVDC	WR, SR	POWERGRID	MO	25-Jul-2023 11:22	Power Regulation	25-Jul-2023 12:31	01:09:00	0
29	+/- 205kV HVDC Bhadravati Pole-II	HVDC	WR, SR	POWERGRID	TS	25-Jul-2023 17:10	Tripped at Bhadravati end on Yph to earth fault	25-Jul-2023 23:28	06:18:00	1
30	+/- 205kV HVDC Bhadravati Pole-I	HVDC	WR, SR	POWERGRID	TS	25-Jul-2023 19:10	Tripped at Bhadravati end on Yph to earth fault	25-Jul-2023 23:09	03:59:00	2
31	+/- 800kV HVDC Raigarh - Pugalur Pole-I	HVDC	WR, SR	POWERGRID	MO	26-Jul-2023 10:46	Power Regulation	27-Jul-2023 01:38	14:52:00	0
32	+/- 800kV HVDC Raigarh - Pugalur Pole-II	HVDC	WR, SR	POWERGRID	MO	26-Jul-2023 10:51	Power Regulation	27-Jul-2023 01:38	14:47:00	0
33	+/- 800kV HVDC Raigarh - Pugalur Pole-III	HVDC	WR, SR	POWERGRID	MO	26-Jul-2023 10:52	Power Regulation	26-Jul-2023 12:08	01:16:00	0
34	+/- 800kV HVDC Raigarh - Pugalur Pole-IV	HVDC	WR, SR	POWERGRID	MO	26-Jul-2023 10:57	Power Regulation	26-Jul-2023 12:06	01:09:00	0
35	+/- 800kV HVDC Raigarh - Pugalur Pole-III	HVDC	WR, SR	POWERGRID	MO	27-Jul-2023 01:04	Due to power regulation by nldc in view of kotra ICTs	27-Jul-2023 12:26	11:22:00	0
36	+/- 800kV HVDC Raigarh - Pugalur Pole-IV	HVDC	WR, SR	POWERGRID	MO	27-Jul-2023 01:10	Power Regulation	27-Jul-2023 12:16	11:06:00	0
37	+/- 800kV HVDC Raigarh - Pugalur Pole-II	HVDC	WR, SR	POWERGRID	MO	27-Jul-2023 11:22	To control line loading of 400kV Kolhapur-Kolhapur Ms-line	27-Jul-2023 23:43	12:21:00	0
38	+/- 800kV HVDC Raigarh - Pugalur Pole-I	HVDC	WR, SR	POWERGRID	MO	27-Jul-2023 11:41	To control line loading of 400kV Kolhapur-Kolhapur Ms-line	27-Jul-2023 23:42	12:01:00	0

23.List of Outages of Elements monitored by NLDC for the month of July-2023

S. No.	Name of Transmission Element	Type of the element	Region Involved	Owner/ Agency	Catagory	Tripping Date and Time	Brief Reason/Relay Indication	Restoration Date and Time	Outage Duration	Number of Cumulative tripping in the month
39	+/- 800kV HVDC Raigarh - Pugalur Pole-III	HVDC	WR, SR	POWERGRID	MO	27-Jul-2023 23:07	Power Regulation	31-Jul-2023 08:19	81:12:00	0
40	+/- 800kV HVDC Raigarh - Pugalur Pole-IV	HVDC	WR, SR	POWERGRID	MO	27-Jul-2023 23:13	Power Regulation	31-Jul-2023 08:24	81:11:00	0
41	+/- 800kV HVDC Champa - Kurukshetra Pole-I	HVDC	WR, NR	POWERGRID	MO	30-Jul-2023 12:51	Voltage Regulation	30-Jul-2023 19:11	06:20:00	0
42	+/- 800kV HVDC Champa - Kurukshetra Pole-II	HVDC	WR, NR	POWERGRID	MO	30-Jul-2023 12:51	Voltage Regulation	30-Jul-2023 19:23	06:32:00	1
43	400KV-BIHARSARIFF(PG)-BALIA-1	Line	ER,NR	ER-I	SO	02-Jul-2023 09:51	CB PIR Overhauling work	02-Jul-2023 18:45	08:54:00	0
44	400KV-BIHARSARIFF(PG)-BALIA-2	Line	ER,NR	ER-I	SO	04-Jul-2023 09:36	overhauling of CB PIR	04-Jul-2023 20:24	10:48:00	0
45	400KV-BIHARSARIFF(PG)-BALIA-2	Line	ER,NR	ER-I	SO	05-Jul-2023 09:25	overhauling of CB PIR	05-Jul-2023 20:00	10:35:00	0
46	220KV-BUDHIPADAR-RAIGARH-1	Line	ER,WR	GRIDCO	SO	05-Jul-2023 10:01	Trimming of tree branches at location no.7-19 and 26-31 of the line	05-Jul-2023 19:28	09:27:00	0
47	220KV-BUDHIPADAR-KORBA-1	Line	ER,WR	GRIDCO	SO	07-Jul-2023 07:54	Trimming of tree branches at location no.7-19 and 26-31 of the line	07-Jul-2023 19:30	11:36:00	0
48	765KV-GAYA-BALIA-1	Line	ER,NR	ER-I	SO	24-Jul-2023 09:48	Rectification of Shutdown Defects	24-Jul-2023 19:58	10:10:00	0
49	132kV Rihand - Garwah Ckt-I	Line	ER, NR	UPPCL/JSEB	TS	04-Jul-2023 13:35	Power failed from Rihand's end	04-Jul-2023 15:01	01:26:00	1
50	132kV Sahupuri - Karmanasa Ckt-I	Line	ER, NR	UPPCL/BSPHCL	TL	11-Jul-2023 21:26	Phase to earth fault B-N Gen trip Z1,B phase GND PU I3=4.81 KA Dist 13.1KM 86 A ,86 B	12-Jul-2023 23:07	25:41:00	1
51	132kV Chandauli - Karmanasa Ckt-I	Line	ER, NR	UPPCL/BSPHCL	TL	11-Jul-2023 21:30	Phase to earth fault R-N line tripped due to heavy rain. SOTF, VCN 80.62kv, VBN 23.74 kv, VAN 77.37 kv, IC 5.286A , IB 1.669 KA, IA 4.254A	12-Jul-2023 17:20	19:50:00	1
52	132kV Rihand - Nagar Untari Ckt-I	Line	ER, NR	UPPCL/JSEB	TS	12-Jul-2023 11:54	132 Kv Rihand end flag is Gen trip, Y Phase B Phase Z-1, ir-23.98a iy-12.74a ib-133.60a in-110.38a	12-Jul-2023 14:12	02:18:00	1
53	400kV Biharshariff - Varanasi Ckt-I	Line	ER, NR	POWERGRID	TL	01-Jul-2023 02:21	DUE TO Y-N FAULT, BSF (AFAS) M1: 122.595 KM, 3.319 KA	01-Jul-2023 03:36	01:15:00	1
54	400KV-BIHARSARIFF(PG)-BALIA-1	Line	ER, NR	ER-I	SO	01-Jul-2023 10:54	OVERHAULLING OF CB PIR (FIXING OF CB PIR FOR O/H PURPOSE)	02-Jul-2023 00:09	13:15:00	0
55	220kV New Karamnasa - Sahupuri	Line	ER, NR	UPPCL/BSPHCL	SO	03-Jul-2023 16:20	jumpfer of R phase and B phase at LILI section at Tower Location No.5 found demange	03-Jul-2023 21:17	04:57:00	0
56	400kV Biharshariff - Varanasi Ckt-II	Line	ER, NR	POWERGRID	TL	07-Jul-2023 16:29	DC earth fault	07-Jul-2023 18:04	01:35:00	1
57	132kV Chandauli - Karmanasa Ckt-I	Line	ER, NR	UPPCL/BSPHCL	TL	12-Jul-2023 17:20	line tripped due to heavy rain on distance relay on 11.07.23 , Charging attempt failed at 14:00 hrs, E/S/D taken to work on breaking of Y phase insulator @	12-Jul-2023 20:47	03:27:00	2
58	132kV Rihand - Nagar Untari Ckt-I	Line	ER, NR	UPPCL/JSEB	SO	13-Jul-2023 12:22	due to line side B phase conductor of said feeder is partially broken	13-Jul-2023 16:53	04:31:00	1
59	132kV Rihand - Garwah Ckt-I	Line	ER, NR	UPPCL/JSEB	TL	17-Jul-2023 13:36	bus bar protection operated	17-Jul-2023 16:15	02:39:00	2
60	132kV Rihand - Nagar Untari Ckt-I	Line	ER, NR	UPPCL/JSEB	TS	17-Jul-2023 13:36	bus bar protection operated	17-Jul-2023 16:27	02:51:00	2
61	132kV Rihand - Nagar Untari Ckt-I	Line	ER, NR	UPPCL/JSEB	SE	26-Jul-2023 01:30	hand tripped for safety reason	26-Jul-2023 03:20	01:50:00	2
62	132kV Rihand - Garwah Ckt-I	Line	ER, NR	UPPCL/JSEB	SE	26-Jul-2023 01:30	hand tripped due to safety reason	26-Jul-2023 03:26	01:56:00	2
63	132kV Rihand - Nagar Untari Ckt-I	Line	ER, NR	UPPCL/JSEB	TL	28-Jul-2023 11:57	Y Phase, B Phase fault, L1: 12.18A,L2: 1090.27A, L3: 1081.3A, N: 119.34A	28-Jul-2023 16:25	04:28:00	3
64	132kV Rihand - Garwah Ckt-I	Line	ER, NR	UPPCL/JSEB	TL	29-Jul-2023 12:06	Zone-1 Ia= 0.51 kA Ib= .01 kA Ic= .02 kA Distance = 53.8 km	29-Jul-2023 15:08	03:02:00	3
65	400KV-JEYPORE-GAIJUWAKA-1	Line	ER, SR	PG (ODISHA PROJECT)	SO	20-Jul-2023 10:47	Retrofitting of Gang operated isolator to individually operated isolators and Line maintenance work	20-Jul-2023 16:32	05:45:00	0
66	400KV-JEYPORE-GAIJUWAKA-2	Line	ER, SR	PG (ODISHA PROJECT)	SO	21-Jul-2023 10:59	Retrofitting of Gang operated isolator to individually operated isolators and Line maintenance work	21-Jul-2023 15:33	04:34:00	0
67	765 kV-JHARSUGUDA-RAIPUR PS (DURG)-II	Line	ER, WR		TL	06-Jul-2023 02:05	Y-phase to earth fault	06-Jul-2023 04:44	02:39:00	1
68	765 kV-JHARSUGUDA-RAIPUR PS (DURG)-I	Line	ER, WR	ER-I	MO	21-Jun-2023 19:17	Voltage regulation	06-Jul-2023 03:56	344:39:00	0
69	220KV-MALANPUR-AURIYA-1	Line	WR,NR	MP	SC	22-Jul-2023 12:41	for modification/shifting work at Loc. No. 40 to 51 due to Construction of Bundelkhand expressway.	01-Aug-2023 00:00	227:19:00	0
70	765kV Dharmjaygarh - Jharsuguda Ckt-II	Line	WR, ER	POWERGRID	TL	04-Jul-2023 17:17	Tripped on B-E fault Dharamjaygarh - 139.37km, 5.45kA Jharsuguda - 12.12km, 25.6kA	04-Jul-2023 19:42	02:25:00	1
71	220kV Raigarh - Budhipadar Ckt-I	Line	WR, ER	POWERGRID/GRIDCO	TL	18-Jul-2023 05:56	At Budhipadar End Z-1, Rph-E, 2.6 kA, 65.5 KM At Raigarh End Z-1, 13.07 kA, 6.26 KM	18-Jul-2023 08:46	02:50:00	1
72	220kV Raigarh - Budhipadar Ckt-I	Line	WR, ER	POWERGRID/GRIDCO	TL	18-Jul-2023 08:46	At Raigarh End Z-1, 13.07 kA, 6.26 KM At Budhipadar End Z-1, Rph-E, 2.6 kA, 65.5 KM	18-Jul-2023 19:52	11:06:00	2
73	765kV Durg - Jharsuguda Ckt-II	Line	WR, ER	POWERGRID	MO	23-Jul-2023 15:50	Voltage Regulation	01-Aug-2023 00:00	200:10:00	0
74	220kV Korba(East) - Budhipadar Ckt-III	Line	WR, ER	CSPTCL/GRIDCO	TL	28-Jul-2023 12:39	breaker electrically fault problem	29-Jul-2023 14:04	25:25:00	1
75	220kV Raigarh - Budhipadar Ckt-I	Line	WR, ER	POWERGRID/GRIDCO	TL	29-Jul-2023 23:03	B Phase to Ground Fault , FAULT CURRENT: 3.011 kA LOCATION: 55.796km (66.420%)	30-Jul-2023 00:01	00:58:00	3

23.List of Outages of Elements monitored by NLDC for the month of July-2023

S. No.	Name of Transmission Element	Type of the element	Region Involved	Owner/ Agency	Catagory	Tripping Date and Time	Brief Reason/Relay Indication	Restoration Date and Time	Outage Duration	Number of Cumulative tripping in the month
76	220kV Mehgaon - Auraiya Ckt-I	Line	WR, NR	MPPTCL/UPPCL	TL	05-Jul-2023 06:03	DISTANCE PROTECTION, IR = 169 A, IY = 228.9 A FAULT CURRENT IB = 2.7 KA, IC = , 58.46 KM FAULT DISTANCE.	05-Jul-2023 07:50	01:47:00	1
77	220kV Mehgaon - Auraiya Ckt-I	Line	WR, NR	MPPTCL/UPPCL	TL	06-Jul-2023 18:52	R-N, ZONE #1, DISTANCE 30.83 KM, FAULT CURRENT I R= 3.329 K A, ZONE 1.	06-Jul-2023 20:32	01:40:00	2
78	132kV Sheopur - Khandar Ckt-I	Line	WR, NR	MPPTCL/RRVP NL	TL	06-Jul-2023 19:19	Bph-Earth fault	06-Jul-2023 22:02	02:43:00	1
79	765kV Gwalior - Agra Ckt-I	Line	WR, NR	POWERGRID	SE	19-Jul-2023 16:34	for attending missing hardware cotter nut and bolts at loc no 206 & 216	19-Jul-2023 21:29	04:55:00	0
80	220KV-MEHGAON-AURIYA-1	Line	WR, NR	MP	SC	22-Jul-2023 12:14	for modification/shifting work at Loc. No. 40 to 51 due to Construction of Bundelkhand expressway.	01-Aug-2023 00:00	227:46:00	0
81	400 KV Vindhyachal (PS)- Rihand ckt-1	Line	WR, NR	POWERGRID WR	SE	29-Jul-2023 12:01	to attend broken jumper subconductor at loc no. 50 middle phase	29-Jul-2023 18:39	06:38:00	0
82	220kV Mudshingi - Chikkodi Ckt-I	Line	WR, SR	MSETCL/KPTCL	SO	01-Jul-2023 11:55	For safety purpose (to avoid heavy induction due to rainy weather during planned activity work of 220 kV Talandge - Mudshingi ckt 1)	01-Jul-2023 19:14	07:19:00	0
83	765kV Solapur - Raichur Ckt-II	Line	WR, SR	POWERGRID	TL	04-Jul-2023 21:22	A/R successful at Solapur end, L/R tripped due to ferro-resonance. Solapur :R-E, FD 180.27kM FC-4.99kA Raichur-R-E, FD 27.72kA FC- 10.48kA	04-Jul-2023 22:33	01:11:00	1
84	400kV Bhadravati - Ramagundam Ckt-I	Line	WR, SR	POWERGRID	TL	25-Jul-2023 19:10	Tripped at Bhadravati end on Yph to earth fault	26-Jul-2023 01:50	06:40:00	1
85	400kV Bhadravati - Ramagundam Ckt-II	Line	WR, SR	POWERGRID	TL	25-Jul-2023 19:10	Tripped at Bhadravati end on Yph to earth fault	25-Jul-2023 21:50	02:40:00	1
86	765kV Wardha - Nizamabad Ckt-II	Line	WR, SR	POWERGRID	TL	27-Jul-2023 05:24	Y Phase to Ground Fault	27-Jul-2023 06:57	01:33:00	1
87	220kV Xeldom - Ambewadi Ckt-I	Line	WR, SR	GOA/KPTCL	TS	27-Jul-2023 09:41	220/110 kV Bus Dead occurred at 220 kV Ambewad	27-Jul-2023 16:36	06:55:00	1
88	765kV Solapur - Raichur Ckt-II	Line	WR, SR	POWERGRID	TS	28-Jul-2023 11:43	Tripped from Raichur end due to DT receipt.	28-Jul-2023 13:03	01:20:00	2

Transnational Elements

1	400kV Binaguri - Malbase Ckt-I	Line	ER, BHUTAN	POWERGRID/T HEP	TL	07-Jul-2023 23:24	Malbase End:Rph-N, Z-1 ,FD-21.25 km,Fault current=6.877kA. Binaguri End: Rph-N,A/R successful FD-112.58km FC-4.032kA.	07-Jul-2023 23:43	00:19:00	1
2	220kV Birpara - Malbase Ckt-I	Line	ER, BHUTAN	POWERGRID/B HUTAN	TL	21-Jul-2023 11:23	Birpara End:Rph-N,Distance 9.46km,Fault current-10.82kA. Malbase End:Rph-N,Z-2,Fault current:2.05kA.	21-Jul-2023 15:37	04:14:00	1
3	220kV Birpara - Malbase Ckt-I	Line	ER, BHUTAN	POWERGRID/B HUTAN	SO	12-Jul-2023 12:00	load at Malbase is internally fed and avoid the back feed from Birpara.	12-Jul-2023 12:25	00:25:00	1
4	400kV Binaguri - Malbase Ckt-I	Line	ER, BHUTAN	POWERGRID/T HEP	TL	13-Jul-2023 03:31	Relay indication, R-N fault, Ir-4.32ka, 104KM from binaguri, A/R successful	13-Jul-2023 04:23	00:52:00	2
5	400kV Binaguri -Tala Ckt-I	Line	ER, BHUTAN	POWERGRID/T HEP	SO	18-Jul-2023 09:09	TO REPLACE THE ONE STRING DECAPPED INSULATOR OF R-PHASE CONDUCTOR	19-Jul-2023 18:53	33:44:00	0
6	400kV Binaguri - Malbase Ckt-I	Line	ER, BHUTAN	POWERGRID/T HEP	TL	21-Jul-2023 04:25	Malbase end relay indication Zone 1 operated,distance-77.12 km,y phase to ground fault.IY-4.599 Kamps. IN-4.889 Kamps	21-Jul-2023 04:46	00:21:00	3
7	220kV Birpara - Malbase Ckt-I	Line	ER, BHUTAN	POWERGRID/B HUTAN	TL	21-Jul-2023 04:42	Relay indication-Rph fault, Zone-2, Ir-2.05ka	21-Jul-2023 05:11	00:29:00	2
8	400kV Binaguri - Malbase Ckt-I	Line	ER, BHUTAN	POWERGRID/T HEP	SO	21-Jul-2023 16:26	for replacement of insulator strings	21-Jul-2023 21:48	05:22:00	3
9	400kV Muzaffarpur - Dhalkebar Ckt-I	Line	ER, NEPAL	POWERGRID/N EPAL	TL	13-Jul-2023 15:38	DEF PROTECTION	13-Jul-2023 16:54	01:16:00	1
10	132kV Kataiya - Kusaha	Line	ER, NEPAL	BSEB/NEPAL	SO	15-Jul-2023 09:40	Maintenance of OPGW cable	17-Jul-2023 20:20	58:40:00	0
11	132kV Kataiya - Kusaha	Line	ER, NEPAL	BSEB/NEPAL	SO	25-Jul-2023 13:20	for jumper connection	25-Jul-2023 18:20	05:00:00	0
12	132kV Salakati - Gelephu Ckt-I	Line	NER, BHUTAN	POWERGRID/B HUTAN	TL	13-Jul-2023 00:00	Gelephu End: Yph-E,Distance protection operated, Zone-1. Salakati End: Yph-E,Distance protection operated, Zone-1,FD: 32.98 KM, Auto Recloser Successful.	18-Jul-2023 16:28	136:28:00	1
13	132kV Salakati - Gelephu Ckt-I	Line	NER, BHUTAN	POWERGRID/B HUTAN	SO	11-Jul-2023 10:22	testing of CB from the SCADA system	11-Jul-2023 13:38	03:16:00	1
14	132kV Salakati - Gelephu Ckt-I	Line	NER, BHUTAN	POWERGRID/B HUTAN	SO	25-Jul-2023 15:56	chance of the line being tripped due to CT failure. The CT was found making noise at the site.	25-Jul-2023 17:45	01:49:00	1

Northern Region Elements

1	+/- 500kV HVDC BALIA - BHIWADI Pole-II	HVDC	NR	POWERGRID	TS	04-Jul-2023 13:53	External protection Trip	04-Jul-2023 14:54	01:01:00	1
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23.List of Outages of Elements monitored by NLDC for the month of July-2023

S. No.	Name of Transmission Element	Type of the element	Region Involved	Owner/ Agency	Catagory	Tripping Date and Time	Brief Reason/Relay Indication	Restoration Date and Time	Outage Duration	Number of Cumulative tripping in the month
2	+/- 500kV HVDC BALIA - BHIWADI Pole-II	HVDC	NR	POWERGRID	SO	10-Jul-2023 13:30	To replace damaged BPI of convertor transformer.	10-Jul-2023 18:22	04:52:00	1
3	+/- 500kV HVDC RIHAND - DADRI Pole-I	HVDC	NR	POWERGRID	SO	26-Jul-2023 16:11	For removing the hanging bird guard at Tower no. 1545.	26-Jul-2023 18:39	02:28:00	0
4	+/- 500kV HVDC RIHAND - DADRI Pole-I	HVDC	NR	POWERGRID	TS	03-Jul-2023 16:41	Rihand :- computer protection sinagl fault	03-Jul-2023 18:11	01:30:00	1
5	+/- 500kV HVDC RIHAND - DADRI Pole-I	HVDC	NR	POWERGRID	SE	11-Jul-2023 08:11	to attend oil leakage from the Line Side DCCT (P1.WL1.T1) of Pole 1	11-Jul-2023 15:07	06:56:00	1
6	240 MVAR BUS REACTOR NO 1 AT 765KV AGRA FATEHBAD	Bus Reactor	NR	STU	SO	17-Jul-2023 09:28	* For routine Maintenance and Testing of Relays and reactor unit at 765kV Substation Fatehabad Agra	17-Jul-2023 16:28	07:00:00	0
7	240 MVAR Line reactor at Rampur of Rampur - Ghatampur	Line Reactor	NR	UPPCL	TS	07-Jul-2023 10:29	RESTRICTED EARTH FAULT	07-Jul-2023 13:30	03:01:00	1
8	240 MVAR Line reactor at Rampur of Rampur - Ghatampur	Line Reactor	NR	UPPCL	TS	07-Jul-2023 13:30	differential protection due the loose CT connection.	07-Jul-2023 18:54	05:24:00	2
9	240 MVAR NON-SWITCHABLE LR ON ANTA-PHAGI (RS) Ckt-II	Line Reactor	NR	SLDC	SO	23-Jul-2023 15:04	Quarterly maintenance work	24-Jul-2023 02:48	11:44:00	0
10	240MVAR 765kV Bus Reactor-I at Agra Fatehabad	Bus Reactor	NR	UPPTCL	TS	17-Jul-2023 16:39	Back-up Impedance protection operated in Y-phase.	18-Jul-2023 00:18	07:39:00	1
11	240MVAR 765kV Bus Reactor-I at Moga	Bus Reactor	NR	POWERGRID	TS	05-Jul-2023 23:43	B-ph String Insulator of jack-bus of 765 kV Bus Reactor-1 found broken	06-Jul-2023 05:37	05:54:00	1
12	240MVAR 765kV Bus Reactor-II at Moga	Bus Reactor	NR	POWERGRID	TS	05-Jul-2023 23:48	For safety purpose work on jack-bus of 765 kV Bus reactor-1	06-Jul-2023 05:50	06:02:00	1
13	240MVAR LR(F) of 765kV Anta - Phagi Ckt-II at Anta	Line Reactor	NR	Rajasthan/RRV PNL	TS	24-Jul-2023 04:05	reactor on Anta End has been tripped	26-Jul-2023 19:03	62:58:00	1
14	240MVAR LR(S) of 765kV Bikaner Ckt-I at Khetri	Line Reactor	NR	BKTL	SE	25-Jul-2023 06:38	Low oil level alarm in R-ph unit of LR	25-Jul-2023 14:17	07:39:00	0
15	240MVAR LR(S) of 765kV Khetri Ckt-II at Jhatikara	Line Reactor	NR	POWERGRID NR	SE	22-Jul-2023 17:14	Hot spot is observed in 710 LR Khetri-2 line reactor	22-Jul-2023 18:33	01:19:00	0
16	765 KV AGRA FATEHBAD(UP)-LALITPUR(LPG) (UP) CKT-2	Line	NR	SLDC	SO	23-Jul-2023 09:45	for charging of 80MVar Spare reactor of 765KV Fatehabad -Lalitpur CKT-II (R-phase unit) at 765KV S/s Fatehabad agra.	23-Jul-2023 13:09	03:24:00	0
17	765 KV ANTA-PHAGI (RS) CKT-2	Line	NR	SLDC	SO	23-Jul-2023 09:48	Quarterly Maintenance work of line bay equipments i.e CVT,WT,LA & line isolator & Maintenance of 3*80 MVAR line reactors at Phagi	24-Jul-2023 04:05	18:17:00	0
18	765 KV BHADLA_2 (PG)-FATEHGARH_II(PG) (PFTL) CKT-1	Line	NR	POWERGRID	SO	24-Jul-2023 17:12	For ladder logic and interlock checking for spare reactor commissioning	25-Jul-2023 06:32	13:20:00	0
19	765 KV BHADLA_2 (PG)-FATEHGARH_II(PG) (PFTL) CKT-2	Line	NR	POWERGRID	SO	17-Jul-2023 17:25	For ladder logic and interlock checking for spare reactor commissioning	18-Jul-2023 05:31	12:06:00	0
20	765 KV BHADLA_2 (PG)-FATEHGARH_II(PG) (PFTL) CKT-2	Line	NR	POWERGRID	SO	31-Jul-2023 17:15	For ladder logic and interlock checking for spare reactor commissioning	01-Aug-2023 00:00	06:45:00	0
21	765/400 KV 1500 MVA ICT 1 AT ALIGARH(PG)	ICT	NR	POWERGRID	SO	11-Jul-2023 11:34	For 765 kV Aligarh-Sikar-II Line 1 Protection testing (Bus PT wiring/Tie LBB testing with DT Ckt/Interlock testing/ A/R testing).	11-Jul-2023 22:21	10:47:00	0
22	765/400 KV 1500 MVA ICT 1 AT BHADLA_2 (PG)	ICT	NR	POWERGRID	SO	03-Jul-2023 20:33	for AMP works	04-Jul-2023 04:55	08:22:00	0
23	765/400 KV 1500 MVA ICT 1 AT BIKANER(PG)	ICT	NR	POWERGRID	SO	25-Jul-2023 17:32	for AMP works	25-Jul-2023 23:43	06:11:00	0
24	765/400 KV 1500 MVA ICT 2 AT AGRA FATEHBAD(UP)	ICT	NR	STU	SO	03-Jul-2023 10:15	* For Charging of 500MVA ICT-II R Phase unit at 765kV Substation Fatehabad Agra. At present Spare Phase unit is charged as R	03-Jul-2023 16:28	06:13:00	0
25	765/400 KV 1500 MVA ICT 2 AT ALIGARH(PG)	ICT	NR	POWERGRID	SO	08-Jul-2023 10:26	for 765 kV Aligarh-Sikar-II Line 2 Protection testing (Bus PT wiring/Tie LBB testing with DT Ckt/Interlock testing/ A/R testing).	09-Jul-2023 17:17	30:51:00	0
26	765/400 KV 1500 MVA ICT 3 AT PHAGI(RS)	ICT	NR	SLDC	SO	08-Jul-2023 13:07	For proper alignment of LV side HCB isolators through M/s TBEA	09-Jul-2023 22:07	33:00:00	0
27	765/400kV 1000MVA ICT-I at Unnao	ICT	NR	UPPCL	SE	02-Jul-2023 19:00	To attend hot spot . To shift the ICT-1 from 400kV Bus-1 to Bus-2, due to heavy sparking on Bus-1 B-phase isolator (400-6-89A)	02-Jul-2023 19:38	00:38:00	0
28	765/400kV 1000MVA ICT-III at Bhiwani	ICT	NR	POWERGRID	TS	03-Jul-2023 14:50	ICT tripped on PRV	04-Jul-2023 18:30	27:40:00	1
29	765/400kV 1000MVA ICT-III at Bhiwani	ICT	NR	POWERGRID	TS	10-Jul-2023 14:14	Buchholz alarm	13-Jul-2023 22:17	80:03:00	2
30	765/400kV 1500MVA ICT-I at Ajmer	ICT	NR	POWERGRID	SE	15-Jul-2023 15:40	to attend Intense Hotspot observed at the connector of 765/400 KV ICT-1 B-phase Knee-type isolator(701-89TB).	15-Jul-2023 19:30	03:50:00	0

23.List of Outages of Elements monitored by NLDC for the month of July-2023

S. No.	Name of Transmission Element	Type of the element	Region Involved	Owner/ Agency	Catagory	Tripping Date and Time	Brief Reason/Relay Indication	Restoration Date and Time	Outage Duration	Number of Cumulative tripping in the month
31	765/400kV 1500MVA ICT-I at Jhatikara	ICT	NR	POWERGRID	SO	28-Jul-2023 17:36	replacement of Terminal Pad clamp of 709-89A isolator	28-Jul-2023 22:12	04:36:00	0
32	765/400kV 1500MVA ICT-II at Moga	ICT	NR	POWERGRID	SE	21-Jul-2023 19:33	to attend hot spot of 765 KV ICT-2 B -Phase tension string	21-Jul-2023 23:55	04:22:00	0
33	765/400kV 800MVA ICT-IV at KOTESHWAR	ICT	NR	POWERGRID	SE	05-Jul-2023 23:03	to attend low oil pressure	06-Jul-2023 02:19	03:16:00	0
34	765kV AGRA - JHATIKARA Ckt-I	Line	NR	POWERGRID	SE	13-Jul-2023 10:32	to attend broken pilot string of B-phase at location no. 553.	13-Jul-2023 14:47	04:15:00	0
35	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	01-Jul-2023 01:29	Voltage regulation	01-Jul-2023 09:03	07:34:00	0
36	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	02-Jul-2023 17:44	Voltage Regulation	03-Jul-2023 08:23	14:39:00	0
37	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	04-Jul-2023 15:55	Voltage Regulation	05-Jul-2023 09:06	17:11:00	0
38	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	06-Jul-2023 16:37	Voltage regulation	07-Jul-2023 09:27	16:50:00	0
39	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	08-Jul-2023 18:28	Voltage Regulation	09-Jul-2023 09:25	14:57:00	0
40	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	10-Jul-2023 17:45	Voltage Regulation	11-Jul-2023 08:43	14:58:00	0
41	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	12-Jul-2023 18:09	Voltage Regulation	13-Jul-2023 08:41	14:32:00	0
42	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	14-Jul-2023 17:43	Voltage Regulation	15-Jul-2023 08:39	14:56:00	0
43	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	16-Jul-2023 17:53	Voltage Regulation	17-Jul-2023 09:09	15:16:00	0
44	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	19-Jul-2023 16:22	Voltage Regulation	20-Jul-2023 08:59	16:37:00	0
45	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	22-Jul-2023 18:01	Voltage regulation	23-Jul-2023 09:04	15:03:00	0
46	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	24-Jul-2023 22:00	Voltage Regulation	25-Jul-2023 09:35	11:35:00	0
47	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	26-Jul-2023 20:15	Voltage Regulation	27-Jul-2023 10:38	14:23:00	0
48	765kV Ajmer - Bhadla II ckt-I	Line	NR	POWERGRID	MO	28-Jul-2023 17:57	Voltage Regulation	29-Jul-2023 09:43	15:46:00	0
49	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	01-Jul-2023 17:17	Voltage Regulation	02-Jul-2023 08:44	15:27:00	0
50	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	03-Jul-2023 17:38	Voltage Regulation	04-Jul-2023 08:38	15:00:00	0
51	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	05-Jul-2023 17:36	Voltage Regulation	06-Jul-2023 09:34	15:58:00	0
52	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	07-Jul-2023 16:21	Voltage Regulation	08-Jul-2023 09:14	16:53:00	0
53	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	09-Jul-2023 17:33	Voltage Regulation	10-Jul-2023 08:42	15:09:00	0
54	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	11-Jul-2023 17:32	Voltage Regulation	12-Jul-2023 08:35	15:03:00	0
55	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	13-Jul-2023 21:30	Voltage Regulation	14-Jul-2023 08:49	11:19:00	0
56	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	15-Jul-2023 18:02	Voltage Regulation	16-Jul-2023 08:51	14:49:00	0
57	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	18-Jul-2023 18:14	Voltage Regulation	19-Jul-2023 09:01	14:47:00	0
58	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	20-Jul-2023 17:53	Voltage Regulation	21-Jul-2023 09:05	15:12:00	0
59	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	23-Jul-2023 17:06	Voltage Regulation	24-Jul-2023 08:51	15:45:00	0

23.List of Outages of Elements monitored by NLDC for the month of July-2023

S. No.	Name of Transmission Element	Type of the element	Region Involved	Owner/ Agency	Catagory	Tripping Date and Time	Brief Reason/Relay Indication	Restoration Date and Time	Outage Duration	Number of Cumulative tripping in the month
60	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	25-Jul-2023 16:26	Voltage Regulation	26-Jul-2023 09:58	17:32:00	0
61	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	28-Jul-2023 01:55	Voltage Regulation	28-Jul-2023 11:14	09:19:00	0
62	765kV Ajmer - Bhadla II ckt-II	Line	NR	POWERGRID	MO	30-Jul-2023 21:17	Voltage Regulation	31-Jul-2023 08:57	11:40:00	0
63	765kV AJMER - CHITTORGARH Ckt-I	Line	NR	POWERGRID	MO	06-Jul-2023 03:40	Voltage regulation	06-Jul-2023 10:26	06:46:00	0
64	765kV AJMER - CHITTORGARH Ckt-I	Line	NR	POWERGRID	MO	01-Jul-2023 16:40	Voltage Regulation	02-Jul-2023 09:24	16:44:00	0
65	765kV AJMER - CHITTORGARH Ckt-I	Line	NR	POWERGRID	MO	09-Jul-2023 16:23	Voltage Regulation	10-Jul-2023 10:07	17:44:00	0
66	765kV AJMER - CHITTORGARH Ckt-I	Line	NR	POWERGRID	MO	11-Jul-2023 16:29	Voltage regulation-790 kv at ajmer	12-Jul-2023 09:18	16:49:00	0
67	765kV AJMER - CHITTORGARH Ckt-I	Line	NR	POWERGRID	MO	14-Jul-2023 16:46	Voltage Regulation	15-Jul-2023 09:18	16:32:00	0
68	765kV AJMER - CHITTORGARH Ckt-I	Line	NR	POWERGRID	MO	17-Jul-2023 16:07	Voltage Regulation	18-Jul-2023 09:24	17:17:00	0
69	765kV AJMER - CHITTORGARH Ckt-I	Line	NR	POWERGRID	MO	18-Jul-2023 16:10	Voltage Regulation	19-Jul-2023 10:13	18:03:00	0
70	765kV AJMER - CHITTORGARH Ckt-I	Line	NR	POWERGRID	MO	20-Jul-2023 16:53	Voltage Regulation	21-Jul-2023 09:49	16:56:00	0
71	765kV AJMER - CHITTORGARH Ckt-I	Line	NR	POWERGRID	MO	21-Jul-2023 17:12	Voltage Regulation	22-Jul-2023 09:45	16:33:00	0
72	765kV AJMER - CHITTORGARH Ckt-I	Line	NR	POWERGRID	MO	22-Jul-2023 16:35	Voltage regulation	23-Jul-2023 09:52	17:17:00	0
73	765kV AJMER - CHITTORGARH Ckt-I	Line	NR	POWERGRID	MO	26-Jul-2023 03:23	Voltage Regulation	29-Jul-2023 11:08	79:45:00	0
74	765kV AJMER - CHITTORGARH Ckt-II	Line	NR	POWERGRID	MO	02-Jul-2023 16:28	Voltage Regulation	03-Jul-2023 08:43	16:15:00	0
75	765kV AJMER - CHITTORGARH Ckt-II	Line	NR	POWERGRID	MO	08-Jul-2023 04:08	Voltage Regulation	09-Jul-2023 11:36	31:28:00	0
76	765kV AJMER - CHITTORGARH Ckt-II	Line	NR	POWERGRID	MO	10-Jul-2023 16:18	Voltage Regulation	11-Jul-2023 10:01	17:43:00	0
77	765kV AJMER - CHITTORGARH Ckt-II	Line	NR	POWERGRID	MO	13-Jul-2023 17:43	Voltage Regulation	14-Jul-2023 09:40	15:57:00	0
78	765kV AJMER - CHITTORGARH Ckt-II	Line	NR	POWERGRID	MO	15-Jul-2023 17:40	Voltage Regulation	16-Jul-2023 09:20	15:40:00	0
79	765kV AJMER - CHITTORGARH Ckt-II	Line	NR	POWERGRID	MO	15-Jul-2023 17:40	Voltage Regulation	17-Jul-2023 09:37	39:57:00	0
80	765kV AJMER - CHITTORGARH Ckt-II	Line	NR	POWERGRID	MO	19-Jul-2023 15:03	Voltage Regulation	20-Jul-2023 10:22	19:19:00	0
81	765kV AJMER - CHITTORGARH Ckt-II	Line	NR	POWERGRID	MO	23-Jul-2023 16:16	Voltage Regulation	24-Jul-2023 10:05	17:49:00	0
82	765kV AJMER - CHITTORGARH Ckt-II	Line	NR	POWERGRID	MO	29-Jul-2023 13:10	Voltage regulation	31-Jul-2023 09:18	44:08:00	0
83	765kV AJMER - CHITTORGARH Ckt-II	Line	NR	POWERGRID	MO	30-Jun-2023 17:15	Voltage Regulation	01-Jul-2023 09:43	16:28:00	0
84	765kV AJMER - CHITTORGARH Ckt-II	Line	NR	POWERGRID	MO	31-Jul-2023 17:08	Voltage Regulation	01-Aug-2023 00:00	06:52:00	0
85	765kV AJMER - PHAGI Ckt-II	Line	NR	POWERGRID/R RVPNL	SO	27-Jul-2023 12:26	for replacement of damaged insulator.	27-Jul-2023 20:08	07:42:00	0
86	765kV ANTA - PHAGI Ckt-II	Line	NR	Rajasthan/RRV PNL	TL	23-Jul-2023 09:48	Quarterly Maintenance work of line bay equipments i.e CVT,WT,LA & line isolator & Maintenance of 3*80 MVAR line reactors at Phagi	24-Jul-2023 08:03	22:15:00	1
87	765kV ANTA - PHAGI Ckt-II	Line	NR	Rajasthan/RRV PNL	TS	26-Jul-2023 16:59	Tripping of 765 KV Anta-Phagi ckt-2 during testing of Line 3*80 MVAR Line reactor Bank-2 at 16:59 hrs dated 26.07.2023	26-Jul-2023 19:04	02:05:00	2
88	765kV BHADLA - BIKANER Ckt-II	Line	NR	POWERGRID	MO	14-Jul-2023 18:14	Voltage Regulation	15-Jul-2023 08:15	14:01:00	0

23.List of Outages of Elements monitored by NLDC for the month of July-2023

S. No.	Name of Transmission Element	Type of the element	Region Involved	Owner/ Agency	Catagory	Tripping Date and Time	Brief Reason/Relay Indication	Restoration Date and Time	Outage Duration	Number of Cumulative tripping in the month
89	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	01-Jul-2023 17:16	Voltage Regulation	02-Jul-2023 09:15	15:59:00	0
90	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	03-Jul-2023 19:02	Voltage Regulation	04-Jul-2023 07:30	12:28:00	0
91	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	06-Jul-2023 16:34	Voltage regulation	07-Jul-2023 08:47	16:13:00	0
92	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	08-Jul-2023 17:21	Voltage Regulation	09-Jul-2023 10:05	16:44:00	0
93	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	09-Jul-2023 16:44	Voltage Regulation	10-Jul-2023 08:38	15:54:00	0
94	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	11-Jul-2023 17:48	Voltage Regulation	12-Jul-2023 08:22	14:34:00	0
95	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	13-Jul-2023 17:07	Voltage Regulation	14-Jul-2023 08:00	14:53:00	0
96	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	16-Jul-2023 17:17	Voltage Regulation	17-Jul-2023 07:56	14:39:00	0
97	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	18-Jul-2023 16:51	Voltage Regulation	19-Jul-2023 09:10	16:19:00	0
98	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	SE	19-Jul-2023 18:28	Heavy sparking was observed in B-Phase Reactor bay isolator	19-Jul-2023 23:41	05:13:00	0
99	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	20-Jul-2023 17:44	Voltage Regulation	21-Jul-2023 08:31	14:47:00	0
100	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	21-Jul-2023 17:22	Voltage Regulation	22-Jul-2023 08:43	15:21:00	0
101	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	23-Jul-2023 16:45	Voltage Regulation	24-Jul-2023 08:11	15:26:00	0
102	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	27-Jul-2023 17:28	Voltage Regulation	28-Jul-2023 08:45	15:17:00	0
103	765kV BIKANER - MOGA Ckt-I	Line	NR	POWERGRID	MO	29-Jul-2023 17:19	Voltage Regulation	30-Jul-2023 09:23	16:04:00	0
104	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	02-Jul-2023 17:32	Voltage Regulation	03-Jul-2023 08:27	14:55:00	0
105	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	05-Jul-2023 17:38	Voltage Regulation	06-Jul-2023 08:48	15:10:00	0
106	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	07-Jul-2023 17:49	Voltage Regulation	08-Jul-2023 09:31	15:42:00	0
107	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	10-Jul-2023 17:30	Voltage Regulation	11-Jul-2023 08:31	15:01:00	0
108	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	12-Jul-2023 17:37	Voltage Regulation	13-Jul-2023 09:02	15:25:00	0
109	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	14-Jul-2023 17:01	Voltage Regulation	14-Jul-2023 19:19	02:18:00	0
110	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	15-Jul-2023 17:40	Voltage Regulation	16-Jul-2023 08:28	14:48:00	0
111	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	17-Jul-2023 16:53	Voltage Regulation	18-Jul-2023 09:17	16:24:00	0
112	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	19-Jul-2023 15:28	Voltage Regulation	19-Jul-2023 18:23	02:55:00	0
113	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	22-Jul-2023 17:00	Voltage regulation	23-Jul-2023 08:26	15:26:00	0
114	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	24-Jul-2023 16:55	Voltage Regulation	25-Jul-2023 08:21	15:26:00	0
115	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	28-Jul-2023 17:54	Voltage Regulation	29-Jul-2023 08:35	14:41:00	0
116	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	30-Jun-2023 18:42	Voltage Regulation	01-Jul-2023 08:12	13:30:00	0

23.List of Outages of Elements monitored by NLDC for the month of July-2023

S. No.	Name of Transmission Element	Type of the element	Region Involved	Owner/ Agency	Catagory	Tripping Date and Time	Brief Reason/Relay Indication	Restoration Date and Time	Outage Duration	Number of Cumulative tripping in the month
117	765kV BIKANER - MOGA Ckt-II	Line	NR	POWERGRID	MO	31-Jul-2023 17:40	Voltage Regulation	01-Aug-2023 00:00	06:20:00	0
118	765kV Fatehgarh II - Bhadla II Ckt-II	Line	NR	PFTL	TL	30-Jun-2023 22:09	B-N persisting fault	01-Jul-2023 00:23	02:14:00	1
119	765kV FATEHPUR - AGRA Ckt-I	Line	NR	POWERGRID	MO	10-Jul-2023 01:32	Voltage Regulation	13-Jul-2023 10:00	80:28:00	0
120	765kV FATEHPUR - AGRA Ckt-I	Line	NR	POWERGRID	MO	23-Jul-2023 16:21	Voltage Regulation	25-Jul-2023 10:27	42:06:00	0
121	765kV FATEHPUR - AGRA Ckt-I	Line	NR	POWERGRID	MO	29-Jul-2023 08:03	Voltage Regulation	29-Jul-2023 22:24	14:21:00	0
122	765kV FATEHPUR - AGRA Ckt-I	Line	NR	POWERGRID	MO	30-Jul-2023 16:43	Voltage Regulation	30-Jul-2023 20:59	04:16:00	0
123	765kV FATEHPUR - AGRA Ckt-II	Line	NR	POWERGRID	MO	07-Jul-2023 05:59	Voltage regulation	09-Jul-2023 21:18	63:19:00	0
124	765kV FATEHPUR - AGRA Ckt-II	Line	NR	POWERGRID	MO	14-Jul-2023 03:14	Voltage Regulation	18-Jul-2023 20:25	113:11:00	0
125	765kV FATEHPUR - AGRA Ckt-II	Line	NR	POWERGRID	MO	25-Jul-2023 16:33	Voltage Regulation	29-Jul-2023 21:13	100:40:00	0
126	765kV FATEHPUR - AGRA Ckt-II	Line	NR	POWERGRID	MO	30-Jul-2023 02:51	Voltage Regulation	30-Jul-2023 20:00	17:09:00	0
127	765kV FATEHPUR - AGRA Ckt-II	Line	NR	POWERGRID	MO	31-Jul-2023 03:11	Voltage Regulation	31-Jul-2023 10:06	06:55:00	0
128	765kV FATEHPUR - AGRA Ckt-II	Line	NR	POWERGRID	MO	31-Jul-2023 17:19	Voltage regulation	01-Aug-2023 00:00	06:41:00	0
129	765kV KOTESHWAR - MEERUT Ckt-I	Line	NR	POWERGRID	TL	11-Jul-2023 07:15	R-N fault	11-Jul-2023 08:33	01:18:00	1
130	765kV MOGA - BHIWANI Ckt-I	Line	NR	POWERGRID	SE	14-Jul-2023 19:46	to attend hotspot	15-Jul-2023 00:20	04:34:00	0
131	765kV MOGA - BHIWANI Ckt-I	Line	NR	POWERGRID	SE	31-Jul-2023 20:20	to attend the Hot Spot (120 Deg. C) in Line dropper Jumper	31-Jul-2023 22:03	01:43:00	0
132	765kV MOGA - MEERUT Ckt-I	Line	NR	POWERGRID	SO	12-Jul-2023 23:03	To provide safety clearance to manpower working on 765KV Bus-1 at Moga S/S	13-Jul-2023 02:54	03:51:00	0
133	765kV MOGA - MEERUT Ckt-I	Line	NR	POWERGRID	MO	25-Jul-2023 17:53	Voltage Regulation	26-Jul-2023 10:51	16:58:00	0
134	765kV ORAI - ALIGARH Ckt-I	Line	NR	POWERGRID	MO	10-Jul-2023 01:36	Voltage Regulation	10-Jul-2023 09:42	08:06:00	0
135	765kV ORAI - ALIGARH Ckt-I	Line	NR	POWERGRID	MO	22-Jul-2023 16:58	Voltage regulation	24-Jul-2023 10:51	41:53:00	0
136	765kV ORAI - ALIGARH Ckt-I	Line	NR	POWERGRID	MO	30-Jun-2023 03:12	Voltage Regulation	01-Jul-2023 21:12	42:00:00	0
137	765kV ORAI - ALIGARH Ckt-II	Line	NR	POWERGRID	MO	08-Jul-2023 11:32	Voltage Regulation	09-Jul-2023 20:54	33:22:00	0
138	765kV ORAI - ALIGARH Ckt-II	Line	NR	POWERGRID	MO	28-Jul-2023 16:35	Voltage Regulation	01-Aug-2023 00:00	79:25:00	0
139	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	06-Jul-2023 03:50	Voltage regulation	06-Jul-2023 11:43	07:53:00	0
140	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	01-Jul-2023 18:54	Voltage Regulation	02-Jul-2023 09:54	15:00:00	0
141	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	07-Jul-2023 17:57	Voltage Regulation	10-Jul-2023 10:54	64:57:00	0
142	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	13-Jul-2023 17:13	Voltage Regulation	14-Jul-2023 11:00	17:47:00	0
143	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	15-Jul-2023 03:29	Voltage Regulation	15-Jul-2023 10:00	06:31:00	0
144	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	15-Jul-2023 16:20	Voltage Regulation	16-Jul-2023 11:10	18:50:00	0

23.List of Outages of Elements monitored by NLDC for the month of July-2023

S. No.	Name of Transmission Element	Type of the element	Region Involved	Owner/ Agency	Catagory	Tripping Date and Time	Brief Reason/Relay Indication	Restoration Date and Time	Outage Duration	Number of Cumulative tripping in the month
145	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	16-Jul-2023 16:53	Voltage Regulation	17-Jul-2023 10:20	17:27:00	0
146	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	17-Jul-2023 18:15	Voltage Regulation	18-Jul-2023 10:06	15:51:00	0
147	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	18-Jul-2023 17:37	Voltage Regulation	19-Jul-2023 10:41	17:04:00	0
148	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	19-Jul-2023 17:31	Voltage Regulation	20-Jul-2023 11:20	17:49:00	0
149	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	20-Jul-2023 18:04	Voltage Regulation	21-Jul-2023 09:47	15:43:00	0
150	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	21-Jul-2023 18:53	Voltage Regulation	24-Jul-2023 11:03	64:10:00	0
151	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	24-Jul-2023 18:31	Voltage Regulation	25-Jul-2023 10:10	15:39:00	0
152	765kV PHAGI - BHIWANI Ckt-I	Line	NR	RRVPNL/PG	MO	31-Jul-2023 17:42	Voltage regulation	01-Aug-2023 00:00	06:18:00	0
153	765kV PHAGI - BHIWANI Ckt-II	Line	NR	RRVPNL/PG	MO	02-Jul-2023 17:10	Voltage Regulation	03-Jul-2023 08:54	15:44:00	0
154	765kV PHAGI - BHIWANI Ckt-II	Line	NR	RRVPNL/PG	MO	03-Jul-2023 19:17	Voltage Regulation	04-Jul-2023 09:20	14:03:00	0
155	765kV PHAGI - BHIWANI Ckt-II	Line	NR	RRVPNL/PG	MO	04-Jul-2023 18:12	Voltage Regulation	05-Jul-2023 09:38	15:26:00	0
156	765kV PHAGI - BHIWANI Ckt-II	Line	NR	RRVPNL/PG	MO	06-Jul-2023 19:10	Voltage Regulation	07-Jul-2023 10:03	14:53:00	0
157	765kV PHAGI - BHIWANI Ckt-II	Line	NR	RRVPNL/PG	MO	10-Jul-2023 17:24	Voltage Regulation	11-Jul-2023 09:29	16:05:00	0
158	765kV PHAGI - BHIWANI Ckt-II	Line	NR	RRVPNL/PG	MO	12-Jul-2023 17:43	Voltage Regulation	13-Jul-2023 09:36	15:53:00	0
159	765kV PHAGI - BHIWANI Ckt-II	Line	NR	RRVPNL/PG	MO	25-Jul-2023 16:32	Voltage Regulation	27-Jul-2023 09:55	41:23:00	0
160	765kV PHAGI - BHIWANI Ckt-II	Line	NR	RRVPNL/PG	MO	27-Jul-2023 17:56	Voltage Regulation	31-Jul-2023 09:50	87:54:00	0
161	765kV PHAGI - BHIWANI Ckt-II	Line	NR	RRVPNL/PG	MO	30-Jun-2023 17:24	Voltage Regulation	01-Jul-2023 10:23	16:59:00	0
162	765kV VARANASI - KANPUR Ckt-I	Line	NR	POWERGRID	MO	10-Jul-2023 11:22	Voltage regulation	10-Jul-2023 15:02	03:40:00	0
163	800KV HVDC pole-4 at Agra	HVDC	NR	POWERGRID NR	SE	11-Jul-2023 17:03	Hotspot in Pole-4 WT bay isolator with a max. temp. of 88 degree C at Agra	11-Jul-2023 18:48	01:45:00	0
Western Region Elements										
1	240MVAR 765kV Bus Reactor-I at KHANDWA PS	Bus Reactor	WR	KTL-STERLITE	SO	16-Jul-2023 11:41	major oil leakage from 765kV 80MVar Bus reactor B-phase Oil conservator tank	16-Jul-2023 16:25	04:44:00	0
2	240MVAR 765kV Bus Reactor-I at KHANDWA PS	Bus Reactor	WR	KTL-STERLITE	TS	22-Jul-2023 21:49	malfunction of Y-Phase PRV-2	23-Jul-2023 00:16	02:27:00	1
3	765/400kV 1500 MVA ICT-I at Lakadiya	ICT	WR	Adani Trans Ind Ltd	TS	22-Jul-2023 05:27	PRD Spare Phase internal contact faulty of ICT-1	22-Jul-2023 08:34	03:07:00	1
4	765/400kV 1500 MVA ICT-I at Lakadiya	ICT	WR	Adani Trans Ind Ltd	TS	28-Jul-2023 12:09	PRD-3 Spare Phase internal contact faulty of ICT-1	28-Jul-2023 13:31	01:22:00	2
5	765/400kV 1500MVA ICT-II at EKTUNI (Aurangabad 3)	ICT	WR	Maharashtra/ MEGPTCL	SE	12-Jul-2023 13:25	ESD for attending hot-spot.	12-Jul-2023 18:46	05:21:00	0
6	765/400kV 1500MVA ICT-II at JABALPUR PS	ICT	WR	POWERGRID	TS	28-Jul-2023 17:27	differential protection	29-Jul-2023 00:51	07:24:00	1
7	765kV AURANGABAD - PHADGHE Ckt-I	Line	WR	POWERGRID	MO	21-Jul-2023 02:40	Voltage Regulation	22-Jul-2023 17:15	38:35:00	0
8	765kV AURANGABAD - PUNE Ckt-I	Line	WR	POWERGRID	MO	22-Jul-2023 17:15	Voltage regulation	26-Jul-2023 06:55	85:40:00	0
9	765kV AURANGABAD - PUNE Ckt-I	Line	WR	POWERGRID	MO	28-Jun-2023 07:59	Voltage Regulation	21-Jul-2023 02:31	546:32:00	0

23.List of Outages of Elements monitored by NLDC for the month of July-2023

S. No.	Name of Transmission Element	Type of the element	Region Involved	Owner/ Agency	Catagory	Tripping Date and Time	Brief Reason/Relay Indication	Restoration Date and Time	Outage Duration	Number of Cumulative tripping in the month
10	765kV AURANGABAD - PUNE Ckt-I	Line	WR	POWERGRID	MO	29-Jul-2023 00:13	Voltage Regulation	01-Aug-2023 00:00	71:47:00	0
11	765kV AURANGABAD - SOLAPUR Ckt-I	Line	WR	POWERGRID	MO	27-Jun-2023 22:06	Voltage Regulation	03-Jul-2023 10:17	132:11:00	0
12	765kV AURANGABAD - SOLAPUR Ckt-II	Line	WR	POWERGRID	MO	17-Jul-2023 01:16	Voltage Regulation	17-Jul-2023 07:59	06:43:00	0
13	765kV AURANGABAD - SOLAPUR Ckt-II	Line	WR	POWERGRID	MO	18-Jul-2023 02:29	Voltage Regulation	01-Aug-2023 00:00	333:31:00	0
14	765kV BINA - JABALPUR PS Ckt-I	Line	WR	POWERGRID	MO	04-Jul-2023 15:54	Voltage Regulation	22-Jul-2023 23:56	440:02:00	0
15	765kV BINA - JABALPUR PS Ckt-II	Line	WR	POWERGRID	MO	28-Jul-2023 17:56	Voltage Regulation	01-Aug-2023 00:00	78:04:00	0
16	765kV DHARAMJAYGARH - JABALPUR PS Ckt-I	Line	WR	POWERGRID	MO	28-May-2023 16:32	Voltage Regulation	27-Jul-2023 11:55	1435:23:00	0
17	765kV JABALPUR PS - BHOPAL Ckt-I	Line	WR	POWERGRID/B DTCL	SO	29-Jul-2023 11:39	one conductor slipped-out of 765KV-BHOPAL-BDTCL-JABALPUR-1 B-phase dropper at BDTCL-Bhopal substation	29-Jul-2023 15:45	04:06:00	0
18	765kV Lakadia - Banaskantha Ckt-I	Line	WR	Adani Trans Ind Ltd	MO	24-Jun-2023 04:40	Voltage Regulation	15-Jul-2023 10:32	509:52:00	0
19	765kV PUNE - SOLAPUR Ckt-I	Line	WR	POWERGRID	TL	26-Jul-2023 05:03	Pune End: FAULT TYPE: B Phase to Ground Fault FAULT CURRENT: 6.105 kA LOCATION: 75.546km (27.030%) Solapur end FAULT TYPE: YB Phase to Ground Fault FAULT CURRENT: 5.462 kA LOCATION: 279.500km (100.000%)	26-Jul-2023 14:23	09:20:00	1
20	765kV RAIGARH-PS(KOTRA) - RAIPUR-PS(DURG) Ckt-II	Line	WR	POWERGRID	SE	19-Jul-2023 10:18	Gantry Jumper is observed to be slipped from its holding plate	19-Jul-2023 16:35	06:17:00	0
21	765kV RAIPUR-PS(DURG) - CHAMPA PS Ckt-I	Line	WR	POWERGRID	MO	08-Jul-2023 13:57	Voltage regulation	11-Jul-2023 11:30	69:33:00	0
22	765kV RAIPUR-PS(DURG) - CHAMPA PS Ckt-II	Line	WR	POWERGRID	MO	20-Jul-2023 08:55	Voltage Regulation	01-Aug-2023 00:00	279:05:00	0
23	765kV RAIPUR-PS(DURG) - CHAMPA PS Ckt-II	Line	WR	POWERGRID	MO	25-Jun-2023 11:45	Voltage Regulation	04-Jul-2023 11:07	215:22:00	0
24	765kV RAIPUR-PS(DURG) - WARDHA Ckt-I	Line	WR	POWERGRID	MO	26-Jul-2023 04:20	Voltage Regulation	01-Aug-2023 00:00	139:40:00	0
25	765kV RAIPUR-PS(DURG) - WARDHA Ckt-III	Line	WR	POWERGRID	MO	05-Jun-2023 20:08	Voltage Regulation	24-Jul-2023 10:54	1166:46:00	0
26	765kV SASAN - SATNA Ckt-I	Line	WR	SASAN UMPP/POWER GRID	SO	03-Jul-2023 12:20	for rectification of broken OPGW wire between tower no- 22 and 23	03-Jul-2023 21:05	08:45:00	0
27	765kV TIRODA - KORADI Ckt-I	Line	WR	APML/MEGPTCL	TL	23-Jul-2023 21:51	Tripped from Koradi end only Tripped due to 86A & 86B operated	24-Jul-2023 00:00	02:09:00	1
28	765kV TIRODA - KORADI Ckt-II	Line	WR	APML/MEGPTCL	SE	23-Jul-2023 21:51	ESD due to monkey trapped on line	24-Jul-2023 01:19	03:28:00	0
29	765kV WARDHA - AURANGABAD Ckt-II	Line	WR	POWERGRID	MO	21-Jul-2023 01:31	Voltage Regulation	01-Aug-2023 00:00	262:29:00	0
30	765kV WARDHA - AURANGABAD Ckt-III	Line	WR	POWERGRID	MO	02-Jul-2023 01:34	Voltage regulation	18-Jul-2023 08:39	391:05:00	0
31	765kV WARDHA - AURANGABAD Ckt-IV	Line	WR	POWERGRID	SO	10-May-2023 02:31	line previously out in planned shutdown now converted into voltage regulation	15-May-2023 11:54	129:23:00	0
32	765KV/132KV GADARWARA-ICT-2	ICT	WR	NTPC	SO	12-Jul-2023 09:25	IBT-2Y shall be taken in service and IBT-2B shall be taken out for annual PM & testing. Spare IBT will serve as IBT-2B	12-Jul-2023 15:55	06:30:00	0
33	765KV/132KV GADARWARA-ICT-2	ICT	WR	NTPC	SO	26-Jul-2023 10:22	IBT-spare shall be taken out and IBT-2B shall be taken in service	26-Jul-2023 18:32	08:10:00	0
34	765KV/400KV BHOPAL-BDTCL-ICT-2	ICT	WR	BDTCL-Sterlite	SO	12-Jul-2023 11:17	Switching operation of ICT-2 R-Phase into service & Spare phase as standby.	12-Jul-2023 13:26	02:09:00	0
35	765KV/400KV EKTUNI-ICT-2	ICT	WR	MSLDC	SO	27-Jul-2023 10:49	"Diagnostic Testing & Routine Maintenance"	27-Jul-2023 22:09	11:20:00	0
36	765KV/400KV GWALIOR-ICT-1	ICT	WR	POWERGRID	SO	28-Jul-2023 10:28	AMP Works.Dia to be kept open at both sides.	28-Jul-2023 17:00	06:32:00	0
37	765KV/400KV KHANDWA-PS-ICT-1	ICT	WR	KTL-Sterlite	SO	10-Jul-2023 09:55	Shutdown required for 2 Y Testing of ICT & Y ph conservator tank Air cell rectification work.	10-Jul-2023 21:35	11:40:00	0

23. List of Outages of Elements monitored by NLDC for the month of July-2023

S. No.	Name of Transmission Element	Type of the element	Region Involved	Owner/ Agency	Catagory	Tripping Date and Time	Brief Reason/Relay Indication	Restoration Date and Time	Outage Duration	Number of Cumulative tripping in the month
38	765KV/400KV KORADI-III-ICT-1	ICT	WR	MEGPTCL	SO	03-Jul-2023 10:01	As per CEA guideline within six month ICT changeover Activity. R-Ph ICT to be taken in service in place of spare Ph ICT.	03-Jul-2023 22:18	12:17:00	0
39	765KV/400KV KORADI-III-ICT-2	ICT	WR	MEGPTCL	SO	13-Jul-2023 12:11	DGA test of Metering Current Transformer	13-Jul-2023 15:47	03:36:00	0
40	765KV/400KV VINDHYACHAL(PS)-ICT-1	ICT	WR	POWERGRID	SO	17-Jul-2023 13:26	ICT 1 Rph to be taken into service and spare to be out. (Spare Rotation Work). Dia to be kept open at both ends.	17-Jul-2023 18:36	05:10:00	0
41	765KV/400KV VINDHYACHAL(PS)-ICT-3	ICT	WR	POWERGRID	SO	19-Jul-2023 10:47	ICT AMP work at Vindhyachal PS.	19-Jul-2023 23:47	13:00:00	0
42	765KV-AKOLA (2)-KORADI-III-1	Line	WR	MEGPTCL	SO	09-Jul-2023 10:59	As per CEA guideline within six month reactor changeover activity, spare reactor to be taken in service in place of B Ph reactor.	09-Jul-2023 16:28	05:29:00	0
43	765KV-AKOLA (2)-KORADI-III-1 L/R@ KORADI-III - 765KV	Line Reactor	WR	MEGPTCL	SO	09-Jul-2023 10:59	As per CEA guideline within six month reactor changeover activity, spare reactor to be taken in service in place of B Ph reactor.	09-Jul-2023 16:28	05:29:00	0
44	765KV-AKOLA (2)-KORADI-III-2	Line	WR	MEGPTCL	SO	15-Jul-2023 10:35	Six monthly Change over of Spare Line Reactor with B-ph Line Reactor as per CEA regualtion at Akola-II SS. Last changeover date - 20.01.2023	15-Jul-2023 16:07	05:32:00	0
45	765KV-BINA-JABALPUR-1	Line	WR	POWERGRID	SO	04-Jul-2023 11:04	For taking spare Reactor unit into service AT Jabalpur PS end (Due date 11.07.2023)	04-Jul-2023 15:54	04:50:00	0
46	765KV-BINA-JABALPUR-1 L/R@ JABALPUR - 765KV	Line Reactor	WR	POWERGRID	SO	04-Jul-2023 11:20	For taking spare Reactor unit into service (Due date 11.07.2023)	04-Jul-2023 17:09	05:49:00	0
47	765KV-DHULE-AURANGABAD-1	Line	WR	BDTCL-Sterlite	SO	01-Jul-2023 11:19	Line shutdown required for the Switching Operation of Line Reactor as per the instruction of NLDC, LR outage will only provide along with line shutdown, after Switching Operation of Line Reactor Y-Phase into service & Spare phase as standby.	01-Jul-2023 13:21	02:02:00	0
48	765KV-DHULE-AURANGABAD-1 L/R@ DHULE - 765KV	Line Reactor	WR	BDTCL-Sterlite	SO	01-Jul-2023 11:19	Switching Operation of Line Reactor, Y-Phase into service & Spare phase as standby.	01-Jul-2023 13:21	02:02:00	0
49	765KV-RAIGARH-PS-RAIPUR-PS (DURG)-2	Line	WR	POWERGRID WR1	SO	04-Jul-2023 11:54	LINE AMP WORK	04-Jul-2023 19:42	07:48:00	0
50	765KV-RAJNANDGAON-RRWTL-WARORA-PG-1	Line	WR	RRWTL	SO	14-Jul-2023 10:02	As per CEA measures related to Safety & Electric supply 2018 Spare Reactor shall be charged within 6 months of approval.	14-Jul-2023 16:36	06:34:00	0
51	765KV-RAJNANDGAON-RRWTL-WARORA-PG-1 L/R@ RAJNANDGAON-RRWTL-WARORA-PG-1	Line Reactor	WR	RRWTL	SO	14-Jul-2023 10:02	As per CEA measures related to Safety & Electric supply 2018 Spare Reactor shall be charged within 6 months of approval.	14-Jul-2023 16:36	06:34:00	0
52	765KV-SEONI-WARDHA-2	Line	WR	POWERGRID WR1	SO	26-Jul-2023 10:58	For spare normalization i.e. taking spare out of service. Last date of spare taken in to service was on 04.05.2023 in place of 765KV warha-seoni-2 B-ph line reactor	26-Jul-2023 20:17	09:19:00	0
53	765KV-SEONI-WARDHA-2 L/R@ WARDHA - 765KV	Line Reactor	WR	POWERGRID WR1	SO	26-Jul-2023 10:58	For spare normalization i.e. taking spare out of service. Last date of spare taken in to service was on 04.05.2023 in place of 765KV warha-seoni-2 B-ph line reactor	26-Jul-2023 20:17	09:19:00	0
54	765KV-TIRODA-KORADI-III-1	Line	WR	MEGPTCL	SO	23-Jul-2023 10:49	Installation of Bird Flight Divertor- In reference to the Hon'zble High Court, Nagpur Order dated 05th January 2022	23-Jul-2023 20:17	09:28:00	0
55	765KV-TIRODA-KORADI-III-1	Line	WR	MEGPTCL	SO	24-Jul-2023 10:26	Installation of Bird Flight Divertor- In reference to the Hon'zble High Court, Nagpur Order dated 05th January 2022	24-Jul-2023 20:52	10:26:00	0
56	765KV-TIRODA-KORADI-III-1	Line	WR	MEGPTCL	SO	25-Jul-2023 10:19	Installation of Bird Flight Divertor- In reference to the Hon'zble High Court, Nagpur Order dated 05th January 2022	25-Jul-2023 21:09	10:50:00	0
57	765KV-TIRODA-KORADI-III-2	Line	WR	MEGPTCL	SO	16-Jul-2023 11:47	As per CEA guideline within six month reactor changeover activity, Y- Phase reactor to be taken in service in place of spare reactor.	16-Jul-2023 17:21	05:34:00	0
58	765KV-TIRODA-KORADI-III-2	Line	WR	MEGPTCL	SO	27-Jul-2023 08:32	Installation of Bird Flight Divertor- In reference to the Hon'zble High Court, Nagpur Order dated 05th January 2022	27-Jul-2023 22:21	13:49:00	0
59	765KV-TIRODA-KORADI-III-2	Line	WR	MEGPTCL	SO	30-Jul-2023 09:34	For bird diverter installation	30-Jul-2023 20:32	10:58:00	0
60	765KV-TIRODA-KORADI-III-2 L/R@ KORADI-III - 765KV	Line Reactor	WR	MEGPTCL	SO	16-Jul-2023 11:48	As per CEA guideline within six month reactor changeover activity, Y- Phase reactor to be taken in service in place of spare reactor.	16-Jul-2023 17:21	05:33:00	0
61	765KV-WARDHA-RAIPUR-PS (DURG)-4	Line	WR	POWERGRID WR1	SO	24-Jul-2023 11:09	For Line Bay Amp Works	24-Jul-2023 22:39	11:30:00	0
62	765KV-WARDHA-RAIPUR-PS (DURG)-4 L/R@ WARDHA - 765KV	Line Reactor	WR	POWERGRID WR1	SC	24-Jul-2023 11:09	For Line Reactor Amp works	24-Jul-2023 22:39	11:30:00	0
63	AURANGABAD - 765KV B/R 2	Bus Reactor	WR	POWERGRID WR1	SO	21-Jul-2023 08:35	BR-2 AMP Work and Tie Bay CDS Tuning.	22-Jul-2023 21:52	37:17:00	0
64	DHARAMJAYGARH - 765KV B/R 3	Bus Reactor	WR	POWERGRID WR1	SO	15-Jul-2023 09:12	For Non-AMP nature of works	15-Jul-2023 17:37	08:25:00	0
65	SOLAPUR - 765KV B/R 2	Bus Reactor	WR	POWERGRID WR1	SO	03-Jul-2023 08:17	Normalisation of Spare Change over as per CEA norms	03-Jul-2023 15:31	07:14:00	0
66	WARDHA - 765KV B/R 3	Bus Reactor	WR	POWERGRID WR1	SO	13-Jul-2023 10:28	For taking spare in to service.Last date of spare taken out of service was on 18.01.2023	13-Jul-2023 19:57	09:29:00	0

23.List of Outages of Elements monitored by NLDC for the month of July-2023

S. No.	Name of Transmission Element	Type of the element	Region Involved	Owner/ Agency	Catagory	Tripping Date and Time	Brief Reason/Relay Indication	Restoration Date and Time	Outage Duration	Number of Cumulative tripping in the month
Eastern Region Elements										
1	330MVAR 765KV B/R-2 AT MEDINIPUR	Bus Reactor	ER	ER-II	SO	13-Jul-2023 10:41	CT SF6 Lockout Logic Modification	13-Jul-2023 13:40	02:59:00	0
2	240MVAR 765KV B/R-1 AT JHARSUGUDA	Bus Reactor	ER	PG (ODISHA PROJECT)	SO	28-Jul-2023 10:00	Yearly AMP and Spare changeover	28-Jul-2023 15:02	05:02:00	0
3	800KV APD pole 4	HVDC	ER	POWERGRID ER	SE	11-Jul-2023 15:40	Hotspot in Pole-4 WT bay isolator with a max. temp. of 88 degree C at Agra	11-Jul-2023 18:50	03:10:00	0
4	765KV/400KV 1500 MVA ICT 1 AT MEDINIPUR	ICT	ER	ER-II	SO	13-Jul-2023 15:07	CT SF6 Lockout Logic Modification	13-Jul-2023 18:14	03:07:00	0
5	765KV/400KV 1500 MVA ICT 2 AT MEDINIPUR	ICT	ER	ER-II	SO	15-Jul-2023 10:39	CT SF6 Lockout Logic Modification	15-Jul-2023 12:49	02:10:00	0
6	765KV/400KV 1500 MVA ICT 2 AT JHARSUGUDA	ICT	ER	PG (ODISHA PROJECT)	SO	17-Jul-2023 09:37	06 Monthly spare changeover	17-Jul-2023 14:57	05:20:00	0
7	765KV/400KV 1500 MVA ICT 1 AT NEW JEERAT	ICT	ER	ER-II	SO	22-Jul-2023 10:29	CT SF6 lockout logic modification	22-Jul-2023 14:40	04:11:00	0
8	765/400kv 1500MVA ICT-II AT ANGUL	ICT	ER	POWERGRID	SE	26-Jul-2023 23:30	Hand tripped due to sparking	27-Jul-2023 01:07	01:37:00	0
9	765KV/400KV 1500 MVA ICT 2 AT NEW JEERAT	ICT	ER	ER-II	SO	31-Jul-2023 11:44	CT SF6 lockout logic modification	31-Jul-2023 16:48	05:04:00	0
10	765kv NEW RANCHI - MEDINIPUR Ckt-I	Line	ER	POWERGRID/P MJTL	SO	27-Jul-2023 09:56	REPLACEMENT OF CLR INSULATOR DAMAGED DUE TO LIGHTING	27-Jul-2023 19:33	09:37:00	0
11	765 kv ANGUL - JHARSUGUDA Ckt-IV	Line	ER	POWERGRID	MO	05-Jul-2023 13:31	Voltage Regulation	01-Aug-2023 00:00	634:29:00	0
12	765kv Medinipur - New Jeerut Ckt-II	Line	ER	PMJTL	SO	09-Jul-2023 06:53	Replacement of damage insulators at Tower No-163	09-Jul-2023 13:28	06:35:00	0
13	765KV-NEW RANCHI-MEDINIPUR-2	Line	ER	ER-II	SO	20-Jul-2023 09:34	REPLACEMENT OF CLR INSULATOR DAMAGED DUE TO LIGHTING	20-Jul-2023 19:33	09:59:00	0
14	765KV-NEW RANCHI-MEDINIPUR-2	Line	ER	ER-II	SO	21-Jul-2023 09:04	REPLACEMENT OF CLR INSULATOR DAMAGED DUE TO LIGHTING	21-Jul-2023 19:39	10:35:00	0
15	765KV-NEW RANCHI-MEDINIPUR-2	Line	ER	ER-II	SO	23-Jul-2023 09:28	REPLACEMENT OF CLR INSULATOR DAMAGED DUE TO LIGHTING	23-Jul-2023 20:55	11:27:00	0
16	765KV-NEW RANCHI-MEDINIPUR-2	Line	ER	ER-II	SO	24-Jul-2023 09:00	REPLACEMENT OF CLR INSULATOR DAMAGED DUE TO LIGHTING	24-Jul-2023 20:12	11:12:00	0
17	765KV-NEW RANCHI-MEDINIPUR-1	Line	ER	ER-II	SO	25-Jul-2023 09:45	REPLACEMENT OF CLR INSULATOR DAMAGED DUE TO LIGHTING	25-Jul-2023 20:02	10:17:00	0
18	765KV-NEW RANCHI-MEDINIPUR-1	Line	ER	ER-II	SO	26-Jul-2023 09:36	REPLACEMENT OF CLR INSULATOR DAMAGED DUE TO LIGHTING	26-Jul-2023 19:28	09:52:00	0
19	765KV-NEW RANCHI-MEDINIPUR-1	Line	ER	ER-II	SO	28-Jul-2023 09:49	REPLACEMENT OF CLR INSULATOR DAMAGED DUE TO LIGHTING	28-Jul-2023 18:29	08:40:00	0
20	765KV-NEW RANCHI-MEDINIPUR-1	Line	ER	ER-II	SO	29-Jul-2023 09:25	REPLACEMENT OF CLR INSULATOR DAMAGED DUE TO LIGHTING	29-Jul-2023 19:54	10:29:00	0
Southern Region Elements										
1	765KV/400KV MAHESHWARAM_PG-ICT-2	ICT	SR	pgsr1_sd	SO	08-Jul-2023 08:03	AMP works of ICT-II, ICT-II bays (765kv and 400kv side)	09-Jul-2023 22:13	38:10:00	0
2	765KV/400KV CUDDAPAH-ICT-2	ICT	SR	pgsr1_sd	SO	13-Jul-2023 08:50	amp works	13-Jul-2023 20:33	11:43:00	0
3	765/400kv 1500MVA ICT-II at VEMAGIRI	ICT	SR	POWERGRID	TS	22-Jul-2023 05:30	to maloperation of gas zone protection (zone 8 in 812 bay).	22-Jul-2023 11:01	05:31:00	1
4	765kv KURNOOL - CUDDAPAH Ckt-I	Line	SR	POWERGRID	MO	05-Jul-2023 20:23	Voltage regulation	21-Jul-2023 11:40	375:17:00	0
5	765KV-THIRUVALAM-ARIYALUR-1 L/R@ THIRUVALAM - 7	Line Reactor	SR	pgsr2	SO	24-Jul-2023 12:25	For connection of jumper between AP 302 to dead end tower by TANTRANSCO	01-Aug-2023 00:00	179:35:00	0
6	765kv CHILAKALURIPETA - CUDDAPAH Ckt-II	Line	SR	POWERGRID	MO	05-Jul-2023 20:36	Voltage regulation	21-Jul-2023 12:05	375:29:00	0
7	765kv NIZAMABAD - MAHESHWARAM Ckt-I	Line	SR	POWERGRID	MO	06-Jul-2023 20:29	Voltage regulation	10-Jul-2023 07:21	82:52:00	0

23.List of Outages of Elements monitored by NLDC for the month of July-2023

S. No.	Name of Transmission Element	Type of the element	Region Involved	Owner/ Agency	Catagory	Tripping Date and Time	Brief Reason/Relay Indication	Restoration Date and Time	Outage Duration	Number of Cumulative tripping in the month
8	765kV NIZAMABAD - MAHESHWARAM Ckt-II	Line	SR	POWERGRID	MO	10-Jul-2023 19:17	Voltage Regulation	01-Aug-2023 00:00	508:43:00	0
9	765kV NIZAMABAD - MAHESHWARAM Ckt-II	Line	SR	POWERGRID	MO	15-Jun-2023 17:46	Voltage Regulation	05-Jul-2023 10:24	472:38:00	0
10	765kV CUDDAPAH - THIRUVALAM Ckt-I	Line	SR	POWERGRID	MO	19-Jun-2023 20:00	Voltage regulation	05-Jul-2023 10:36	374:36:00	0
11	765kV CHILAKALURIPETA - CUDDAPAH Ckt-I	Line	SR	POWERGRID	MO	19-Jun-2023 20:46	Voltage Regulation	05-Jul-2023 11:10	374:24:00	0
12	765kV VEMAGIRI - CHILAKALURIPETA Ckt-I	Line	SR	POWERGRID	MO	21-Jul-2023 21:24	Voltage Regulation	01-Aug-2023 00:00	242:36:00	0
13	765 kV-KURNOOL_(PG)-MAHESHWARAM_PG-II	Line	SR	POWERGRID SR	MO	21-Jul-2023 23:19	Voltage Regulation	23-Jul-2023 14:42	39:23:00	0
14	765kV CUDDAPAH - THIRUVALAM Ckt-II	Line	SR	POWERGRID	MO	23-Jul-2023 23:55	Voltage regulation	24-Jul-2023 08:14	08:19:00	0
15	765 kV-KURNOOL_(PG)-MAHESHWARAM_PG-I	Line	SR	POWERGRID SR	MO	24-Jul-2023 18:00	Voltage Regulation	01-Aug-2023 00:00	174:00:00	0
16	765kV CUDDAPAH - THIRUVALAM Ckt-II	Line	SR	POWERGRID	MO	24-Jul-2023 22:52	Voltage Regulation	01-Aug-2023 00:00	169:08:00	0
17	765kV KURNOOL - CUDDAPAH Ckt-II	Line	SR	POWERGRID	MO	31-Jul-2023 23:03	Voltage regulation	01-Aug-2023 00:00	00:57:00	0

North Eastern Region Elements

Nil

	TL= Tripping due to Line Fault
	TS= Tripping due to S/S Fault
	MO= Opened due to system constraints
	TV= Tripped on Overvoltage
	SC= Planned Shutdown for construction activities
	SO= Planned Shutdown for O&M activities
	SE= Emergency Shutdown
	TP= Protection Maloperation
	ME= Miscellaneous outage

ALL TIME HIGHEST

31-07-2023

	Maximum Demand Met during the day (MW)	Demand Met during Evening Peak hrs(MW)	Energy Met (MU)	Hydro Gen. (MU)	Wind Gen. (MU)	Solar Gen. (MU)
NR	77569 21-07-2023 (14:25)	76072 23-06-2023	1737 28-06-2022	441 30-07-2023	78 09-06-2022	147 10-05-2023
WR	71321 27-12-2022	63593 18-04-2023	1539 18-04-2023	167 18-12-2014	272 22-05-2022	70.69 12-05-2023
SR	64314 15-03-2023	52366 18-04-2023	1329 19-04-2023	208 31-08-2018	298 21-07-2023	142.7 21-03-2023
ER	29641 28-07-2023 (20:35)	28579 18-04-2023	639 08-06-2023	157 14-09-2022	-	7 23-03-2023
NER	3603 17-08-2022	3510 12-08-2022	69 11-08-2022	41 01-08-2022	-	2 22-06-2022
All India	223235 09-06-2023	211250 22-06-2023	4914 13-06-2023	877 30-08-2022	555 22-05-2022	351 10-05-2023

Regions	States	Max. Demand Met during the day (MW)		Energy Consumption (MU)	
			As on date		As on date
NR	Punjab	15237	23-06-2023	343.5	24-06-2023
	Haryana	12540	28-06-2022	266.2	07-07-2021
	Rajasthan	17097	23-01-2023	332.2	13-06-2023
	Delhi	7770	29-06-2022	153.5	28-06-2022
	UP	28553	25-07-2023 (21:00)	577.5	24-07-2023
	Uttarakhand	2594	14-06-2022	55.8	17-06-2023
	HP	2071	06-01-2023	39.6	05-06-2022
	J&K(UT) and Ladakh(UT)	3636	29-06-2023	65.4	20-01-2023
	Chandigarh	426	08-07-2021	8.4	08-07-2021
	Railways NR ISTS	-	-	-	-
WR	Chhattisgarh	5642	18-04-2023	129.8	19-04-2023
	Gujarat	22225	09-06-2023	462.3	09-06-2023
	MP	17238	13-01-2023	329.3	10-01-2023
	Maharashtra	28846	14-04-2022	636.1	21-06-2023
	Goa	776	07-06-2023	17.1	08-06-2023
	DD & DNH	1311	21-06-2023	30.7	24-06-2023
	AMNSIL	977	18-08-2021	21.0	31-05-2022
	Balco	-	-	-	-
SR	Andhra Pradesh*	12900	17-06-2023	263.8	16-06-2023
	Telangana*	15497	30-03-2023	303.4	03-03-2023
	Karnataka	16110	20-04-2023	309.5	20-04-2023
	Kerala	5024	18-04-2023	103.3	19-04-2023
	Tamil Nadu	19387	20-04-2023	415.4	20-04-2023
	Pondy	512	17-05-2023	11.6	18-05-2023
ER	Bihar	7578	27-07-2023 (22:46)	161.3	28-07-2023
	DVC	3751	18-04-2023	81.2	22-04-2022
	Jharkhand	1923	16-06-2023	43.3	17-06-2023
	Odisha	7192	19-04-2023	145.5	25-09-2022
	West Bengal	11868	18-04-2023	245.7	18-04-2023
	Sikkim	137	03-02-2022	2.5	28-01-2020
	Railways ER ISTS	-	-	-	-
NER	Arunachal Pradesh	172	31-05-2023	3.1	07-06-2023
	Assam	2413	31-07-2023 (22:07)	47.6	07-06-2023
	Manipur	258	31-01-2022	3.7	10-02-2022
	Meghalaya	415	09-02-2022	7.8	31-01-2022
	Mizoram	157	16-01-2023	2.3	18-01-2023
	Nagaland	170	08-06-2023	3.14	29-07-2023
	Tripura	370	07-06-2023	7.0	06-06-2023
*SR	Andhra Pradesh (Undivided)	13162	23-03-2014	284.8	22-03-2014

**GRID CONTROLLER OF INDIA LIMITED
NATIONAL LOAD DESPATCH CENTRE
NEW DELHI**

Date: 04-Aug-2023

System Reliability Indices Report for the month of July 2023

Percentage (%) of times ATC was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	74	18.50	2.49
2	ER-NR	0	0.00	0.00
3	Import of NR	0	0.00	0.00
4	NEW-SR	0	0.00	0.00
5	NER Import	0	0.00	0.00

Percentage(%) of times (N-1) Criteria was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	25	6.25	0.84
2	ER-NR	0	0.00	0.00
3	Import of NR	0	0.00	0.00
4	NEW-SR	0	0.00	0.00
4	NER Import	0	0.00	0.00

Remarks: Flows crossing Total Transfer Capability (TTC) on interregional corridors has been worked out as a proxy for (N-1) violation.



POWER SYSTEM OPERATION CORPORATION LIMITED

पावर सिस्टम ऑपरेशन कारपोरेशन लिमिटेड

NATIONAL LOAD DESPATCH CENTRE – राष्ट्रीय भार प्रेषण केन्द्र

First floor, B-9, Qutab Institutional Area,
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